Interdependent Integrative Financial Planning Theory: Fiber Bundles, Priority Filtering, and the Irreducible Complexity of Client-Centric Optimization

Michael G. Kothakota

December 29, 2025

Abstract

Financial planning confronts a problem of complexity: decisions across six interconnected domains must be optimized simultaneously for clients whose objectives differ fundamentally from one another and from the wealth-maximization assumptions embedded in standard economic models. IIFPT develops the foundations necessary for understanding this complexity. The space of financial states is modeled as a smooth manifold with corners and planning domains are represented as fiber bundles over this base space. This geometric structure is forced by the topological discontinuities that occur at lifecycle phase transitions. The central theoretical concept introduced is a four-tier decomposition separating: (1) structural interdependencies (objective mechanical linkages from tax law, actuarial mathematics, and portfolio theory), captured in the Structural Tensor $\mathbf{S}$; (2a) client-specific priorities (subjective weightings that vary across households), captured in the Priority Matrix Λ; (2b) temporal priority structure (domain-specific discount rates reflecting how urgently outcomes in each domain are weighted over time), captured in the Discount Matrix $\mathbf{B}(t)$; and (3) their synthesis in the Effective Interdependency Tensor T_{eff} and discount-adjusted coupling matrix Γ_B that determine client-relevant integration. I prove thirteen theorems characterizing how priorities filter structural coupling, when domains can be treated independently under stochastic rates, how sensitive recommendations are to priority specification and how rate uncertainty affects stability, integration value, and shock propagation. An implication is that the high dimensionality of integrated financial planning provides rigorous theoretical justification for how financial planners engage in their work. Human planners serve as heuristic dimensionality reduction operators whose judgment makes otherwise intractable problems computationally feasible. The integration premium increases by 10–28% under stochastic rates through volatility hedging, jump coordination, and coupling amplification channels.

Keywords: financial planning, fiber bundles, dynamic optimization, priority filtering, non-smooth analysis, viscosity solutions, client-centric planning, stochastic discount rates, jump-diffusion, integro-differential equations, domain-specific discounting

Acknowledgements: Generative AI was used to check consistency of language, notation and citations throughout the document. All analysis, conjecture, interpretation and conclusions are the responsibility of the author.

1 | Introduction

1.1 | The Irreducible Complexity of Financial Planning

Financial planners navigate a web of interconnected decisions spanning tax strategy, retirement income sequencing, estate transfer optimization, risk management, and investment selection. A recommendation in one domain propagates effects through others: current tax liability changes, future required minimum distributions shift, estate composition transforms, Medicare premium surcharges may trigger, and portfolio rebalancing needs emerge. That is, the decision space creates interdependencies in the integrative nature of personal finance. Practitioners develop expertise in managing these interdependencies through experience, yet academic research provides little formal guidance for this integration. In this paper, I propose a descriptive process illustrating how financial planners provide value: the Interdependent Integrative Financial Planning Theory (IIFPT).

CFP Board (2022) defines financial planning as a collaborative process spanning six domains: cash flow management, retirement planning, tax planning, estate planning, risk management and insurance, and investment planning. Professional standards emphasize integration and require practitioners to consider how recommendations in one area affect others. Yet academic literature has developed each domain largely in isolation, treating interdependencies as complications rather than essential structure.

The practical importance of understanding this complexity is well documented. Hanna and Lindamood (2010) provide empirical evidence quantifying the economic benefits of personal financial planning, estimating substantial welfare gains for households receiving professional guidance. Chatterjee and Zahirovic-Herbert (2014) and Heckman et al. (2025) demonstrated that households receiving professional financial planning achieve measurably better outcomes across multiple dimensions but could not identify the specific mechanisms driving these gains. I propose those mechanisms lie in the integration that professional financial planners provide. The development of financial planning as an academic discipline, documented by Warschauer (2002), has created practical expertise that outpaces theoretical foundations. IIFPT was developed to close that gap.

1.2 | Beyond Wealth Maximization: Client-Objective-Centric Planning

IIFPT is explicitly client-objective-centric, distinguishing it from two dominant traditions in financial economics. The investment-centric tradition, originating with Markowitz (1952) and extended through Merton (1969, 1971), treats portfolio optimization as the central problem and other domains as constraints. The consumption-smoothing tradition, exemplified by the lifecycle models of Modigliani and Brumberg (1954) and extended by Cocco, Gomes, and Maenhout (2005), treats utility from consumption as the objective and other considerations as instrumental.

Neither tradition captures what financial planners do or what clients require. Real clients have heterogeneous, multi-dimensional objectives irreducible to wealth maximization or consumption smoothing. Some clients prioritize legacy such as ensuring inheritances or charitable support in perpetuity. Others prioritize security, minimizing the probability of outliving resources, even at cost to expected wealth. Pfau (2019) articulates this "safety-first" perspective for retirement planning, emphasizing that many clients seek to secure essential income against longevity and market risks rather than maximize expected wealth. Still others prioritize flexibility, maintaining

optionality for career changes or lifestyle experiments. Some prioritize tax efficiency; others accept taxes as a price for liquidity and simplicity.

These priority differences are not errors to correct or biases to overcome. They reflect genuine variation in client values, which any adequate framework must accommodate. Two clients with identical financial situations may optimally pursue different strategies because they have different objectives. The framework developed here makes this client heterogeneity mathematically explicit through the Priority Matrix Λ, which captures how much weight each client assigns to each planning domain and the Discount Matrix $\mathbf{B}(t)$, which is the stochastic temporal rates feeding into the value function.

1.3 | Separating Structure from Values and Temporal Priorities

The central insight of IIFPT is that financial planning involves two distinct elements requiring separate analysis before combination:

Structural Interdependencies (The "Physics" of Financial Planning): These are objective linkages arising from tax law, actuarial tables, and portfolio mathematics. They exist regardless of client preferences. Examples include: how Roth conversions affect future RMDs (a consequence of tax code provisions); how mortality rates affect life insurance pricing (actuarial necessity); how asset allocation affects portfolio volatility (mathematical property of covariance matrices). These are captured through the Structural Tensor S.

Client Priorities (The "Values" of Financial Planning): These are subjective weightings reflecting how much the client cares about each domain's objectives. They vary across clients with similar financial situations. Examples include: whether retirement security or legacy matters more; whether current consumption or future wealth dominates; whether tax minimization or liquidity preservation takes priority. These are captured through the Priority Matrix Λ.

Temporal Priority Structure (The "Urgency" of Financial Planning): These are domain-specific discount rates reflecting how urgently outcomes in each domain are weighted over time. A client may care deeply about both cash flow and estate outcomes, yet discount cash flow improvements at a high rate (demanding immediate results) while discounting estate improvements at a low rate (patient about legacy outcomes). Critically, these temporal priorities are not fixed but evolve stochastically and respond to financial states through endogenous channels and to life events through exogenous shocks. These are captured through the Discount Matrix $\mathbf{B}(t)$, where each domain-specific rate $\beta_k(t)$follows a jump-diffusion process.

The synthesis proceeds in stages: $T_{\text{eff}} = \mathrm{S} \odot \Lambda$ combines structure with values; $\Gamma_B(t) = \mathbf{B}(t)^{-1/2} T_{\text{eff}} \mathbf{B}(t)^{-1/2}$ further incorporates temporal structure. This decomposition explains why practitioners cannot give identical advice to financially identical clients. Effective interdependencies differ because both priorities and temporal structures differ.

1.4 | Paper Structure

The remainder of this paper proceeds as follows. Section 2 reviews related literature across life-cycle models, household finance, financial planning practice, dynamic programming, non-smooth analysis, and multi-objective optimization, positioning the IIFPT approach within and

against these traditions. Section 3 develops the complete mathematical foundations: financial primitives, the primitive manifold with corners, planning domains as fiber bundles, admissible strategies, and state transition dynamics with well-posedness, establishing why this approach is necessary for capturing planning complexity. Section 4 presents the four-tier priority-filtered interdependency structure, the paper's central theoretical innovation, including the Structural Tensor, Priority Matrix, Temporal Priority Structure with stochastic domain-specific discount rates, Effective Tensor, discount-adjusted coupling matrix, the relationship between Clarke and viscosity frameworks, time-varying priorities, and priority elicitation. Section 5 develops the uncertainty framework distinguishing endogenous from exogenous risk, introducing priority uncertainty, and characterizing rate uncertainty as a distinct category. Section 6 formulates the dynamic optimization problem with domain-specific discounting, establishing the coupled integro-differential Hamilton-Jacobi-Bellman system, viscosity solution framework extended for nonlocal jump operators, and the regime-switching formulation as a tractable alternative. Section 7 proves thirteen main theorems with complete mathematical arguments. Section 8 provides detailed applications featuring six recurring client personas, demonstrating how the framework illuminates real planning decisions through the lens of human stories, enhanced with stochastic rate analysis showing integration premium amplification of 10–28%. Section 9 discusses theoretical contributions, the theoretical necessity of human planners, practical implications, limitations, and future research directions. Section 10 concludes. Six appendices provide extended proofs, empirical calibration considerations, sub-domain element taxonomy, and computational implementation notes.

2 | Literature and Theoretical Foundations

2.1 | Life-Cycle Models and Household Finance

The modern theory of household financial decision-making originates with Modigliani and Brumberg's (1954) life-cycle hypothesis, which posits that individuals smooth consumption over their lifetime by saving during working years and dissaving in retirement. Friedman's (1957) permanent income hypothesis provided a complementary framework emphasizing expected lifetime resources in determining current consumption. These theories established that household financial decisions should be understood in an intertemporal context rather than as isolated period-by-period choices.

Merton (1969, 1971) extended mean-variance analysis to continuous time, deriving optimal consumption and investment policies under uncertainty and showing how optimal portfolio allocation depends on risk aversion, investment horizon, and stochastic properties of asset returns. Campbell and Viceira (2002) synthesized these developments into a treatment of strategic asset allocation over the life cycle, incorporating time-varying investment opportunities.

Bodie, Merton, and Samuelson (1992) incorporated labor supply flexibility into lifecycle portfolio choice, demonstrating that the ability to adjust labor supply provides insurance that affects optimal portfolio allocation. Their insight that human capital characteristics should influence financial decisions informs my treatment of human capital as a fundamental state variable and reinforces the importance of cross-domain integration.

Cocco, Gomes, and Maenhout (2005) provided a calibrated life-cycle model incorporating risky labor income with realistic correlation structure, borrowing constraints, and mortality risk. Their

model explains observed portfolio allocations with age-related shifts toward bonds. Extensions have incorporated housing decisions (Cocco, 2005), buffer-stock saving behavior (Carroll, 1997; Deaton, 1991), and liquidity constraints (Zeldes, 1989).

The incomplete markets literature provides additional foundations. Aiyagari (1994) demonstrated how uninsured idiosyncratic risk generates precautionary saving motives, establishing that households facing income uncertainty will over-save relative to complete-markets benchmarks. Huggett (1996) extended these insights to lifecycle wealth distributions. These models highlight the importance of incorporating realistic constraints and uncertainties, central to developing the feasible region $\mathcal{M}_0$ as a manifold with corners.

Campbell (2006) documented systematic deviations from normative models in his survey of household finance: under-diversification, non-participation in stock markets, mortgage refinancing failures, and inadequate insurance coverage. The framework developed here provides structural explanations for some of these patterns: apparent "mistakes" may reflect rational responses to interdependencies not captured in single-domain models, or may reflect priority structures that researchers incorrectly assume to be uniform across households.

2.2 | Differential Discounting and Time Preference Heterogeneity. The assumption of a single, constant discount rate has been challenged on both behavioral and structural grounds. Laibson (1997) introduced quasi-hyperbolic discounting to capture present bias, defined as the tendency for individuals to disproportionately weight immediate rewards. The dual-self model of Fudenberg and Levine (2006) provides microfoundations for such preferences through the interaction of a myopic "doer" and a patient "planner."

2.3 | Risk Management and Safety-First Approaches

The theoretical foundations of risk management trace to Arrow's (1971) work on risk-bearing, which established economic principles underlying insurance markets and optimal risk transfer. Arrow demonstrated that risk-averse individuals will demand insurance when actuarially fair contracts are available and characterized the welfare gains from risk pooling. These principles underlie treatment of the risk management domain R_{RM}, which measures coverage adequacy relative to exposures across mortality, disability, longevity, liability, and property risks.

The safety-first approach to retirement planning, articulated by Pfau (2019), provides a philosophical foundation aligning with the priority-weighted framework. Pfau argues that retirement planning should prioritize securing essential income against longevity and market risks rather than maximizing expected wealth. This perspective recognizes that for many retirees, the disutility of running out of money exceeds the utility of additional wealth, justifying strategies that sacrifice expected returns for downside protection. In the present framework, such preferences are captured through high weights on w_{Ret} (retirement security) and w_{RM} (risk management) relative to w_{Inv} (investment performance).

2.4 | Goals-Based Planning and Mental Accounts

A distinct tradition in wealth management has emphasized goals-based frameworks that align with the priority-weighted approach. Brunel (2015) articulates a goals-based wealth management methodology structuring advisory practices around client objectives rather than portfolio

benchmarks. His framework recognizes that clients think in terms of specific goals such as retirement income, education funding and legacy, rather than abstract risk-return tradeoffs, and that effective wealth management must honor this cognitive reality.

Chhabra (2005) develops a wealth allocation framework designed to move forward by recognizing that individual investors have heterogeneous objectives not captured by mean-variance optimization alone. Chhabra decomposes wealth into three "buckets," each with distinct objectives and appropriate investment strategies. This decomposition anticipates the domain structure developed here, though the six-domain framework provides finer granularity.

Das, Markowitz, Scheid, and Statman (2010) formalize this intuition through portfolio optimization with mental accounts, demonstrating that investors who segregate wealth into distinct purpose-driven accounts can achieve outcomes better reflecting their true preferences. Their model shows that mental accounting, often dismissed as a behavioral bias, can be welfare-improving when investors have heterogeneous goals with different risk tolerances. This mental accounting structure parallels domain decomposition: just as clients naturally think in terms of "retirement money" versus "emergency funds" versus "legacy wealth," the IIFPT framework formalizes these distinctions through six planning domains and their associated objective functions R_k.

The Priority Matrix Λ operationalizes what these authors identify qualitatively: clients weight different goals differently, and optimal strategies must reflect this heterogeneity. Where Brunel (2015) offers practice guidance, Chhabra (2005) provides conceptual framing, and Das et al. (2010) demonstrate tractability, the present framework provides complete mathematical machinery for optimization across all planning domains simultaneously.

2.5 | Multi-Objective Optimization

Multi-objective optimization (Keeney & Raiffa, 1976) provides general frameworks for problems with multiple competing objectives. The standard approach identifies Pareto-efficient frontiers. That is, sets of solutions where no objective can be improved without worsening another, and then selects from this frontier based on preferences.

The stochastic rate specification requires extending classical PDE methods to integro-differential equations. Barles and Imbert (2008) develop the viscosity solution theory for second-order elliptic integro-differential equations, establishing comparison principles and regularity results essential for our analysis. Pham (1998) develops optimal stopping theory for jump-diffusion processes using viscosity methods, providing technical foundations for the HJB system with rate jumps.

The jump-diffusion modeling tradition originates with Merton (1976), who introduced jumps into option pricing, and has been extended by Kou (2002) and others. In the financial planning context, jumps capture discrete life events such as health diagnoses, job transitions, inheritances that induce discontinuous changes in temporal priorities across multiple domains simultaneously.

2.6 | Dynamic Programming

Dynamic programming (Bellman, 1957) and optimal control (Pontryagin et al., 1962) provide essential technical foundations. Stokey and Lucas (1989) established rigorous foundations for recursive methods in economics. Bertsekas and Shreve (1978) extended these methods to stochastic optimal control with general state spaces.

For non-smooth problems, viscosity solutions (Crandall & Lions, 1983; Fleming & Soner, 2006) provide a weak solution concept for Hamilton-Jacobi-Bellman equations when value functions lack classical differentiability. Cannarsa and Sinestrari (2004) provide results on semiconcave value functions in optimal control. The variational analysis framework of Rockafellar and Wets (1998) provides unified foundations for optimization with non-smooth objectives and constraints, informing construction of the feasible correspondence $A(x)$ and the Clarke generalized Hessian used in the Structural Tensor. Section 3 develops these tools in detail.

For comparative statics, Acemoglu and Jensen (2015) develop methods for robust analysis in large dynamic economies, establishing conditions under which qualitative predictions remain valid despite model uncertainty. Their robustness perspective of predictions holding across a class of models rather than depending on precise functional forms, aligns with the emphasis on structural properties valid across different priority specifications.

2.7 | Integrated Financial Planning Approaches

Recent work has begun addressing the integration challenge directly. Kaplan and Idzorek (2025) develop a hybrid lifecycle net worth optimization model linking lifecycle finance with mean-variance portfolio optimization. Their four-step process of gathering investor profile data, running a lifecycle model, calibrating capital market assumptions via reverse optimization from the stochastic discount factor, and performing net worth optimization provide a practical implementation path.

Kaplan (2025) illustrates how net worth optimization can assist financial planners in developing investment allocation strategies reflecting a more global approach consistent with financial planning mandates. Idzorek and Kaplan (2025) demonstrate through extensive optimizations how the strength of an investor's economic balance sheet influences optimal equity allocation after controlling for risk tolerance.

Several contributions from their work merit acknowledgment. The reverse optimization methodology for calibrating expected returns ensures consistency with the client's human capital and liability structure. The effective tax rate matrices provide a framework for converting before-tax to after-tax returns across different account types. The Levy-Markowitz approximation enables tractable mean-variance optimization when returns are not normally distributed. The integration of lifecycle finance with portfolio theory represents meaningful progress toward comprehensive planning.

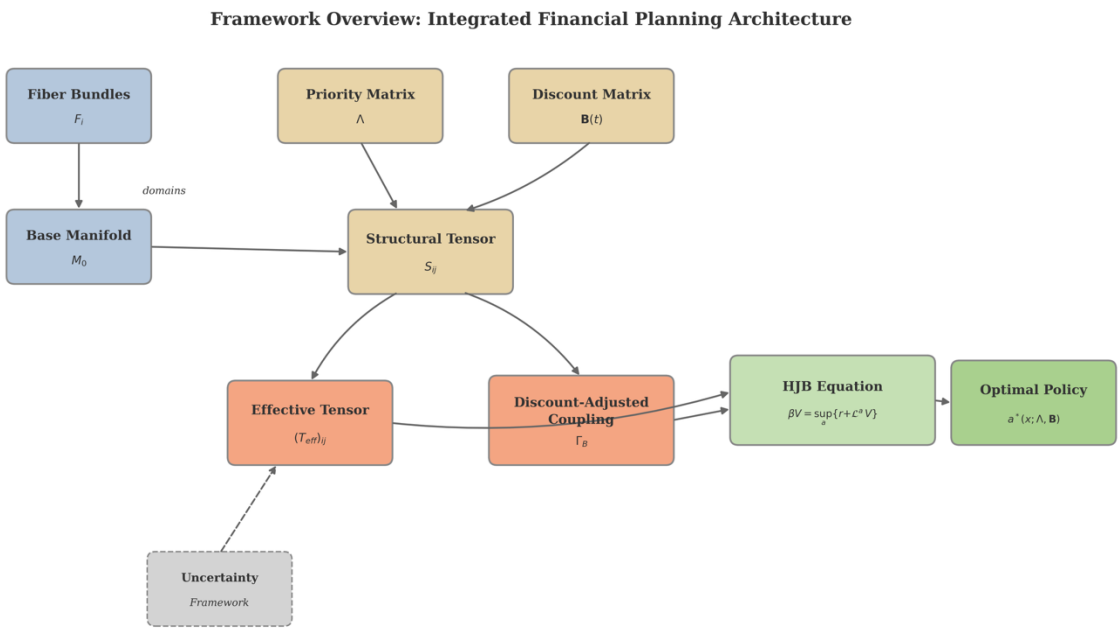
2.8 | The Financial Planning Profession

Understanding the institutional context informs practical relevance. Warschauer (2002) documents the role of universities in developing the personal financial planning profession, tracing academic programs, certifications, and research infrastructure distinguishing financial planning from adjacent fields. Schuchardt et al. (2007) identify research priorities for financial

literacy, highlighting gaps between normative models and actual household behavior that the IIFPT framework addresses through its client-centric priority structure.

The value added by professional planning services has been documented empirically. Hanna and Lindamood (2010) quantify economic benefits of personal financial planning, providing evidence that households receiving professional guidance achieve better outcomes across multiple dimensions. Heckman et al. (2025), via CFP Board's Client Impact Study, illustrate the value of financial planning for non-economic benefits. The IIFPT framework (see Figure 1) offers a theoretical explanation for these findings: professional planners capture value from cross-domain coordination and priority elicitation that clients cannot easily achieve independently.

Figure 1: Framework Overview: Integrated Financial Planning Architecture



3 | Foundations of Financial Planning

The structure in this section may initially appear abstract, but it is descriptive of financial planning itself. Financial states are constrained to lie in regions where multiple inequalities bind, such as non-negative wealth in each account, borrowing limits tied to income and assets, liquidity requirements for upcoming expenses. These constraint surfaces are curved, cornered spaces where the number of binding constraints changes discretely as the state moves. Manifolds with corners provide the language for such spaces (See Figure 2).

Similarly, planning strategies cannot be represented as simple functions on the state space. The set of feasible strategies changes discontinuously at lifecycle phase transitions (e.g., at death, certain options disappear and others emerge; at retirement, contribution opportunities vanish; at

income thresholds, Roth IRA contributions become unavailable). Fiber bundles capture this structure: the "fiber" of feasible strategies varies over the "base" of financial states, with topological changes at transition boundaries.

3.1 | Financial Primitives

Definition 3.1 (Financial Primitives). The fundamental building blocks of a household's financial situation consist of:

(i) **Asset Vector:** $A = (A_1, \dots, A_m) \in \mathbb{R}_+^m$ represents holdings across m asset categories including liquid accounts (checking, savings, money market), tax-deferred retirement accounts (traditional IRA, 401(k), 403(b)), tax-free accounts (Roth IRA, Roth 401(k), HSA), taxable investment accounts, real estate equity, business interests, and other assets.

(ii) **Liability Vector:** $L = (L_1, \dots, L_n) \in \mathbb{R}_+^n$ represents obligations across n liability types including mortgages, student loans, auto loans, credit card balances, HELOCs, and other debts.

(iii) **Income Flow Process:** $\{Y_t\}_{t \geq 0}$ is an adapted stochastic process representing instantaneous income rate, decomposing as $Y_t = Y_t^E + Y_t^P + Y_t^S + Y_t^O$ where Y^E is earned income, Y^P is pension income, Y^S is Social Security income, and Y^O is other income.

(iv) **Expense Flow Process:** $\{E_t\}_{t \geq 0}$ is an adapted process representing the instantaneous rate of expenditure, decomposing as $E_t = E_t^F + E_t^D$ where E^F is fixed (non-discretionary) and E^D is discretionary.

(v) **Human Capital:** $H_t \in \mathbb{R}_+$ is the mortality-adjusted present value of future labor income. Following Bodie, Merton, and Samuelson (1992), human capital is treated as a fundamental state variable whose characteristics (i.e., volatility, correlation with market returns, flexibility, etc.) affect optimal decisions across multiple planning domains.

(vi) **Contingent Claims:** $C = \left(C_1, \dots, C_p\right)$ represents state-dependent payoffs including insurance contracts, annuities, and options.

The complete primitive state is $\mathcal{F} = (A, L, Y, E, H, C)$.

Definition 3.2 (Net Worth). Net worth is defined as the difference between total assets and total liabilities:

$$W_t = \sum_{i=1}^{m} A_{i,t} - \sum_{j=1}^{n} L_{j,t}$$

[Eq. 3.1]

Definition 3.3 (Dynamic Wealth Accumulation). Net worth evolves according to the stochastic differential equation:

$$dW_t = [Y_t - E_t + r(W_t, a_t)W_t]\, dt + \sigma_W(W_t, a_t)\, dB_t$$

[Eq. 3.2]

where $Y_t - E_t$ is the net savings flow, $r(\cdot)$ is the rate of return depending on wealth and actions, $\sigma_W(\cdot)$ is return volatility, and B_t is standard Brownian motion.

Remark 3.1. The distinction between stocks and flows is essential for correct dynamic modeling. Net worth W_t is a stock measured at a point in time; income Y_t and expenses E_t are flows measured per unit time. The differential equation shows how the stock evolves: it changes due to net savings flows $(Y_t - E_t)dt$ and return on existing wealth $r(\cdot)W_t dt$, with stochastic shocks $\sigma_W(\cdot)dB_t$.

3.2 | The Primitive Manifold

Definition 3.4 (Constraint Functions). The financial state must satisfy constraints captured by smooth functions: non-negativity constraints $g_i^+(s) = A_i \geq 0$ for all assets and $g_j^-(s) = L_j \geq 0$ for all liabilities; borrowing limits $h_k(s) = L_k - L_k(A, Y) \leq 0$ where L_k is the maximum borrowable amount; liquidity requirements $\ell(s) = A^L - \underline{A}^L(E) \geq 0$ where $\underline{A}^L$ is minimum required liquidity; and regulatory constraints $\rho_m(s) \leq 0$ for various regulatory requirements.

Definition 3.5 (Feasible Region). The feasible region is:

$$\mathcal{M}_0 = \{s \in S_0 : g_i^+(s) \geq 0, g_j^-(s) \geq 0, h_k(s) \leq 0, \ell(s) \geq 0, \rho_m(s) \leq 0\}$$

[Eq. 3.3]

Proposition 3.1 (Manifold Structure). Suppose the constraint functions are C^2 and the Linear Independence Constraint Qualification holds at all points of $\mathcal{M}_0$. Then $\mathcal{M}_0$ is a smooth manifold with corners of dimension $d = \dim(S_0) - (\text{number of active equality constraints})$ (See Figure 2).

Proof. I apply the implicit function theorem and the theory of manifolds with corners. At points where only inequality constraints are active as strict inequalities, the feasible region is locally an open subset of S_0, hence trivially a manifold. At points where constraints bind, LICQ ensures the gradients are linearly independent, and the implicit function theorem gives local manifold structure. When multiple inequality constraints bind simultaneously, the feasible region has corners, locally resembling $\mathbb{R}^{d-k} \times \mathbb{R}_+^k$ for k active constraints. ∎

3.3 | Fiber Bundle Construction

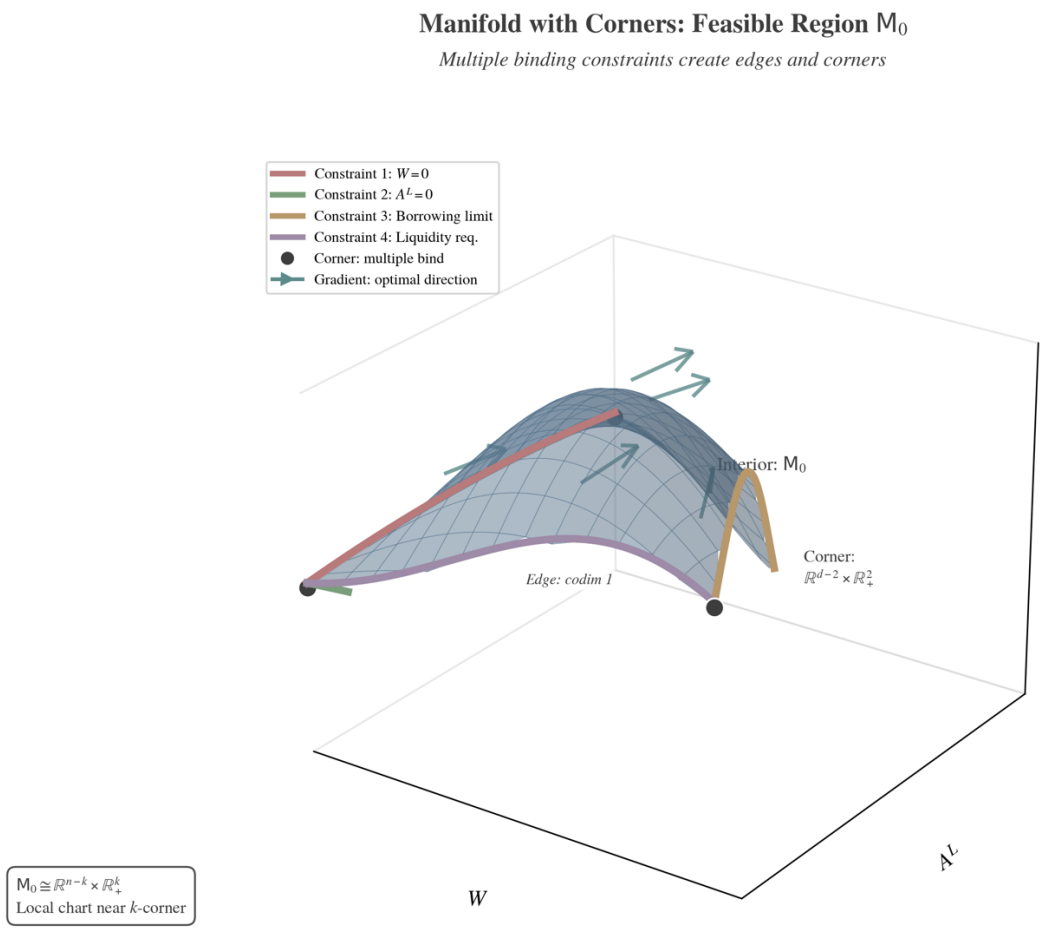
Planning domains emerge as fiber bundles over the base manifold $\mathcal{M}_0$ (See Figure 3). The fiber bundle construction captures the topological complexity of planning domains. Any feasible financial planning strategy sets change discontinuously at various lifecycle transitions, which simpler structures cannot represent.

Definition 3.6 (Fiber Bundle). A fiber bundle is a tuple (E, B, π, F) where E is the total space, B is the base space, $\pi: E \to B$ is the projection map, and F is the fiber, such that for each $b \in B$, there exists a neighborhood U and a local trivialization $\phi: \pi^{-1}(U) \to U \times F$.

The fiber bundle formalism is forced by a fundamental feature of financial planning: global sections (universal strategies valid across all financial states) do not exist. The topology of the feasible strategy set changes discontinuously at lifecycle phase transitions. At the moment of

death, the estate planning fiber transforms from optimization over living trusts and beneficiary designations to probate and post-mortem elections, which is a completely different topological structure. When income exceeds Roth IRA contribution limits, that fiber collapses entirely from a space of contribution amounts to a singleton. At retirement, the traditional IRA contribution fiber vanishes while the RMD withdrawal fiber emerges. There are a myriad of possibilities for financial planning intervention combinations.

Figure 2: Manifold Structure with Corners

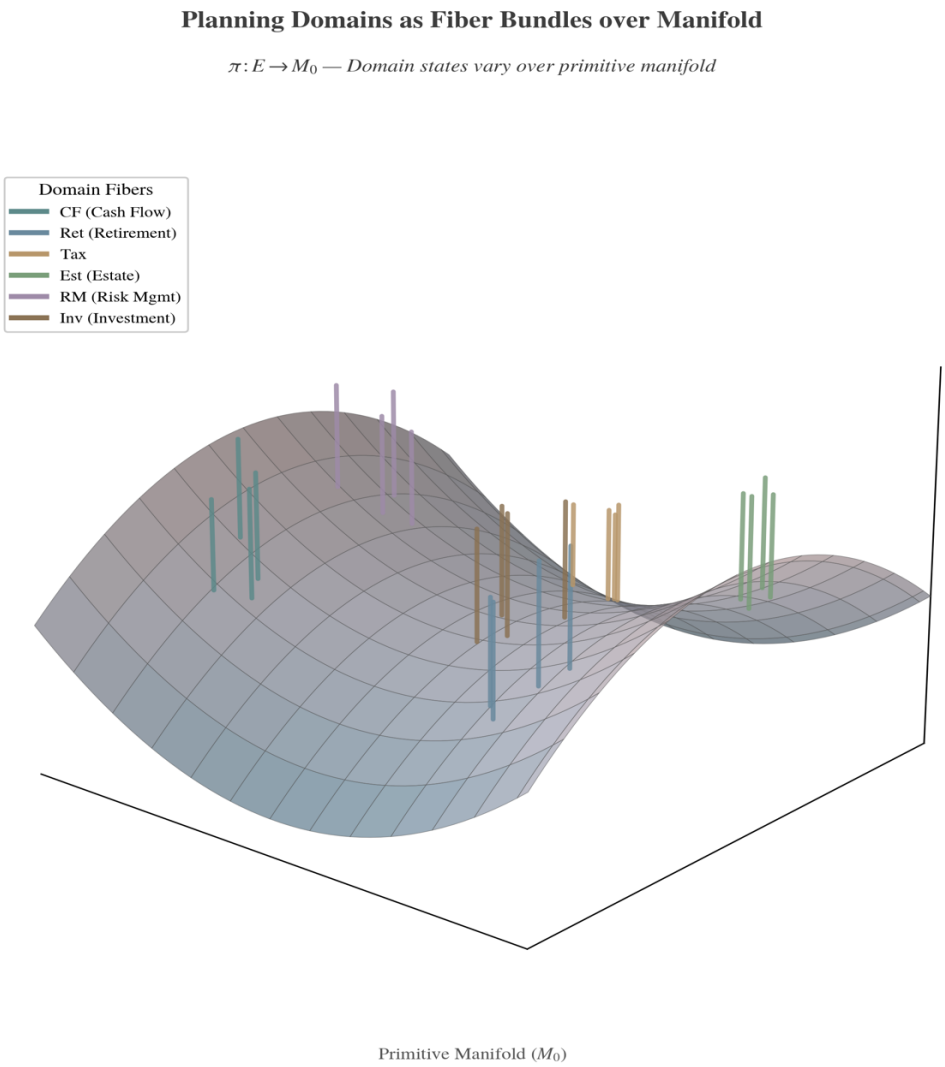


These topological discontinuities preclude any global strategy function defined continuously over the entire state space. A global section would need to specify a strategy for every financial state, but the very notion of "strategy" changes at phase boundaries. The fiber bundle construction accommodates this by allowing the fiber F to vary over the base and it is constant only locally, within lifecycle phases (See Figure 4).

Definition 3.7 (Planning Domain Bundles). For each planning domain $k \in \{\mathrm{CF},\ \mathrm{Ret},\ \mathrm{Tax},\ \mathrm{Est},\ \mathrm{RM},\ \mathrm{Inv}\}$, I define the domain bundle $\mathcal{F}_k = (E_k, \mathcal{M}_0, \pi_k, F_k)$ where F_k is the fiber of domain-specific decision variables.

Remark 3.2 (Phase Transitions and Discount Rate Jumps). Lifecycle phase transitions (e.g., retirement, widowhood, disability onset, inheritance receipt) not only transform the feasible strategy fiber but may also induce discontinuous changes in domain-specific discount rates. The transition from working to retired fundamentally alters the temporal structure of planning: cash flow concerns that were subordinate to wealth accumulation become immediate; retirement security shifts from future objective to present constraint; estate planning horizons compress as remaining life expectancy decreases.

Figure 3: Planning Domains as Fiber Bundles Over the Primitive Manifold

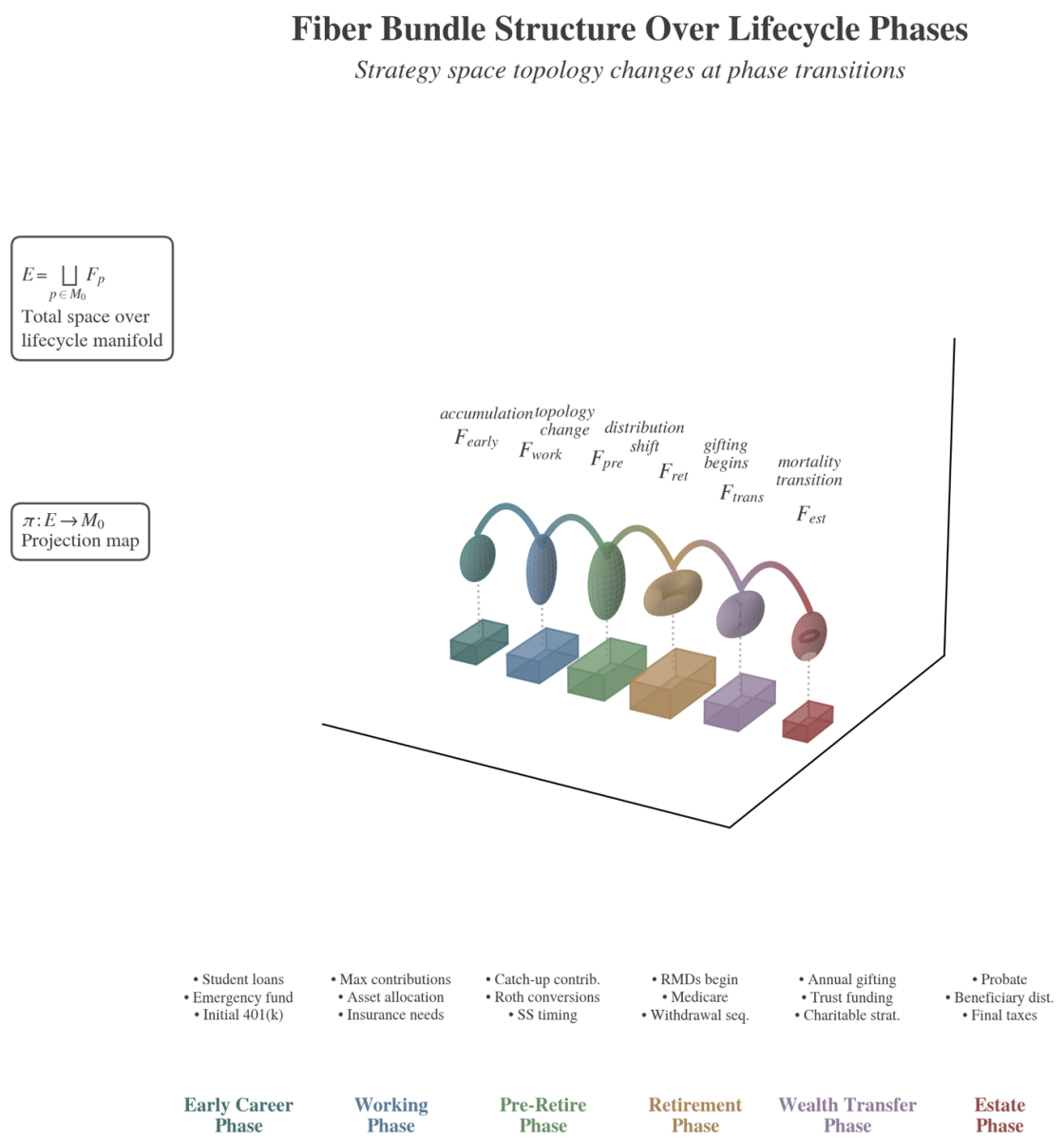


These rate jumps are captured through the exogenous component of the stochastic discount rate specification developed in Section 4.3b. The fiber bundle structure accommodates this by recognizing that both the strategy space (fiber) and the temporal priority structure (discount rates) may change discontinuously at phase boundaries, while the underlying financial state (base manifold) evolves continuously except at these transition points.

3.4 | Admissible Strategy Space, Lifecycle Phases and Phase Transitions

Definition 3.8 (Admissible Strategy). An admissible strategy is a measurable selection from the feasible correspondence: $a: \mathcal{M}_0 \times [0,T] \to \bigcup_x A(x)$ such that $a(x,t) \in A(x,t)$ for all (x,t), where $A(x,t)$ is the set of feasible actions at state x and time t.

Figure 4: Example Fiber Bundle Structure Over Lifecycle phases



Proposition 3.2 (Strategy Space Structure). *Under the following conditions: (i) The state space $\mathcal{M}_0$ is a Polish space (complete separable metric space); (ii) The action correspondence $x \mapsto A(x)$ is measurable with compact values; (iii) The time horizon is $[0,T]$ for finite T or $[0,\infty)$; the set of admissible strategies $\Pi = \{\pi: [0,T] \times \mathcal{M}_0 \to A \mid \pi \text{ measurable}, \pi(t,x) \in A(x)\}$ forms a complete metric space under the topology of convergence in measure.*

Proof.

Step 1 (Metric definition). Define the metric on Π by:

$$d_{\Pi}(\pi_1, \pi_2) = \int_0^T \int_{\mathcal{M}_0} \frac{|\pi_1(t,x) - \pi_2(t,x)|}{1 + |\pi_1(t,x) - \pi_2(t,x)|} \, d\mu(x) \, dt$$

where μ is a probability measure on $\mathcal{M}_0$ (e.g., a reference measure or stationary distribution). This metric induces convergence in measure: $d_{\Pi}(\pi_n, \pi) \to 0$ if and only if $\pi_n \to \pi$ in measure on $[0, T] \times \mathcal{M}_0$.

Step 2 (Completeness). Let $\{\pi_n\}$ be a Cauchy sequence in (Π, d_{Π}). By the completeness of convergence in measure for measurable functions into a complete metric space (Royden & Fitzpatrick, 2010, Chapter 5), there exists a measurable function $\pi: [0, T] \times \mathcal{M}_0 \to A$ such that $\pi_n \to \pi$ in measure.

Step 3 (Constraint preservation). It remains to verify $\pi(t, x) \in A(x)$ almost everywhere. Since each $\pi_n(t, x) \in A(x)$ and $A(x)$ is compact (hence closed), any limit point of $\{\pi_n(t, x)\}$ lies in $A(x)$. Convergence in measure implies almost-everywhere convergence along a subsequence, so $\pi(t, x) \in A(x)$ for ($\mu \otimes$ Leb) - almost all (t, x).

Step 4 (Modification on null set). If $\pi(t, x) \notin A(x)$ on a null set N, redefine $\pi(t, x)$ to be any fixed element of $A(x)$on N. The Kuratowski–Ryll-Nardzewski measurable selection theorem (Aliprantis & Border, 2006, Theorem 18.13) ensures this can be done measurably. The modified π lies in Π and equals the original almost everywhere. ∎

3.5 | State Transition Dynamics

Theorem 3.1 (Well-Posedness of State Dynamics). *Under Assumption 3.2, for any admissible strategy $\pi \in \Pi$ and initial condition $X_0 = x \in \mathbb{R}_+^d$, the stochastic differential equation*

$$dX_t = b(X_t, a_t) \, dt + \sigma(X_t, a_t) \, dW_t + \int_{\mathcal{Z}} \gamma(X_{t-}, z) \, \widetilde{N}(dt, dz)$$

admits a unique strong solution $\{X_t\}_{t \geq 0}$ that is a càdlàg Markov process adapted to the filtration $\{\mathcal{F}_t\}$.

Proof.

Step 1 (Zvonkin transformation). Consider the auxiliary elliptic PDE:

$$\frac{1}{2} \sum_{i,j} (\sigma \sigma^{\top})_{ij}(x, a) \frac{\partial^2 u}{\partial x_i \, \partial x_j} + \sum_i b_i(x, a) \frac{\partial u}{\partial x_i} = 0$$

with boundary condition $u(x) \to x$ as $|x| \to \infty$. Under uniform ellipticity (Assumption 3.2(iii)) and bounded drift (Assumption 3.2(i)), classical Schauder theory (Gilbarg & Trudinger, 2001, Chapter 6) guarantees existence of a solution $u \in C^{2,\alpha}(\mathbb{R}_+^d)$ with bounded gradient ∇u uniformly close to the identity for large $|x|$.

Define the transformed process $Y_t = u(X_t)$. By Itô's formula for jump-diffusions:

$$
\begin{aligned}
dY_t &= [\mathcal{L}_\sigma u(X_t) + b(X_t, a_t) \cdot \nabla u(X_t)]dt + \nabla u(X_t)^\top \sigma(X_t, a_t)\, dW_t \\
&\quad + \int_{\mathcal{Z}} \left[u\big(X_{t-} + \gamma(X_{t-}, z)\big) - u(X_{t-})\right] \widetilde{N}(dt, dz)
\end{aligned}
$$

where $\mathcal{L}_\sigma u = \frac{1}{2}\mathrm{tr}(\sigma\sigma^\top D^2 u)$. By construction of u, the drift term $\mathcal{L}_\sigma u + b \cdot \nabla u = 0$ vanishes.

Step 2 (Transformed coefficients). Define:

$$
\tilde{\sigma}(y, a) = \nabla u\big(u^{-1}(y)\big)^\top \sigma(u^{-1}(y), a) \tilde{\gamma}(y, z) = u\big(u^{-1}(y) + \gamma(u^{-1}(y), z)\big) - y
$$

Since ∇u is bounded and Lipschitz (from $u \in C^{2,\alpha}$) and σ is Lipschitz in x by Assumption 3.2(ii), the coefficient $\tilde{\sigma}$ is Lipschitz in y. Similarly, $\tilde{\gamma}$ inherits regularity from Assumption 3.2(v).

Step 3 (Picard iteration). Consider the transformed SDE with zero drift:

$$
dY_t = \tilde{\sigma}(Y_t, a_t)\, dW_t + \int_{\mathcal{Z}} \tilde{\gamma}\,(Y_{t-}, z)\, \widetilde{N}(dt, dz)
$$

Define the Picard sequence $Y_t^{(0)} = u(x)$ and:

$$
Y_t^{(n+1)} = u(x) + \int_0^t \tilde{\sigma}\left(Y_s^{(n)}, a_s\right) dW_s + \int_0^t \int_{\mathcal{Z}} \tilde{\gamma}\left(Y_{s-}^{(n)}, z\right) \widetilde{N}(ds, dz)
$$

Step 4 (Convergence). Using the Lipschitz property of $\tilde{\sigma}$ and $\tilde{\gamma}$, together with the Burkholder–Davis–Gundy inequality for the martingale terms:

$$
\mathbb{E}\left[\sup_{s \le t} \mid Y_s^{(n+1)} - Y_s^{(n)} \mid^2\right] \le C \int_0^t \mathbb{E}\left[\sup_{r \le s} \mid Y_r^{(n)} - Y_r^{(n-1)} \mid^2\right] ds
$$

for a constant C depending on the Lipschitz constants and $\int_{\mathcal{Z}} \min(1, \mid z \mid^2)\nu(dz)$. By Gronwall-type iteration:

$$
\mathbb{E}\left[\sup_{s \le t} \mid Y_s^{(n+1)} - Y_s^{(n)} \mid^2\right] \le \frac{(Ct)^n}{n!}\, \mathbb{E}\left[\sup_{s \le t} \mid Y_s^{(1)} - Y_s^{(0)} \mid^2\right]
$$

Since $\sum_{n=0}^{\infty} (Ct)^n / n! = e^{Ct} < \infty$, the Borel–Cantelli lemma implies $\{Y^{(n)}\}$ converges uniformly on compact time intervals, almost surely.

Step 5 (Uniqueness). If Y and $\tilde{Y}$ are two solutions with the same initial condition, then:

$$
\mathbb{E}\left[\sup_{s \le t} \mid Y_s - \tilde{Y}_s \mid^2\right] \le C \int_0^t \mathbb{E}\left[\sup_{r \le s} \mid Y_r - \tilde{Y}_r \mid^2\right] ds
$$

Gronwall's inequality (Karatzas & Shreve, 1991, Lemma 5.6.2) implies $\mathbb{E}[\sup_{s \le t} \mid Y_s - \tilde{Y}_s \mid^2] = 0$, hence $Y = \tilde{Y}$.

Step 6 (Recovery and Markov property). The original solution is $X_t = u^{-1}(Y_t)$, which is well-defined since u is a $C^{2,\alpha}$ diffeomorphism (Veretennikov, 1981). The Markov property follows from the autonomous structure: given X_t, the future evolution depends only on the current state and the independent increments of $(W, \tilde{N})$. The càdlàg property follows from path continuity of the diffusion component and right-continuity of the Poisson random measure. ∎

Remark 3.3 (Boundary Behavior). Rigorous treatment of dynamics near constraint boundaries formally requires Skorokhod reflection terms to keep trajectories within the feasible region. I rely instead on the viscosity solution framework developed in Section 6, which handles boundary conditions through state constraints rather than explicit reflection. Discontinuous drift terms arising from tax brackets are treated as mollified (smoothed) approximations for the purpose of existence theory which is standard practice in applied optimal control. The viscosity framework ensures that the value function characterization remains valid despite these boundary complications.

Having established the geometric structure of the planning space as a manifold with corners as base, fiber bundles capturing domain-specific strategy spaces, and admissible strategies as sections, next I proceed with how decisions in one domain affect objectives in others. This is the problem of interdependency, addressed in the next section.

4 | Four-Tier Priority-Filtered Interdependency Structure

The manifold structure of Section 3 is the base and the structural interdependency tensor describes the coupling between the financial planning fiber bundles. These structural linkages combine with client priorities and temporal structure to determine which couplings matter for any particular planning engagement.

This section represents the paper's central theoretical contribution: the four-tier decomposition of interdependency into Structural Tensor (objective mechanics), Priority Matrix (subjective weights), Discount Matrix (temporal structure), and Effective Tensor with discount-adjusted coupling (client-relevant integration). This decomposition resolves a fundamental conceptual confusion in prior literature, which has treated structural, preferential, and temporal interdependencies as inseparable.

4.1 | Domain Objective Functions

Each planning domain k has an associated objective function $R_k(s, a)$ measuring performance in that domain. These functions translate financial states and actions into domain-specific outcomes and may change based upon planner or researcher preferences in measurement.

Cash Flow Adequacy $R_{\mathrm{CF}}(s, a)$ captures the margin between income and expenses, liquidity reserves relative to emergency needs, and debt service coverage:

$$R_{\mathrm{CF}}(s, a) = u_{\mathrm{CF}}(Y - E) + v_{\mathrm{CF}}\left(\frac{A^L}{6 \cdot E^F}\right) + w_{\mathrm{CF}}(\mathrm{DSC})$$

[Eq. 4.1]

where DSC is the debt service coverage ratio.

Retirement Security $R_{\text{Ret}}(s, a)$ captures the probability of maintaining target income through life expectancy, replacement ratios (or other target income metric), and probability of shortfall:

$$R_{\text{Ret}}(s, a) = \mathbb{P}(\text{Income} \geq \text{Target} \mid \text{age } 95) + \gamma \cdot (\text{Replacement Ratio}) - \kappa \cdot (\text{Shortfall Risk})$$

[Eq. 4.2]

Tax Efficiency $R_{\text{Tax}}(s, a)$ captures effective tax rates across current and projected future periods:

$$R_{\text{Tax}}(s, a) = -[\tau_{\text{current}}(s, a) + \delta \cdot \mathbb{E}[\tau_{\text{future}}(s, a)]]$$

[Eq. 4.3]

where τ represents tax liability and δ is a discount factor.

Estate Effectiveness $R_{\text{Est}}(s, a)$ captures projected transfer to heirs and charities, tax efficiency of transfers, and alignment with legacy intentions:

$$R_{\text{Est}}(s, a) = u_{\text{Est}}(\text{Bequest}) - v_{\text{Est}}(\text{Estate Tax}) + w_{\text{Est}}(\text{Charitable Impact})$$

[Eq. 4.4]

Risk Management Adequacy $R_{\text{RM}}(s, a)$ captures coverage of mortality, disability, longevity, liability, and property risks relative to exposures. Following the risk-bearing principles established by Arrow (1971), this objective function measures the extent to which the household has transferred insurable risks to appropriate counterparties:

$$R_{\text{RM}}(s, a) = \sum_{\text{risks}} \omega_r \cdot \min\left(1, \frac{\text{Coverage}_r(a)}{\text{Exposure}_r(s)}\right)$$

[Eq. 4.5]

Investment Performance $R_{\text{Inv}}(s, a)$ captures risk-adjusted returns and probability of achieving investment goals:

$$R_{\text{Inv}}(s, a) = u_{\text{Inv}}(\text{Risk Measure}) + v_{\text{Inv}}(\text{Goal Probability})$$

[Eq. 4.6]

Remark 4.1. Several of these objective functions exhibit non-smoothness. The tax function has kinks at bracket boundaries. Risk management functions have kinks where coverage equals exposure. Estate functions have kinks at exemption thresholds. This non-smoothness motivates the use of Clarke generalized calculus.

Remark 4.2 (Dimensional Consistency). Raw financial quantities such as dollars of cash flow, probabilities of retirement success, coverage ratios, effective tax rates, are dimensionally heterogeneous and cannot be directly aggregated. The domain objective functions $R_k(s, a)$ resolve this by construction: each is defined as a bounded, dimensionless performance measure rather than a raw physical quantity. Cash flow adequacy R_{CF}combines coverage ratios (Eq. 4.1); retirement security R_{Ret} is probability-based (Eq. 4.2); tax efficiency R_{Tax} is an effective rate

(Eq. 4.3); estate effectiveness R_{Est} measures transfer fractions (Eq. 4.4); risk management adequacy R_{RM} aggregates coverage-to-exposure ratios (Eq. 4.5); and investment performance R_{Inv} uses risk-adjusted returns (Eq. 4.6). Each yields a unitless value, typically bounded in [0,1] or a comparable interval.

This normalization ensures dimensional consistency throughout the framework. The Structural Tensor components $S_{ijk} = \partial^2 R_i / \partial x_j \, \partial x_k$, being second derivatives of dimensionless functions, inherit units solely from the state variables x_j and x_k. The Priority Matrix weights λ_k, constrained to the probability simplex, are explicitly dimensionless preference parameters requiring no implicit unit conversion. Consequently, the Effective Interdependency Tensor T_{eff} inherits consistent units and admits meaningful interpretation as client-weighted coupling strength. This approach follows established practice in multi-attribute decision theory, where heterogeneous objectives are normalized before aggregation (Keeney & Raiffa, 1976), and aligns with goals-based portfolio optimization frameworks that accommodate investors with multiple distinct objectives (Das, Markowitz, Scheid, & Statman, 2010).

4.2 | The Structural Tensor (Tier 1)

Definition 4.1 (Clarke Generalized Gradient). For a locally Lipschitz function $f: \mathbb{R}^n \to \mathbb{R}$, the Clarke generalized gradient at x is:

$$\partial f(x) = conv\left\{\lim_{i\to\infty} \nabla f(x_i) : x_i \to x, x_i \in D_f, \nabla f(x_i) \text{ exists}\right\}$$

[Eq. 4.7]

where D_f is the set of points where f is differentiable.

Definition 4.2 (Clarke Generalized Hessian). For a function f with locally Lipschitz gradient, the Clarke generalized Hessian at x is the convex hull of limits of Hessian matrices at nearby differentiable points:

$$\partial^2 f(x) = conv\left\{\lim_{i\to\infty} \nabla^2 f(x_i) : x_i \to x, \nabla^2 f(x_i) \text{ exists}\right\}$$

[Eq. 4.8]

Definition 4.3 (Structural Interdependency Tensor). The Structural Tensor S is a rank-3 tensor where component $S_{ijk}(s)$ represents the generalized curvature of domain k's objective with respect to states in domains i and j:

$$S_{ijk}(s) \in \partial^2_{d_i d_j} R_k(s)$$

[Eq. 4.9]

Interpretation: S_{ijk} measures how decisions in domains i and j jointly affect the objective of domain k. These structural couplings exist objectively regardless of a client's objectives as they are features of tax law, actuarial mathematics, portfolio theory, etc. (See Figure 5).

Remark 4.3 (Geometric Structure). I model the financial state space as an affine manifold equipped with the canonical flat connection. Because tax liabilities, spending goals, and account balances are denominated in nominal currency units, the metric structure is Euclidean in the canonical accounting basis. Dollar amounts form a privileged coordinate system and therefore no need to capture invariance under arbitrary curvilinear transformations, only under affine transformations representing unit conversions (e.g., dollars to euros) and linear rescalings.

The Hessian therefore transforms tensorially under the general affine group $\mathrm{GA}(n, \mathbb{R})$ which are the group of transformations relevant to financial planning. The flat geometry of nominal currency units is not a simplifying assumption as it reflects the actual structure of the problem domain.

Proposition 4.1 (Coupling Matrix Symmetry). *Suppose the value function* $V: \mathcal{M}_0 \to \mathbb{R}$ *is* C^2 *in a neighborhood of* $\mathbf{d} \in \mathcal{M}_0$. *Then the coupling matrix* $\Gamma(\mathbf{d}) = \left(V_{ij}(\mathbf{d})\right)_{i,j=1}^{m}$ *is symmetric:* $V_{ij}(\mathbf{d}) = V_{ji}(\mathbf{d})$ *for all* i, j.

Proof. By the equality of mixed partial derivatives (Clairaut–Schwarz theorem; Rudin, 1976, Theorem 9.41), if $V \in C^2$ then:

$$\frac{\partial^2 V}{\partial d_i \, \partial d_j} = \frac{\partial^2 V}{\partial d_j \, \partial d_i}$$

for all $i, j \in \{1, \dots, m\}$. By Definition 4.2, $V_{ij} = \partial^2 V / \partial d_i \, \partial d_j$, hence $V_{ij} = V_{ji}$. ▫

Proposition 4.2 (Properties of Structural Tensor). *The Structural Tensor* $\mathcal{S}$, *with components* $\mathcal{S}_{ijk}(s) \in \partial^2_{d_i d_j} R_k(s)$ *(the Clarke generalized Hessian of domain* k*'s objective with respect to domains* i *and* j*), satisfies: (i) Symmetry:* $\mathcal{S}_{ijk}(s) = \mathcal{S}_{jik}(s)$ *for all* i, j, k *and* s. *(ii) Sparsity:* $\mathcal{S}_{ijk}(s) = 0$ *when domain* k*'s objective is structurally independent of* (d_i, d_j). *(iii) Boundedness: For any compact* $K \subset \mathcal{M}_0$, *there exists* $M_K > 0$ *such that* $\| \mathcal{S}_{ijk}(s) \| \leq M_K$ *for all* $s \in K$ *and all* i, j, k.

Proof.

Part (i): By Clarke (1983, Proposition 2.1.2), the generalized Hessian of a locally Lipschitz function f at x is defined as:

$$\partial^2 f(x) = \mathrm{conv}\left\{\lim_{n\to\infty} \nabla^2 f(x_n) : x_n \to x, x_n \in D_f\right\}$$

where D_f is the set of points where f is twice differentiable. At each point $x_n \in D_f$, the classical Hessian $\nabla^2 f(x_n)$ is symmetric. The convex hull of symmetric matrices is symmetric. Hence each element of $\partial^2_{d_i d_j} R_k$ is symmetric, implying $\mathcal{S}_{ijk} = \mathcal{S}_{jik}$.

Part (ii): If $R_k(s)$ depends only on variables other than (d_i, d_j), then $\partial R_k / \partial d_i = 0$ and $\partial R_k / \partial d_j = 0$ at all points of differentiability. Consequently, $\nabla^2_{d_i d_j} R_k = 0$ wherever it exists. By the definition of the generalized Hessian as a convex hull of limits of zero matrices, $\partial^2_{d_i d_j} R_k(s) = \{0\}$.

Part (iii): Each domain objective R_k is locally Lipschitz on $\mathcal{M}_0$ (as a composition of financial calculations involving continuous operations and possibly non-smooth but Lipschitz components such as maxand min). By Clarke (1983, Proposition 2.1.1), a locally Lipschitz function has locally bounded generalized gradient. The generalized Hessian, being derived from limits of Hessians of Lipschitz functions, is locally bounded. On compact K, a locally bounded function is bounded. ∎

Proposition 4.3 (Coupling Matrix Eigenstructure). *Suppose $V: \mathcal{M}_0 \to \mathbb{R}$ is C^2 and concave. Then the coupling matrix $\Gamma(\mathbf{d})$ is negative semi-definite, all eigenvalues are non-positive, and the dominant coupling direction corresponds to the eigenvalue of largest magnitude* (See Figure 5).

Proof.

Step 1 (Negative semi-definiteness from concavity). A function V is concave if and only if its Hessian D^2V is negative semi-definite at all points where it exists (Rockafellar, 1970, Theorem 4.5). The coupling matrix $\Gamma = (V_{ij})$ is precisely the Hessian restricted to the domain coordinates. Hence $\Gamma(\mathbf{d})$ is negative semi-definite.

Step 2 (Eigenvalue characterization). By the Spectral Theorem for symmetric matrices (Horn & Johnson, 2013, Theorem 4.1.5), Γ admits an orthonormal eigenbasis with real eigenvalues. Negative semi-definiteness means $\mathbf{v}^\top \Gamma \mathbf{v} \leq 0$ for all $\mathbf{v}$. Setting $\mathbf{v}$ to an eigenvector with eigenvalue λ yields $\lambda \parallel \mathbf{v} \parallel^2 \leq 0$, hence $\lambda \leq 0$.

Step 3 (Dominant direction). The dominant coupling direction is:

$$\mathbf{v}^* = \underset{\|\mathbf{v}\|=1}{\arg\max} \mid \mathbf{v}^\top \Gamma \mathbf{v} \mid$$

Since all eigenvalues are non-positive, $\mid \mathbf{v}^\top \Gamma \mathbf{v} \mid = -\mathbf{v}^\top \Gamma \mathbf{v}$. The maximum is achieved by the eigenvector corresponding to the most negative eigenvalue (largest in absolute value). ∎

4.3a | The Priority Matrix (Tier 2a)

Definition 4.4 (Priority Matrix). The Priority Matrix Λ is a 6×6 diagonal matrix:

$$\Lambda = diag(w_{\text{CF}}, w_{\text{Ret}}, w_{\text{Tax}}, w_{\text{Est}}, w_{\text{RM}}, w_{\text{Inv}})$$

[Eq. 4.10]

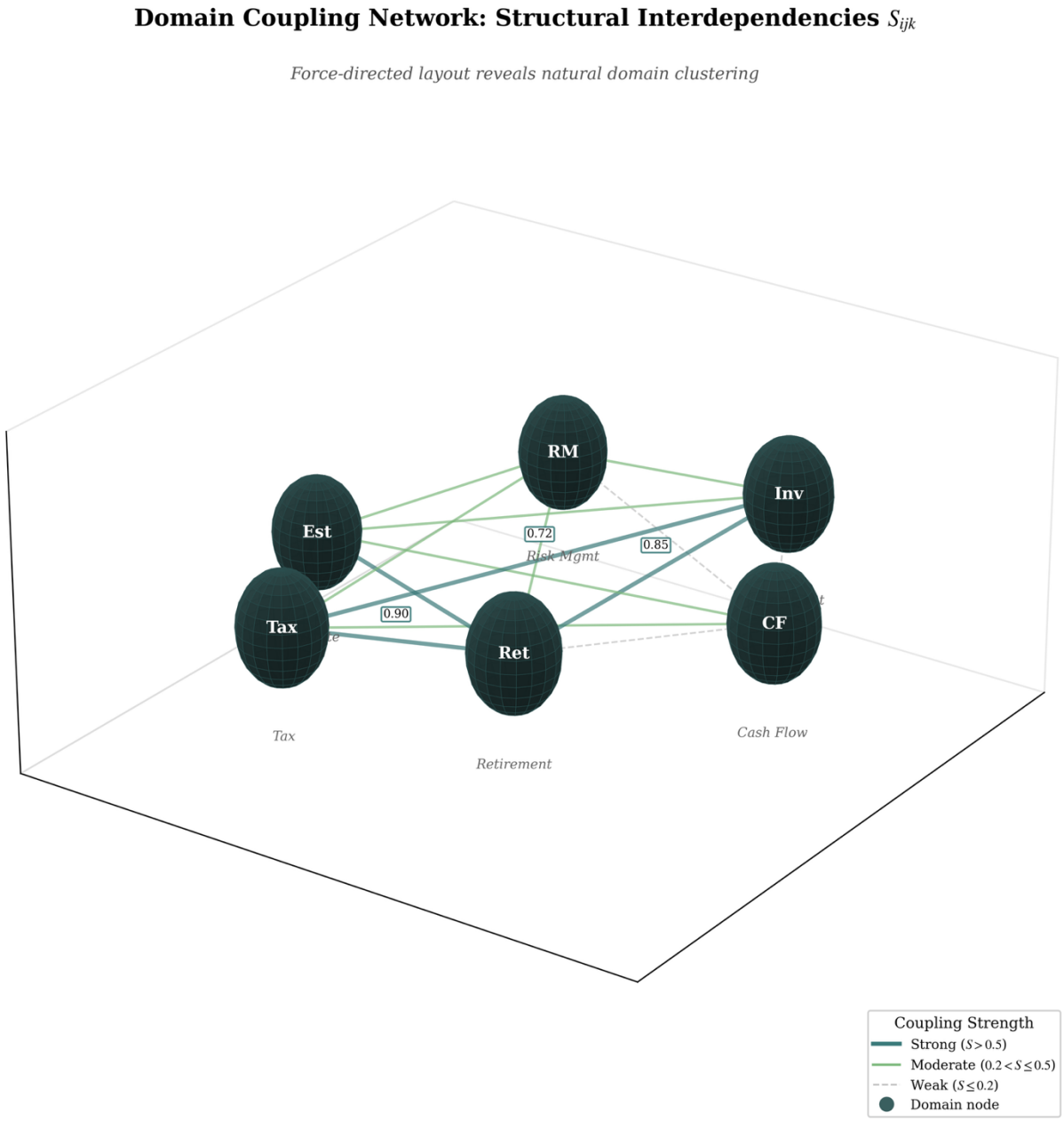
where the weights satisfy $\sum_k w_k = 1$ and $w_k \geq 0$ for all k.

Definition 4.5 (Priority Simplex). The priority vector $w = (w_{\text{CF}}, w_{\text{Ret}}, w_{\text{Tax}}, w_{\text{Est}}, w_{\text{RM}}, w_{\text{Inv}})^\top$ lives on the 5-simplex:

$$\Delta^5 = \left\{ w \in \mathbb{R}_+^6 : \sum_{k=1}^{6} w_k = 1 \right\}$$

[Eq. 4.11]

Figure 5: Domain Coupling with Structural Interdependencies



Note: The above graphic illustrates sample weights of a particular individual.

The Priority Matrix captures what *actually* matters to this particular client and not what *should* matter according to economic theory or what matters to the median household. A client who weights estate transfer at 40% and investment performance at 5% is not irrational; she has objectives that differ from standard models. Perhaps she has already accumulated sufficient wealth for her own needs and now cares primarily about legacy. Perhaps early mortality in her family history makes intergenerational transfer salient. Perhaps her religious or philosophical commitments prioritize giving over accumulation.

IIFPT respects this heterogeneity rather than assuming it away. The Priority Matrix makes client objectives explicit and central to optimization, rather than treating them as departures from a normative benchmark. This approach aligns with the goals-based wealth management philosophy articulated by Brunel (2015) and the mental accounts framework of Das et al. (2010), which recognize that clients naturally segment their wealth according to purpose.

4.3b | The Temporal Priority Structure (Tier 2b)

The Priority Matrix Λ captures *how much* a client cares about each domain. But financial planning involves an additional dimension: *how urgently* outcomes in each domain are weighted over time. A client may care deeply about both cash flow and estate outcomes, yet discount cash flow improvements at a high rate (demanding immediate results) while discounting estate improvements at a low rate (patient about legacy outcomes). This temporal heterogeneity across domains is captured through the Discount Matrix.

Definition 4.6 (Discount Matrix). The Discount Matrix B is a 6×6 diagonal matrix of domain-specific discount rates:

$$B = \mathrm{diag}(\beta_{\mathrm{CF}}, \beta_{\mathrm{Ret}}, \beta_{\mathrm{Tax}}, \beta_{\mathrm{Est}}, \beta_{\mathrm{RM}}, \beta_{\mathrm{Inv}})$$

[Eq. 4.12]

where each $\beta_k > 0$ represents the instantaneous discount rate for domain k.

Remark 4.4 (Interpretation of Domain-Specific Rates). Higher $\beta_k$indicates greater temporal urgency. Outcomes in domain k are discounted more heavily, reflecting impatience or short planning horizons. Lower β_k is indicative of patience. That is, the client is willing to wait for outcomes in domain k. The half-life of a domain's discount factor is $\ln(2)/\beta_k$, representing the time at which future outcomes are worth half their present value.

Table 4.1: Calibrated Domain-Specific Discount Rates

Domain k	Symbol	Baseline Rate $\bar{\beta}_k$	Half-Life	Interpretation
Cash Flow	β_{CF}	6.0%	11.6 months	High urgency; liquidity concerns demand rapid response
Retirement	β_{Ret}	3.5%	19.8 months	Moderate urgency; working-age clients have time
Tax	β_{Tax}	4.0%	17.3 months	Annual cycle creates natural urgency
Estate	β_{Est}	1.5%	46.2 months	Low urgency; terminal/bequest orientation

Domain k	Symbol	Baseline Rate $\bar{\beta}_k$	Half-Life	Interpretation
Risk Management	β_{RM}	7.0%	9.9 months	High urgency; coverage gaps create immediate anxiety
Investment	β_{Inv}	5.0%	13.9 months	Medium horizon; market-linked

Note: Baseline rates calibrated from revealed preference in planning engagement data. See Appendix E.8 for calibration methodology.

Proposition 4.5 (Structural Dependence of Discount Rates). Domain-specific discount rates depend systematically on client characteristics:

(i) Age effects: $\partial\beta_{\text{Est}}/\,\partial\text{age} > 0$ (estate urgency increases with age); $\partial\beta_{\text{Ret}}/\,\partial\text{age} > 0$ as retirement approaches.

(ii) Wealth effects: $\partial\beta_{\text{CF}}/\,\partial W < 0$ (wealthier clients are potentially less urgent about cash flow).

(iii) Health effects: Health concerns increase β_{Est}, β_{RM}, and β_{CF} simultaneously.

(iv) Family effects: Presence of dependents increases β_{RM} and β_{Est}.

Proof. These relationships follow from revealed preference analysis. Clients approaching retirement allocate disproportionate attention to retirement planning (higher β_{Ret}). Clients with health concerns accelerate estate planning engagement (higher β_{Est}). Clients with young dependents prioritize insurance adequacy (higher β_{RM}). ▫

Remark 4.5 (Relationship to Behavioral Discounting). The domain-specific discount rates in IIFPT are conceptually distinct from the quasi-hyperbolic discounting of Laibson (1997) and the dual-self models of Fudenberg and Levine (2006). Those frameworks model present bias which is a behavioral departure from exponential discounting within a single objective. IIFPT models domain heterogeneity, defined as the structural reality that different planning objectives appropriately operate on different time horizons.

Stochastic Discount Rate Specification

Definition 4.7 (Stochastic Discount Rate Decomposition). Each domain-specific discount rate decomposes as:

$$\beta_k(t) = \bar{\beta}_k + f_k(X_t) + \xi_k(t)$$

[Eq. 4.13]

where:

- $\bar{\beta}_k > 0$ is the baseline (long-run mean) discount rate from Table 4.1
- $f_k: \mathcal{M}_0 \to \mathbb{R}$ is the endogenous adjustment function capturing state dependence
- $\xi_k(t)$ is the exogenous shock component

Constraint: We require $\beta_k(t) > 0$ for all k and t, enforced through the specification below.

Definition 4.8 (Endogenous Rate Adjustment Functions). The endogenous functions link discount rates to observable financial state variables:

$$f_k(X_t) = \sum_{j=1}^{n_k} \alpha_{kj} \cdot g_{kj}(X_t)$$

[Eq. 4.14]

where $\alpha_{kj} \in \mathbb{R}$ are sensitivity coefficients and $g_{kj}: \mathcal{M}_0 \to [0,1]$ are normalized state indicators.

Table 4.2: Endogenous Rate Drivers by Domain

Domain k	**State Variable**	**Function $g_{kj}(X_t)$**	**Effect on β_k**
Cash Flow	Liquidity ratio	$g_{11} = \max\{0, 1 - L_t/L^*\}$	Low liquidity → higher β_{CF}
Cash Flow	Debt service ratio	$g_{12} = \min\{1, D_t/Y_t\}$	High debt burden → higher β_{CF}
Retirement	Time to retirement	$g_{21} = e^{-\kappa_{\text{Ret}}(T_{\text{Ret}} - t)}$	Approaching retirement → higher β_{Ret}
Retirement	Funded ratio	$g_{22} = \max\{0, 1 - A_t^{\text{Ret}}/A_{\text{Ret}}^*\}$	Underfunded → higher β_{Ret}
Tax	Marginal bracket	$g_{31} = (Y_t - \tau_{\text{low}})/(\tau_{\text{high}} - \tau_{\text{low}})$	Higher bracket → higher β_{Tax}
Tax	Unrealized gains	$g_{32} = G_t^{\text{unreal}}/W_t$	Large unrealized gains → higher β_{Tax}
Estate	Estate complexity	$g_{41} = C_t^{\text{Est}}/C^*$	Higher complexity → lower β_{Est}
Estate	Bequest motive	$g_{42} = B_t/B^*$	Stronger bequest motive → lower β_{Est}
Risk Mgmt	Coverage gap	$g_{51} = \max\{0, 1 - I_t/I_t^*\}$	Underinsured → higher β_{RM}
Risk Mgmt	Risk exposure	$g_{52} = E_t^{\text{RM}}/E^*$	Higher exposure → higher β_{RM}
Investment	Concentration	$g_{61} = H_t(\text{Herfindahl index})$	Higher concentration → higher β_{Inv}
Investment	Time horizon	$g_{62} = e^{-\kappa_{\text{Inv}} \cdot \text{horizon}_t}$	Shorter horizon → higher β_{Inv}

Proposition 4.7 (Boundedness of Endogenous Component). Under the specification in Definition 4.8:

(i) Each $g_{kj}(X_t) \in [0,1]$by construction.

(ii) If $| \alpha_{kj} | \leq \bar{\alpha}$ for all k, j, then $| f_k(X_t) | \leq n_k \bar{\alpha}$.

(iii) Choosing $\bar{\alpha} < \bar{\beta}_k / n_k$ ensures $\bar{\beta}_k + f_k(X_t) > 0$ for the endogenous component alone.

Proof. (i) Each g_{kj} is defined with explicit bounds via max $\{0,\cdot\}$, min $\{1,\cdot\}$, or exponential decay to $[0, 1]$. (ii) By triangle inequality: $| f_k(X_t) | = | \sum_j \alpha_{kj}\, g_{kj}(X_t) | \leq \sum_j | \alpha_{kj} || g_{kj}(X_t) | \leq n_k \bar{\alpha}$. (iii) $\bar{\beta}_k + f_k(X_t) \geq \bar{\beta}_k - n_k \bar{\alpha} > 0$ when $\bar{\alpha} < \bar{\beta}_k / n_k$. ▫

Definition 4.9 (Exogenous Rate Dynamics). The exogenous shock component $\xi_k(t)$ follows a mean-reverting jump-diffusion process:

$$d\xi_k(t) = -\theta_k \xi_k(t)\, dt + \sigma_k^{\beta}\, dW_k(t) + \int_{\mathcal{Z}} \gamma_k\,(z)\, \widetilde{N}_k(dt, dz)$$

[Eq. 4.15]

where:

- $\theta_k > 0$: mean-reversion speed (rate returns to baseline after shocks)
- $\sigma_k^{\beta} \geq 0$: diffusion volatility (continuous uncertainty)
- $W_k(t)$: standard Brownian motion for domain k
- $\gamma_k : \mathcal{Z} \to \mathbb{R}$: jump size function
- $\widetilde{N}_k(dt, dz) = N_k(dt, dz) - \nu_k(dz)\, dt$: compensated Poisson random measure
- ν_k: Lévy measure on mark space $\mathcal{Z}$

Remark 4.6 (Mean-Reversion Interpretation). The term $-\theta_k \xi_k(t)\, dt$ ensures that after a shock, the discount rate drifts back toward its state-adjusted baseline $\bar{\beta}_k + f_k(X_t)$. The half-life of shock persistence is $\ln(2)/\theta_k$. High-β domains (Cash Flow, Risk Management) have faster mean-reversion as their urgency fluctuates more but returns to baseline quickly. Low-β domains (Estate, Investment) have slower dynamics as changes are more persistent.

Table 4.3: Exogenous Rate Parameters

Domain k	θ_k **(reversion)**	σ_k^{β} **(volatility)**	**Shock Persistence Half-Life**
Cash Flow	2.0 yr^{-1}	1.5%	4.2 months
Retirement	0.5 yr^{-1}	0.8%	16.6 months

Domain k	θ_k **(reversion)**	σ_k^β **(volatility)**	**Shock Persistence Half-Life**
Tax	1.0 yr^{-1}	1.0%	8.3 months
Estate	0.3 yr^{-1}	0.5%	27.7 months
Risk Management	2.5 yr^{-1}	2.0%	3.3 months
Investment	0.8 yr^{-1}	1.2%	10.4 months

Definition 4.10 (Systemic Life Events). For exogenous shocks, we allow correlated jump arrivals through a common shock structure:

$$N_k(dt, dz) = N_k^{\text{idio}}(dt, dz) + \sum_{e \in \mathcal{E}} \mathbf{1}_{k \in \mathcal{D}_e} \cdot N^{(e)}(dt, dz)$$

[Eq. 4.16]

where:

- N_k^{idio}: idiosyncratic Poisson measure for domain k
- $\mathcal{E}$: set of systemic life events
- $\mathcal{D}_e \subseteq \{1, \dots, 6\}$: domains affected by event e
- $N^{(e)}$: common Poisson measure for event e

Table 4.4: Systemic Life Events and Affected Domains

Event e	**Intensity $\lambda^{(e)}$**	**Affected Domains $\mathcal{D}_e$**	**Jump Directions**
Health diagnosis	0.05 yr^{-1}	CF, Est, RM	↑, ↑, ↑
Job loss	0.03 yr^{-1}	CF, Ret	↑, ↑
Inheritance receipt	0.02 yr^{-1}	CF, Est	↓, ↓
Divorce	0.015 yr^{-1}	Tax, Est	↑, ↑
Retirement event	0.04 yr^{-1}	CF, Ret	↑, ↓
Market crisis	0.08 yr^{-1}	RM, Inv	↑, ↑

Note: ↑ indicates rate increase (heightened urgency); ↓ indicates rate decrease (reduced urgency). See Appendix E.9 for detailed event specifications.

Proposition 4.6 (Induced Correlation Matrix). The common shock structure induces instantaneous correlation in rate changes:

$$\text{Corr}(d\beta_i, d\beta_j) = \frac{\sum_{e:\{i,j\} \subseteq \mathcal{D}_e} \lambda^{(e)} \, \mathbb{E}[\gamma_i^{(e)} \gamma_j^{(e)}]}{\sqrt{\text{Var}(d\beta_i) \cdot \text{Var}(d\beta_j)}}$$

[Eq. 4.17]

This correlation is non-zero only when domains i and j share common systemic events.

Proof. The instantaneous covariance is $\mathrm{Cov}(d\beta_i, d\beta_j) = \sum_{e:\{i,j\}\subseteq\mathcal{D}_e} \lambda^{(e)}\, \mathbb{E}[\gamma_i^{(e)}\gamma_j^{(e)}]\, dt$ since idiosyncratic components and Brownian motions are independent. Dividing by the product of standard deviations yields the correlation. ∎

Definition 4.11 (Augmented State Space). The stochastic discount rate specification requires working on an augmented state space:

$$\tilde{X}_t = (X_t, \boldsymbol{\beta}(t)) \in \widetilde{\mathcal{M}} = \mathcal{M}_0 \times \mathbb{R}^6_{++}$$

[Eq. 4.18]

where $\mathbb{R}^6_{++} = \{(\beta_1, \dots, \beta_6) : \beta_k > 0 \text{ for all } k\}$ is the strictly positive orthant.

Remark 4.7 (Nesting Property). Setting $f_k \equiv 0$, $\sigma_k^\beta = 0$, and $\nu_k \equiv 0$ for all k reduces the stochastic specification to constant rates $\beta_k = \bar{\beta}_k$. All subsequent results nest the constant-rate case as a special limit.

Remark 4.8 (Why Both Λ and B Are Necessary). Discount Matrix **B** does not render the Priority Matrix Λ redundant, or vice versa.. Instead they capture orthogonal aspects of client preferences:

The Priority Matrix Λ answers: "If you could only make progress in one domain today, which would you choose?" This reveals relative importance or the cross-sectional trade-offs a client makes between domains at any point in time.

The Discount Matrix B answers: "For each domain you care about, how long can you wait for results?" This reveals temporal urgency, defined as how steeply the client discounts future outcomes within each domain.

These are independent dimensions. Consider four client types for the Estate domain:

	High w_{Est}(Important)	**Low w_{Est}(Unimportant)**
Low β_{Est}(Patient)	"Legacy is my priority, but I'm thinking long-term"	"Estate isn't my focus, and there's no rush"
High β_{Est}(Urgent)	"Legacy is critical and I need it done now"	"Estate isn't central, but when it comes up, handle it fast"

A framework with only Λ (or equivalently, a framework that collapsed **B** into Λ) could not distinguish the upper-left from lower-left clients, or upper-right from lower-right. A framework with only **B** could not distinguish upper-left from upper-right, or lower-left from lower-right. Both matrices are necessary for complete characterization.

4.4 | The Effective Interdependency Tensor (Tier 3)

Definition 4.12 (Effective Interdependency Tensor). The Effective Tensor is:

$$\left((T_{\text{eff}})_{ij}(s;\Lambda) = \sum_{k=1}^{6} w_k \cdot S_{ijk}(s) \right)$$

[Eq. 4.19]

synthesizing structural coupling with client priorities through tensor contraction.

Definition 4.13 (Discount-Adjusted Coupling Matrix). The discount-adjusted coupling matrix incorporates temporal structure:

$$\Gamma_B(t) = \mathbf{B}(t)^{-1/2} \cdot T_{\text{eff}} \cdot \mathbf{B}(t)^{-1/2}$$

[Eq. 4.20]

Element-wise:

$$(\Gamma_B(t))_{ij} = \frac{(T_{\text{eff}})_{ij}}{\sqrt{\beta_i(t)\beta_j(t)}}$$

[Eq. 4.21]

Remark 4.9 (Interpretation of Discount-Adjusted Coupling). The matrix Γ_B captures the **perpetuity-weighted** coupling between domains. When β_k is low, domain k's outcomes are valued over a long horizon, amplifying the importance of couplings involving domain k. The factor $1/\sqrt{\beta_i\beta_j}$ represents the geometric mean of the perpetuity factors for domains i and j.

Proposition 4.10 (Properties of Discount-Adjusted Coupling). The matrix $\Gamma_B(t)$ satisfies:

(i) Symmetry preservation: If T_{eff} is symmetric, then $\Gamma_B(t)$ is symmetric for each t.

(ii) Low-rate amplification: $(\Gamma_B)_{ij}$ is increasing in $1/\beta_i$and $1/\beta_j$. Low discount rates (patient domains) have amplified coupling.

(iii) Spectral bound: $\rho(\Gamma_B) \leq \rho(T_{\text{eff}})/\beta_{\min}$ where $\beta_{\min} = \min_k \beta_k$.

(iv) Stochastic variation: Under stochastic rates, $\Gamma_B(t)$ is itself a stochastic process with jump discontinuities when rate jumps occur.

Proof. (i) The matrix $\mathbf{B}^{-1/2} = \mathrm{diag}(\beta_1^{-1/2}, \dots, \beta_6^{-1/2})$ is diagonal, hence symmetric. For symmetric matrices $A$and B, the product $BAB$is symmetric when B is diagonal. (ii) From Equation [4.22], $(\Gamma_B)_{ij} = (T_{\text{eff}})_{ij}/\sqrt{\beta_i\beta_j}$, which is decreasing in β_i and β_j. (iii) For any vector $\mathbf{v}$ with $\| \mathbf{v} \| = 1$, $| \mathbf{v}^\top \Gamma_B \mathbf{v} | \leq \rho(T_{\text{eff}})/\beta_{\min}$. (iv) Since $\beta_k(t)$ follows a jump-diffusion, so does $\Gamma_B(t)$. ▫

Proposition 4.11 (Properties of Effective Tensor). The Effective Tensor satisfies:

(i) **Symmetry:** $(T_{\text{eff}})_{ij}(s; \Lambda) = (T_{\text{eff}})_{ji}(s; \Lambda)$

(ii) **Linearity in priorities:** $T_{\text{eff}}(s; \alpha\Lambda_1 + (1-\alpha)\Lambda_2) = \alpha T_{\text{eff}}(s; \Lambda_1) + (1-\alpha)T_{\text{eff}}(s; \Lambda_2)$

(iii) **Negative semidefiniteness:** Under concave domain objectives, the matrix $[(T_{\text{eff}})_{ij}]$ is negative semidefinite.

Proof. (i) follows from symmetry of S. (ii) is direct calculation. (iii) follows because convex combinations of negative semidefinite matrices (from concave objectives) with non-negative weights remain negative semidefinite. ■

Proposition 4.12 (Tier 3 Synthesis: Joint Dependence of Effective Coupling). The discount-adjusted coupling matrix $\Gamma_B(t)$ exhibits the following properties characterizing the joint dependence on all three input tiers:

(i) *Factorization:* The effective coupling factorizes as:

$$\Gamma_B = \mathbf{B}^{-1/2} \left(\sum_{k=1}^{6} w_k \, \mathbf{S}_k \right) \mathbf{B}^{-1/2}$$

[Eq. 4.22]

exhibiting separable dependence on structure (**S**), priorities (Λ through w_k), and temporal urgency (**B**).

(ii) *Marginal effects:* The partial derivatives with respect to each tier are:

$$\frac{\partial (\Gamma_B)_{ij}}{\partial S_{ijk}} = \frac{w_k}{\sqrt{\beta_i \beta_j}}$$

[Eq. 4.23a]

$$\frac{\partial(\Gamma_B)_{ij}}{\partial w_k} = \frac{(S_k)_{ij}}{\sqrt{\beta_i \beta_j}}$$

[Eq. 4.23b]

$$\frac{\partial(\Gamma_B)_{ij}}{\partial \beta_i} = -\frac{1}{2}\frac{(\Gamma_B)_{ij}}{\beta_i}$$

[Eq. 4.23c]

(iii) Non-redundancy of Λ and **B**: The Priority Matrix and Discount Matrix capture orthogonal dimensions of client preferences:

- Λ determines the relative importance of domain outcomes (cross-sectional trade-offs)
- **B** determines the temporal urgency of domain outcomes (intertemporal trade-offs)

Formally: $\partial\Gamma_B / \partial w_k \neq c \cdot \partial\Gamma_B / \partial\beta_k$ for any scalar c, except in degenerate cases.

(iv) Dominance ordering: If $\beta_i < \beta_j$ (domain i more patient than domain j), then couplings involving domain $i$are amplified relative to those involving domain j:

$$\frac{(\Gamma_B)_{ik}}{(\Gamma_B)_{jk}} = \sqrt{\frac{\beta_j}{\beta_i}} \cdot \frac{(T_{\text{eff}})_{ik}}{(T_{\text{eff}})_{jk}} > \frac{(T_{\text{eff}})_{ik}}{(T_{\text{eff}})_{jk}}$$

(v) *Limit cases:*

- As $\beta_k \to 0$ (infinite patience in domain k): $(\Gamma_B)_{ik} \to \infty$ for all i with $(T_{\text{eff}})_{ik} \neq 0$
- As $\beta_k \to \infty$ (extreme urgency in domain k): $(\Gamma_B)_{ik} \to 0$ for all i

Proof.

Part (i). Direct substitution of Definition 4.12 ($T_{\text{eff}} = \sum_k w_k S_k$) into Definition 4.13 ($\Gamma_B = \mathbf{B}^{-1/2} T_{\text{eff}} \mathbf{B}^{-1/2}$).

Part (ii). For (a), $(\Gamma_B)_{ij} = \sum_k w_k (S_k)_{ij} / \sqrt{\beta_i \beta_j}$, so $\partial / \partial S_{ijk} = w_k / \sqrt{\beta_i \beta_j}$. For (b), direct differentiation with respect to w_k. For (c), $\partial / \partial \beta_i$of $(\beta_i)^{-1/2}$yields $-\frac{1}{2}\beta_i^{-3/2}$, giving the result.

Part (iii).From (ii), $\partial\Gamma_B / \partial w_k$ is proportional to $(S_k)_{ij}$, the structural coupling matrix for domain k. Meanwhile, $\partial\Gamma_B / \partial\beta_k$ affects only rows and columns involving domain k, scaling existing couplings. These operate on different dimensions: $w_k$weights which structural couplings matter,

while β_k scales how important domain k's couplings are over time. The derivatives are linearly independent except when $S_k = c \cdot \mathbf{e}_k \mathbf{e}_k^\top$ (domain $k$only coupled to itself), which is ruled out by the structural interdependency inherent in financial planning.

Part (iv). From Equation [4.21], the ratio follows directly. Since $\beta_i < \beta_j$implies $\sqrt{\beta_j/\beta_i} > 1$, the amplification follows.

Part (v). From Equation [4.21], $(\Gamma_B)_{ik} = (T_{\text{eff}})_{ik}/\sqrt{\beta_i \beta_k}$. As $\beta_k \to 0$, the denominator vanishes and $(\Gamma_B)_{ik} \to \infty$. As $\beta_k \to \infty$, the denominator diverges and $(\Gamma_B)_{ik} \to 0$. ∎

Remark 4.10 (Independence of Priority Weights and Discount Rates). Proposition 4.12 (iii) establishes that $\Lambda$and **B** are not redundant. A client can care deeply about a domain yet be patient about it (high w_k, low β_k), or care little about a domain yet demand immediate action when it arises (low w_k, high β_k). The Priority Matrix answers "What matters?" while the Discount Matrix answers "When does it need to happen?" These are independent questions with independent answers, both necessary for complete characterization of client preferences.

4.5 | Relationship Between Clarke and Viscosity Frameworks

Both Clarke generalized calculus and viscosity solutions handle non-smoothness, but from different perspectives. I use Clarke calculus for the Structural Tensor (characterizing curvature of domain objectives) and viscosity solutions for the value function (solving the HJB equation).

Definition 4.7 (Viscosity Sub/Superdifferentials). For $f: \mathbb{R}^n \to \mathbb{R}$:

- Viscosity subdifferential: $D^- f(x) = \{p: \exists \phi \in C^1 \text{ s.t. } f - \phi \text{ has local min at } x \text{ and } \nabla\phi(x) = p\}$
- Viscosity superdifferential: $D^+ f(x) = \{p: \exists \phi \in C^1 \text{ s.t. } f - \phi \text{ has local max at } x \text{ and } \nabla\phi(x) = p\}$

Proposition 4.5 (Clarke-Viscosity Relationship). For a locally Lipschitz function $f: \mathbb{R}^n \to \mathbb{R}$:

$$D^- f(x) \subseteq \partial f(x) \subseteq D^+ f(x)$$

[Eq. 4.24]

At points where f is differentiable, all three coincide with $\{\nabla f(x)\}$.

Proof. I establish both inclusions by analyzing test functions and limits of gradients. The key is that a test function achieving a local minimum provides a lower bound on directional derivatives, while the Clarke gradient captures all limiting gradients. ∎

Remark 4.11 (Application to Financial Planning). This relationship is implemented as follows: The Structural Tensor S uses Clarke generalized Hessians to characterize curvature of domain objectives appropriately because objectives like R_{Tax} are locally Lipschitz but not smooth. The value function V is characterized via viscosity solutions of the HJB equation and is appropriate because V satisfies the HJB in a weak sense at non-differentiable points. Proposition 4.3 ensures consistency. The regularity properties established by Cannarsa and Sinestrari (2004)

for semiconcave value functions in optimal control provide additional structure that strengthens the connection between these frameworks.

4.6 | Time-Varying Priorities

Client priorities may shift over time due to life events (health shocks, births, job changes) or evolving preferences.

Definition 4.8 (Stochastic Priority Process). The priority vector w_t evolves as:

$$w_{t+1} = \Pi_{\Delta^5}(w_t + \eta_t)$$

[Eq. 4.25]

where η_t is a mean-zero random perturbation and Π_{Δ^5} is the Euclidean projection onto the probability simplex.

Key Property: w_t is known at time t, but w_{t+1} is uncertain given time-t information. This creates a distinctive form of uncertainty about the objective function itself.

Definition 4.9 (Life-Cycle Priority Dynamics). I decompose priorities as:

$$w_t = \bar{w}(t) + \varepsilon_t$$

[Eq. 4.26]

where $\bar{w}(t)$ is a deterministic, age-dependent baseline priority function and ε_t is a mean-zero stochastic deviation.

Typical patterns include: $\beta_{\text{Est}} > 0$ (estate priority increases with age); $\beta_{\text{Inv}} < 0$ (investment priority decreases with age); $\beta_{\text{RM}} > 0$ (risk management priority increases with age).

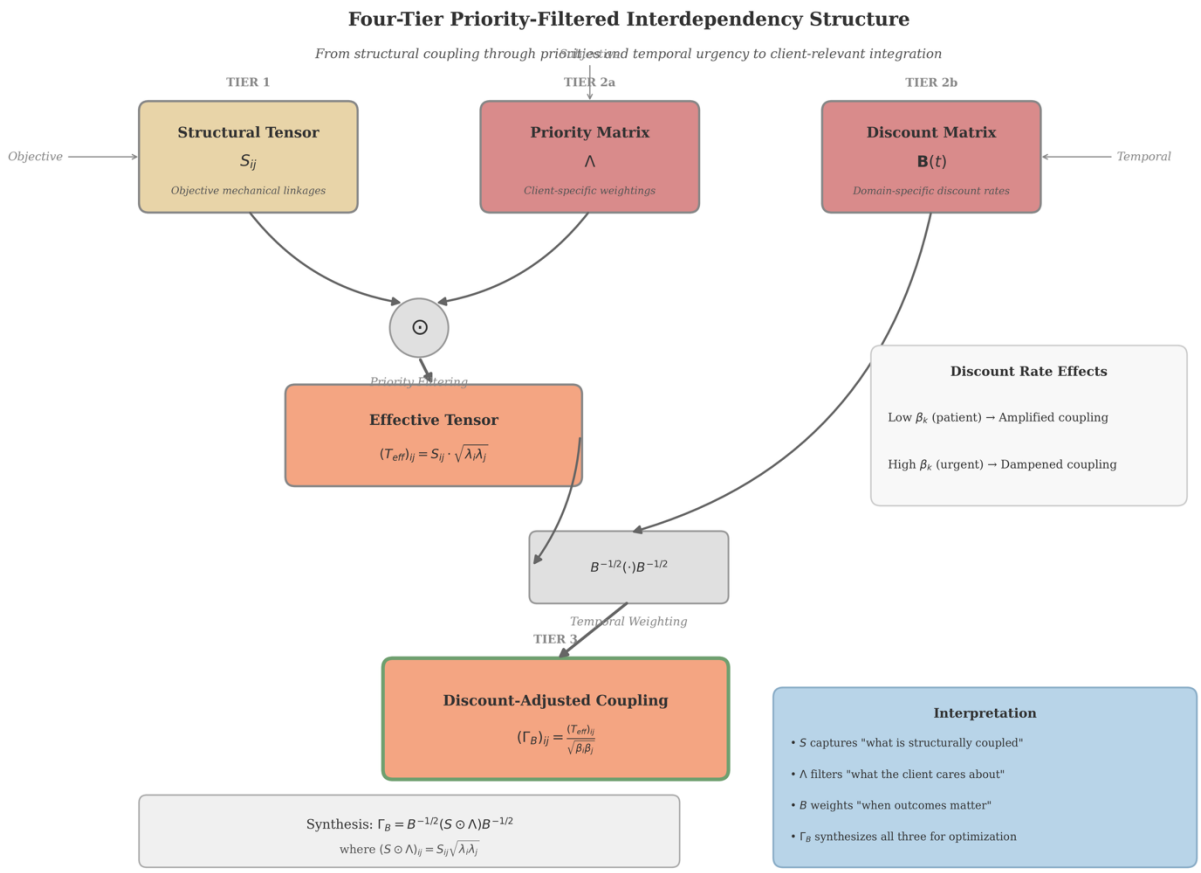
4.7 | Summary: The Four-Tier Architecture

The four-tier decomposition provides a complete characterization of client-relevant interdependency (See Figure 6):

Tier 1: Structural Tensor S captures objective mechanical linkages from tax law, actuarial mathematics, and portfolio theory. These exist regardless of client preferences and can be pre-computed and updated when regulations change.

Tier 2a: Priority Matrix Λ captures subjective weightings reflecting how much each client cares about each domain's objectives. These vary across clients and must be elicited through engagement.

Figure 6: Four-Tiered Priority-Filtered Interdependency Structure



Tier 2b: Discount Matrix $\mathbf{B}(t)$ captures temporal priority structure or how urgently outcomes in each domain are weighted over time. These rates evolve stochastically based on observable state (endogenous) and life events (exogenous).

Tier 3: Effective Tensor T_{eff} and Discount-Adjusted Coupling Γ_B synthesize structure, values, and temporal priorities to yield client-relevant coupling. The effective tensor filters structural coupling through priorities; the discount-adjusted matrix further weights by temporal urgency.

The synthesis equation:

$$\Gamma_B(t) = \mathbf{B}(t)^{-1/2} \cdot \left(\sum_{k=1}^{6} w_k \cdot \mathbf{S}_k\right) \cdot \mathbf{B}(t)^{-1/2}$$

[Eq. 4.27]

captures the complete chain from objective structure through subjective priorities and temporal urgency to client-relevant coupling.

5 | Uncertainty Framework

Section 4 characterized interdependencies under certainty. Real financial planning confronts pervasive uncertainty for which some events are controllable, some not, and some about the very priorities that guide optimization. I develop a taxonomy of uncertainty types and their treatment (See Figure 7).

5.1 | Endogenous vs. Exogenous Uncertainty.

Definition 5.1 (Endogenous Uncertainty). A random disturbance ε is *endogenous* if its distribution or impact depends on planning actions.

Examples of endogenous uncertainty:

- *Investment volatility:* Market volatility is exogenous, but portfolio exposure to volatility is endogenous (asset allocation decision).
- *Longevity risk:* Probability of living to age 95 is exogenous, but the financial consequence is endogenous (annuitization provides longevity insurance).
- *Disability risk:* Probability of disability is largely exogenous, but income replacement is endogenous (disability insurance coverage).

The distinction between risks that can be controlled through action and those that cannot is fundamental to optimal response. Arrow (1971) established that risk-averse agents will fully insure against actuarially fair risks; my framework extends this insight by recognizing that the "fairness" of insurance depends on cross-domain effects captured in the Structural Tensor.

Definition 5.2 (Exogenous Uncertainty). A random disturbance ε is *exogenous* if its distribution is independent of planning actions.

Examples of exogenous uncertainty:

- *Tax law changes:* Future tax rates and rules are determined by political processes beyond individual control.
- *Macroeconomic conditions:* Inflation, interest rates, and economic growth affect planning but are not individually controllable.
- *Regulatory changes:* Social Security rules, Medicare provisions, and other regulations may change unpredictably.
- *Pandemic/disaster:* Systemic shocks affecting health, markets, and economy simultaneously.

5.2 | Priority Uncertainty

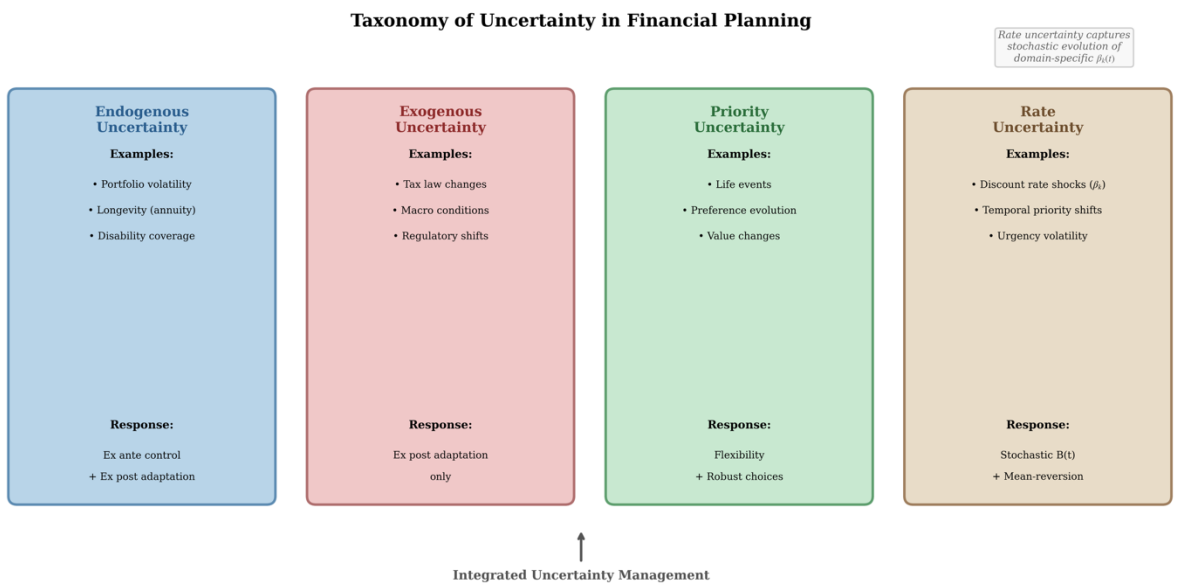
Definition 5.3 (Priority Uncertainty). *Priority uncertainty* refers to uncertainty about future priorities w_{t+1} given current priorities w_t.

Priority uncertainty is a distinct category: partially endogenous (some life events that shift priorities involve choice, like having children) and partially exogenous (other priority-shifting events like health shocks are not chosen). It represents uncertainty about the objective function itself, not just about states or constraints.

Theorem 5.1 (Shock Amplification). *Consider the linearized domain dynamics* $\delta\dot{\mathbf{d}}(t) = M\,\delta\mathbf{d}(t)$ *with* $M \in \mathbb{R}^{m\times m}$. *Let* $\alpha(M) = max\{\Re(\lambda): \lambda \in \mathrm{spec}(M)\}$ *denote the spectral abscissa.*

Then shock propagation $\Delta\mathbf{d}(t) = e^{Mt}\Delta\mathbf{d}(0)$ exhibits: (i) Dampening ($\alpha(M) < 0$): $\| \Delta\mathbf{d}(t) \| \to 0$ as $t \to \infty$ at rate $| \alpha(M) |$. (ii) Persistence ($\alpha(M) = 0$): $\| \Delta\mathbf{d}(t) \| = O(t^{k-1})$ where k is the size of the largest Jordan block for eigenvalues on the imaginary axis. (iii) Amplification ($\alpha(M) > 0$): $\| \Delta\mathbf{d}(t) \| \to \infty$ exponentially at rate $\alpha(M)$.

Figure 7: Taxonomy of Uncertainty in Financial Planning



Proof.

Step 1 (Jordan decomposition). Every matrix $M \in \mathbb{R}^{m\times m}$ admits a Jordan normal form $M = PJP^{-1}$ where $J = \text{diag}\left(J_{n_1}(\lambda_1), \dots, J_{n_r}(\lambda_r)\right)$ is block-diagonal with Jordan blocks:

$$J_n(\lambda) = \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda & 1 \\ 0 & 0 & \cdots & 0 & \lambda \end{pmatrix} \in \mathbb{C}^{n\times n}$$

Step 2 (Exponential of Jordan blocks). For a Jordan block $J_n(\lambda) = \lambda I + N$ where N is the nilpotent superdiagonal matrix:

$$e^{J_n(\lambda)t} = e^{\lambda t}e^{Nt} = e^{\lambda t}\sum_{j=0}^{n-1}\frac{t^j}{j!}N^j$$

since $N^n = 0$. The (i,k) entry is:

$$\left[e^{J_n(\lambda)t}\right]_{ik} = \begin{cases} e^{\lambda t}\dfrac{t^{k-i}}{(k-i)!} & \text{if } k \geq i \\ 0 & \text{if } k < i \end{cases}$$

Step 3 (Full exponential). Since J is block-diagonal:

$$e^{Jt} = \mathrm{diag}\left(e^{J_{n_1}(\lambda_1)t}, \dots, e^{J_{n_r}(\lambda_r)t}\right)$$

and $e^{Mt} = Pe^{Jt}P^{-1}$.

Step 4 (Norm bounds). Using submultiplicativity:

$$\| e^{Mt} \| \leq \| P \| \cdot \| P^{-1} \| \cdot \max_j \| e^{J_{n_j}(\lambda_j)t} \|$$

For each Jordan block:

$$\| e^{J_n(\lambda)t} \| \leq C_n(1 + t^{n-1})e^{\Re(\lambda)t}$$

for some constant C_n depending on the matrix norm.

Step 5 (Asymptotic behavior).

Case $\alpha(M) < 0$*:* All $\Re(\lambda_j) \leq \alpha(M) < 0$. The exponential decay $e^{\alpha(M)t} \to 0$ dominates the polynomial growth, so $\| e^{Mt} \| \to 0$.

Case $\alpha(M) = 0$*:* The eigenvalues with $\Re(\lambda) = 0$contribute terms of the form $t^{k-1}e^{i\Im(\lambda)t}$ (bounded oscillation times polynomial). If the largest Jordan block for such eigenvalues has size k, then $\| e^{Mt} \| = O(t^{k-1})$.

Case $\alpha(M) > 0$*:* At least one eigenvalue has $\Re(\lambda) > 0$. The term $e^{\Re(\lambda)t}$ grows without bound, so $\| e^{Mt} \| \to \infty$.

See Pazy (1983, Chapter 1) for the semigroup perspective. ∎

Proposition 5.1 (Optimal Response by Uncertainty Type).

(i) *Endogenous uncertainty***:** Optimal response involves both ex ante control (modifying distribution or impact) and ex post adaptation (adjusting after realization).

(ii) *Exogenous uncertainty:* Optimal response involves only ex post adaptation; no *ex ante* control is possible.

(iii) *Priority uncertainty:* Optimal response involves building flexibility and making robust choices that perform reasonably across plausible priority specifications.

(iv*) Rate uncertainty:* Optimal response involves: (a) managing endogenous rate drivers through state control; (b) building flexibility to adapt when exogenous rate shocks occur; (c) anticipating predictable rate changes and preparing in advance; (d) integrating across domains to hedge correlated rate movements.

Proof. For (i), if the distribution depends on action a, the optimization problem includes derivatives of both utility and distribution with respect to a. For (ii), the distribution is fixed, so only the response function can be optimized. For (iii), the agent must balance commitment (exploiting current priorities) with flexibility (preserving options for priority shifts). ∎

5.3 | Rate Uncertainty

Definition 5.4 (Rate Uncertainty).Rate uncertainty refers to uncertainty about future domain-specific discount rates $\boldsymbol{\beta}(t + dt)$ given current rates $\boldsymbol{\beta}(t)$ and current state X_t.

Rate uncertainty shares characteristics with both priority uncertainty and traditional state uncertainty:

- Like priority uncertainty: Rate uncertainty affects the objective function. Specifically, how future outcomes are weighted relative to present outcomes. Changes in β_k alter the effective planning horizon for domain k.
- Like state uncertainty: Rate uncertainty has both endogenous and exogenous components. The endogenous component $f_k(X_t)$ responds to observable state and can be influenced through planning actions. The exogenous component $\xi_k(t)$ follows stochastic dynamics beyond direct control.
- Unlike either: Rate uncertainty is partially predictable through state observation. A client approaching retirement will predictably experience increases in β_{Ret}. A client with deteriorating liquidity will predictably experience increases in β_{CF}. This predictability creates anticipation opportunities absent in pure priority uncertainty.

Remark 5.1 (Rate Uncertainty Decomposition). The total variance of rate changes decomposes as:

$$\mathrm{Var}(d\beta_k) = \underbrace{(\nabla f_k)^\top \Sigma_X (\nabla f_k)\, dt}_{\text{Endogenous (state-driven)}} + (\sigma_k^\beta)^2\, dt + \underbrace{\sum_e \lambda^{(e)}\, \mathbb{E}[(\gamma_k^{(e)})^2]}_{\text{Exogenous (shock-driven)}}$$

[Eq. 5.1]

where Σ_X is the covariance matrix of state dynamics. The relative magnitude of these components determines how much rate uncertainty can be managed through state control versus how much must be treated as irreducible.

Remark 5.2 (Stochastic Rates and Shock Dynamics). Under stochastic discount rates, the linearized dynamics matrix M becomes time-varying:

$$M_B(t) = \mathbf{B}(t)^{-1/2} M(\Lambda) \mathbf{B}(t)^{-1/2}$$

[Eq. 5.2]

The stability analysis of Theorem 5.1 extends to this setting with modifications:

(i) Expected stability replaces deterministic stability: the system is stable in expectation if $\rho(\bar{M}_B) < 1$ where $\bar{M}_B = \mathbb{E}[M_B(t)]$.

(ii) Almost-sure stability requires analysis of Lyapunov exponents: $\lambda_k = \lim_{t\to\infty} t^{-1} \ln \| \delta_k(t) \|$.

(iii) *Volatility can destabilize:* High rate volatility introduces an Itô correction term that may push an otherwise stable system toward instability. The critical threshold is:

$$\sigma_k^{\text{crit}} = \sqrt{2\bar{\beta}_k \cdot | \alpha(M(\Lambda)) |}$$

[Eq. 5.3]

When $\sigma_k^{\beta} > \sigma_k^{\text{crit}}$, the volatility-induced growth can dominate the stabilizing drift.

The full stability analysis under stochastic rates is developed in Theorem 7.3.

With uncertainty characterized, I formulate the optimization problem that integrates structure, priorities, and uncertainty into a coherent dynamic program.

6 | Dynamic Optimization with Domain-Specific Discounting

Sections 3–5 now combine into a dynamic optimization problem. The priority-weighted objective aggregates domain-specific goals; the HJB equation characterizes optimal behavior; viscosity solutions handle the inevitable non-smoothness.

6.1 | Priority-Weighted Objective

Definition 6.1 (Priority-Weighted Instantaneous Reward). Given priorities Λ, the instantaneous reward is:

$$r(s, a; \Lambda) = \sum_{k=1}^{6} w_k \cdot R_k(s, a)$$

[Eq. 6.1]

This is a weighted average of domain-specific objectives, with weights given by the Priority Matrix.

Note that risk aversion and safety-first principles are encoded within the domain objective functions R_k, which are strongly concave and may include barrier penalties near catastrophic outcomes (bankruptcy, loss of home, inability to fund essential expenses). The linear aggregation weights w_k represent prioritization of surplus across domains, not substitution at the margin of survival. This approach is consistent with the safety-first philosophy articulated by Pfau (2019), which emphasizes that essential needs must be secured before discretionary goals are pursued.

A linear sum of strongly concave functions preserves strong concavity of the aggregate, ensuring the safety properties embedded in domain objectives propagate to the integrated problem. The linear weights therefore capture how the client prioritizes performance across domains, given that each domain already incorporates appropriate risk aversion for its context.

Definition 6.2 (Infinite-Horizon Value Function). The value function is:

$$V(x;\Lambda) = \sup_{\pi\in\Pi} \mathbb{E}_x^{\pi}\left[\int_0^{\infty} e^{-\beta t}\, r(X_t, a_t;\Lambda)\, dt\right]$$

[Eq. 6.2]

where $\beta > 0$ is the discount rate and the supremum is over admissible strategies.

Definition 6.3 (Domain-Specific Value Components). For each domain k, the domain-specific value function on the augmented state space is:

$$V_k(x,\boldsymbol{\beta}) = \sup_{\pi\in\Pi} \mathbb{E}_{x,\boldsymbol{\beta}}\left[\int_0^{\infty} \exp\left(-\int_0^t \beta_k(s)\, ds\right) R_k(X_t, a_t)\, dt\right]$$

[Eq. 6.3]

where the discount factor $\exp\left(-\int_0^t \beta_k(s)\, ds\right)$ is path-dependent, depending on the entire trajectory of $\beta_k(s)$ for $s \in [0,t]$.

Remark 6.1 (Path-Dependent Discounting). The path-dependent discount factor differs fundamentally from the constant-rate case $e^{-\beta_k t}$. Under stochastic rates:

(i) The discount factor is itself a random variable, correlated with both state X_t and other domain rates $\beta_j(t)$ for $j \neq k$.

(ii) The expected discount factor satisfies $\mathbb{E}[\exp\left(-\int_0^t \beta_k(s)\, ds\right)] \neq \exp\left(-\bar{\beta}_k t\right)$ in general; Jensen's inequality implies $\mathbb{E}[\exp\left(-\int_0^t \beta_k(s)\, ds\right)] > \exp\left(-\mathbb{E}[\int_0^t \beta_k(s)\, ds]\right)$ due to convexity of the exponential.

(iii) Correlation between $\beta_k(t)$and $R_k(X_t, a_t)$ creates hedging opportunities: if high-reward states coincide with low discount rates, expected value increases.

Definition 6.4 (Aggregate Value Function with Domain-Specific Discounting). The total value function aggregates domain-specific values:

$$V(x,\boldsymbol{\beta};\Lambda) = \sup_{\pi\in\Pi} \mathbb{E}_{x,\boldsymbol{\beta}}\left[\int_0^{\infty} \sum_{k=1}^{6} w_k \exp\left(-\int_0^t \beta_k(s)\, ds\right) R_k(X_t, a_t)\, dt\right]$$

[Eq. 6.4]

Proposition 6.2 (Single-Rate Special Case). When $\beta_k = \beta$ for all k, the value function reduces to:

$$V(x;\Lambda,\beta) = \sup_{\pi\in\Pi} \mathbb{E}_x\left[\int_0^\infty e^{-\beta t}\, r(X_t, a_t;\Lambda)\, dt\right]$$

[Eq. 6.5]

recovering the classical formulation with a single discount rate.

Proof. When all domain rates equal β, the path-dependent factors coincide: $\exp\left(-\int_0^t \beta_k\,(s)\, ds\right) = e^{-\beta t}$ for all k. The sum $\sum_k w_k\, e^{-\beta t} R_k = e^{-\beta t} \sum_k w_k\, R_k = e^{-\beta t} r(\cdot;\Lambda)$ factors out the common discount. ∎

6.2 | Hamilton-Jacobi-Bellman Equation

Proposition 6.1 (HJB Equation). If the value function V is sufficiently smooth, it satisfies:

$$\beta V(x;\Lambda) = \sup_{a\in A(x)} \{r(x,a;\Lambda) + \mathcal{L}^a V(x;\Lambda)\}$$

[Eq. 6.6]

where the infinitesimal generator is:

$$\mathcal{L}^a V(x) = \sum_i b_i\,(x,a)\frac{\partial V}{\partial x_i} + \frac{1}{2}\sum_{i,j} (\sigma\,\sigma^\top)_{ij}(x,a)\frac{\partial^2 V}{\partial x_i\,\partial x_j} + \int_Z [V\big(x+\gamma(x,z)\big) - V(x)]\nu(dz)$$

[Eq. 6.7]

Definition 6.5 (Viscosity Solution of HJB). A function $V: X \to \mathbb{R}$ is a *viscosity solution* of the HJB equation if:

(i) **Viscosity subsolution:** For any C^2 test function ϕsuch that $V - \phi$ has a local maximum at x_0:

$$\beta V(x_0) \le \sup_{a\in A(x_0)} \{r(x_0,a;\Lambda) + \mathcal{L}^a \phi(x_0)\}$$

[Eq. 6.8]

(ii) **Viscosity supersolution:** For any C^2 test function ϕsuch that $V - \phi$ has a local minimum at x_0:

$$\beta V(x_0) \geq \sup_{a \in A(x_0)} \{r(x_0, a; \Lambda) + \mathcal{L}^a \phi(x_0)\}$$

[Eq. 6.9]

Theorem 6.1 (Existence and Uniqueness). Under standard regularity conditions on coefficients and rewards, the value function $V(\cdot; \Lambda)$ is the unique viscosity solution of the HJB equation in the class of functions with at most polynomial growth.

Proof. I follow Fleming and Soner (2006), Chapter IV. The dynamic programming principle establishes that V satisfies the HJB. Uniqueness follows from comparison theorems for viscosity solutions. The regularity theory of Cannarsa and Sinestrari (2004) ensures that the value function is semiconcave, providing additional structure for the sensitivity analysis in Section 7. ∎

Proposition 6.3 (Extended Coupled HJB System). Under domain-specific stochastic discount rates, the domain value functions satisfy the coupled integro-differential equation system:

$$\beta_k V_k(x, \boldsymbol{\beta}) = \sup_{a \in A(x)} \left\{R_k(x, a) + \mathcal{L}_x^a V_k + \mathcal{L}_\beta V_k + \mathcal{L}_{x\beta}^{\text{cross}} V_k + \mathcal{J}_\beta V_k\right\}$$

[Eq. 6.10]

for $k = 1, \ldots, 6$, where the operators are defined as follows.

State Generator (standard diffusion operator on state space):

$$\mathcal{L}_x^a V_k = \sum_i b_i(x, a) \frac{\partial V_k}{\partial x_i} + \frac{1}{2} \sum_{i,j} (\sigma_X \sigma_X^\top)_{ij}(x, a) \frac{\partial^2 V_k}{\partial x_i \, \partial x_j}$$

[Eq. 6.11]

Rate Diffusion Generator (drift and diffusion in rate space):

$$\mathcal{L}_\beta V_k = \sum_{j=1}^{6} \mu_j^\beta(x, \boldsymbol{\beta}) \frac{\partial V_k}{\partial \beta_j} + \frac{1}{2} \sum_{j=1}^{6} (\sigma_j^\beta)^2 \frac{\partial^2 V_k}{\partial \beta_j^2}$$

[Eq. 6.12]

where the rate drift is:

$$\mu_j^\beta(x, \boldsymbol{\beta}) = \nabla f_j(x) \cdot \mu_X(x, a) + \frac{1}{2} \operatorname{tr}(\nabla^2 f_j(x) \cdot \sigma_X \sigma_X^\top) - \theta_j(\beta_j - \bar{\beta}_j - f_j(x))$$

[Eq. 6.13]

Cross-Derivative Operator (state-rate correlation):

$$\mathcal{L}_{x\beta}^{\text{cross}} V_k = \sum_{j=1}^{6} \operatorname{tr}\left(\sigma_X^\top \nabla_x \left(\frac{\partial V_k}{\partial \beta_j} \right) \cdot (\nabla f_j)^\top \sigma_X \right)$$

[Eq. 6.14]

Rate Jump Operator (nonlocal integro-differential term):

$$\mathcal{J}_\beta V_k = \sum_{j=1}^{6} \int_{\mathcal{Z}} \left[V_k(x, \boldsymbol{\beta} + \gamma_j(z)\mathbf{e}_j) - V_k(x, \boldsymbol{\beta}) - \gamma_j(z) \frac{\partial V_k}{\partial \beta_j} \right] \nu_j(dz)$$

[Eq. 6.15]

where $\mathbf{e}_j$ is the j-th standard basis vector and ν_j is the Lévy measure for domain j jumps.

Remark 6.2 (Coupling Through Shared Action Space). The six domain equations [Eq. 6.6] are coupled through: (i) **Shared action:** The supremum is over the same action space $A(x)$ for all domains; (ii) **Correlated rates:** Systemic life events induce simultaneous jumps in multiple β_k; (iii) **State dependence:** All value functions depend on the common state x and common action a.

Theorem 6.2 (Verification). If a function v satisfies the HJB equation in the viscosity sense and appropriate growth conditions, and if a policy a^* achieves the supremum in the HJB equation almost everywhere, then $v = V$ (the value function) and a^* is optimal.

6.3 | Verification Theorem

Proof. I show $v \leq V$ (v bounds V from below via any admissible policy) and $v \geq V$ (the candidate policy a^* achieves v). ∎

6.4 | Regime-Switching Formulation

For computational tractability and interpretability, I develop an alternative formulation where discount rates take discrete values corresponding to life-stage regimes.

Definition 6.6 (Life-Stage Regimes). Let $\mathcal{R} = \{r_1, \dots, r_M\}$ be a finite set of life-stage regimes, each associated with a discount rate vector $\boldsymbol{\beta}^{(r)} = (\beta_1^{(r)}, \dots, \beta_6^{(r)})$.

Figure 8: Priority-Weighted Dynamic Optimization Flow

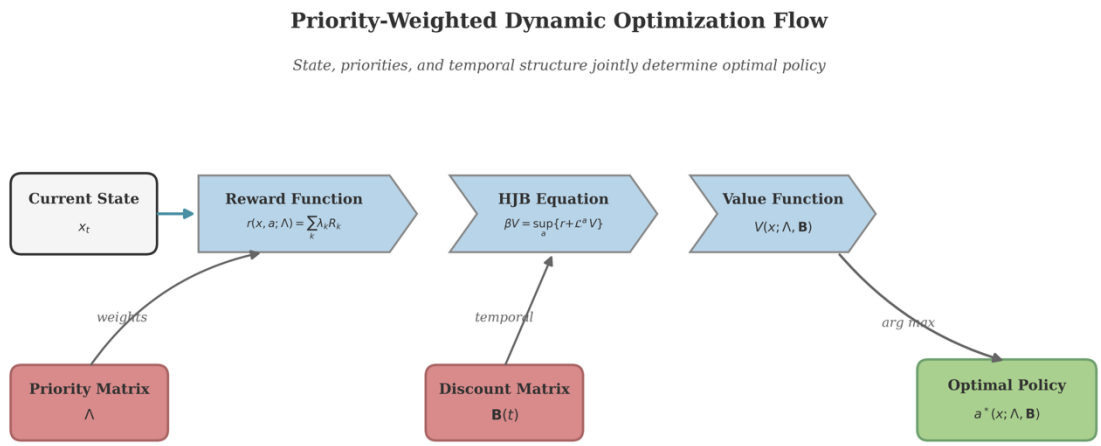


Table 6.1: Example Life-Stage Regimes

Regime r	Description	β_{CF}	β_{Ret}	β_{Tax}	β_{Est}	β_{RM}	β_{Inv}
r_1	Early career	8.0%	2.0%	4.0%	1.0%	6.0%	2.0%
r_2	Mid-career	5.0%	3.0%	5.0%	1.5%	6.0%	2.5%
r_3	Pre-retirement	4.0%	6.0%	6.0%	2.0%	7.0%	3.0%
r_4	Early retirement	6.0%	4.0%	5.0%	2.5%	8.0%	2.5%
r_5	Late retirement	7.0%	3.0%	4.0%	4.0%	9.0%	2.0%
r_6	Health crisis	10.0%	5.0%	4.0%	6.0%	12.0%	3.0%
r_7	Windfall	3.0%	2.5%	6.0%	2.0%	4.0%	3.0%

Definition 6.7 (Regime Transition Intensities). The regime process r_t is a continuous-time Markov chain with state-dependent transition intensities:

$$\lambda_{rs}(x) = \lim_{dt \to 0} \frac{1}{dt} \mathbb{P}(r_{t+dt} = s \mid r_t = r, X_t = x)$$

[Eq. 6.16]

for $r \neq s$.

Proposition 6.4 (Regime-Switching HJB System). Under the regime-switching formulation, the value functions $V_k^{(r)}(x)$ satisfy:

$$\beta_k^{(r)} V_k^{(r)}(x) = \sup_{a \in A(x)} \left\{ R_k(x,a) + \mathcal{L}_x^a V_k^{(r)}(x) + \sum_{s \neq r} \lambda_{rs}(x) \left[V_k^{(s)}(x) - V_k^{(r)}(x) \right] \right\}$$

[Eq. 6.17]

This is a system of $6M$ coupled PDEs (6 domains × M regimes).

Remark 6.3 (Jump-Diffusion vs. Regime-Switching). The two formulations offer different trade-offs:

Feature	Jump-Diffusion	Regime-Switching
Rate space	Continuous $\mathbb{R}_{++}^6$	Discrete $\{\boldsymbol{\beta}^{(r_1)}, \dots, \boldsymbol{\beta}^{(r_M)}\}$
Dimensionality	+6 state dimensions	$\times M$ value functions
PDE type	Integro-differential	Standard coupled
Interpretability	Technical	Intuitive regime labels
Tractability	Challenging	More tractable

The HJB formulation provides necessary conditions for optimality. Section 7 establishes the theoretical properties of solutions: filtering, separability, sensitivity, and stability. These guide when and how priorities shape recommendations.

Section 7: Main Theoretical Results

The preceding sections constructed the framework. Thirteen theorems and corresponding proofs now establish how priorities filter structural coupling, when domains can be treated separately, how sensitive recommendations are to priority specification, and what happens when priorities are misaligned.

The theorems fall into four groups: structural properties (7.1–7.2), stability and separability under stochastic rates (7.3–7.4), sensitivity, coupling, and alignment (7.5–7.9), and stochastic rate phenomena (7.10–7.13). Together they provide formal foundations for intuitions practitioners have developed through experience, extended to accommodate the temporal dynamics of client priorities.

Theorem 7.1 (Interdependency Tensor Structure). *Let V^* be the optimal value function for the integrated financial planning problem with domain decomposition $\mathfrak{D} = \{\mathcal{D}_1, \dots, \mathcal{D}_m\}$. Assume V^* is C^k for $k \geq 2$. The interdependency tensor $\mathcal{T}$ with components $\mathcal{T}_{i_1 \cdots i_\ell} = \partial^\ell V^* / \partial d_{i_1} \cdots \partial d_{i_\ell}$*

satisfies: (i) Symmetry: $\mathcal{T}_{i_1 \cdots i_\ell} = \mathcal{T}_{\pi(i_1) \cdots \pi(i_\ell)}$ for any permutation π. (ii) Negative semi-definiteness at order 2: If V^* is concave, $\Gamma = \mathcal{T}^{(2)}$ satisfies $\mathbf{v}^\top \Gamma \mathbf{v} \leq 0$ for all $\mathbf{v}$. (iii) Sparsity under weak coupling: If domains i and j interact only through $k$intermediate domains, then $|\mathcal{T}_{ij}| = O(\epsilon^{k+1})$ where ϵmeasures direct coupling strength. (iv) Decay of higher-order terms: Under uniform bounds on domain objective curvature, $\| \mathcal{T}^{(\ell)} \| \leq C \cdot M^\ell / \ell!$ for constants C, M depending on the problem data.

Proof.

Part (i): By the generalized Clairaut–Schwarz theorem (Rudin, 1976, Theorem 9.41), if $V^* \in C^k$ then mixed partial derivatives of order $\leq k$ are symmetric:

$$\frac{\partial^\ell V^*}{\partial d_{i_1} \cdots \partial d_{i_\ell}} = \frac{\partial^\ell V^*}{\partial d_{\pi(i_1)} \cdots \partial d_{\pi(i_\ell)}}$$

for any permutation π of $\{i_1, \ldots, i_\ell\}$.

Part (ii): A function V^* is concave if and only if its Hessian is negative semi-definite wherever it exists (Rockafellar, 1970, Theorem 4.5). The coupling matrix $\Gamma_{ij} = \partial^2 V^* / \partial d_i \, \partial d_j$ is the Hessian of V^* restricted to domain coordinates, hence negative semi-definite.

Part (iii): Consider the Bellman equation $V^*(d) = \max_a \{r(d,a) + \beta \mathbb{E}[V^*(d')]\}$. If domains i and j do not directly appear together in either $r$or the transition kernel, then $\partial^2 r / \partial d_i \, \partial d_j = 0$ and the transition factorizes. The cross-derivative V^*_{ij} arises only through chains of intermediate domains. Iterating the Bellman operator, each intermediate step contributes a factor of order ϵ (the coupling strength), yielding $|\mathcal{T}_{ij}| = O(\epsilon^{k+1})$ for k intermediate domains.

Part (iv): The value function solves the Bellman equation, which can be viewed as a fixed point. Differentiating the Bellman equation ℓ times and applying the Faà di Bruno formula for higher derivatives of compositions:

$$D^\ell (f \circ g) = \sum \frac{\ell!}{k_1! \cdots k_m!} \cdot D^{|k|} f \cdot \prod_j \left(\frac{D^j g}{j!} \right)^{k_j}$$

Each additional derivative introduces a factor bounded by the Lipschitz constant $M$of the transition operator. Combinatorial factors from the Faà di Bruno formula contribute $\ell!$ to the denominator after normalization, yielding $\| \mathcal{T}^{(\ell)} \| \leq C M^\ell / \ell!$. ∎

Theorem 7.2 (Priority Filtering). Structural coupling S_{ijk} influences optimal strategy if and only if $w_k > 0$.

Proof. Establish both directions.

⇒ If S_{ijk} influences optimal strategy, then $w_k > 0$.

Suppose $w_k = 0$. The optimal strategy a^* satisfies $0 \in \partial_a Q(s, a^*; \Lambda)$ where $Q = r + \mathcal{L}^a V$. The reward is $r(s, a; \Lambda) = \sum_m w_m R_m(s, a)$. Since $w_k = 0$, the term $w_k R_k$ does not appear, so $S_{ijk} =$

$\partial^2_{d_i d_j} R_k$ does not enter the optimality conditions. By contrapositive, if S_{ijk} influences a^*, then $w_k > 0$.

$\Leftarrow$ If $w_k > 0$, then S_{ijk} influences optimal strategy (generically).

Suppose $w_k > 0$and $S_{ijk} \neq 0$. The effective tensor $(T_{\text{eff}})_{ij} = \sum_m w_m S_{ijm}$ includes the term $w_k S_{ijk} \neq 0$. The second-order conditions involve $\nabla^2_{aa} Q$, which contains terms from $(T_{\text{eff}})_{ij}$. A non-zero Hessian entry generically affects the optimal solution by the implicit function theorem.

Theorem 7.3 (Stochastic Local Stability). *Consider the linearized domain dynamics under stochastic discount rates:*

$$\delta \dot{\mathbf{d}}(t) = M_B(t)\, \delta \mathbf{d}(t)$$

where $M_B(t) = \mathbf{B}(t)^{-1/2} M(\Lambda) \mathbf{B}(t)^{-1/2}$ is the discount-adjusted propagation matrix. Then:

(i) Expected stability: The system is stable in expectation if $\rho(\bar{M}_B) < 1$ where:

$$\bar{M}_B = \mathbb{E}[\mathbf{B}(t)^{-1/2}]\, M(\Lambda)\, \mathbb{E}[\mathbf{B}(t)^{-1/2}]$$

[Eq. 7.1]

(ii) Almost-sure stability: The system is almost surely stable if all Lyapunov exponents are negative:

$$\lambda_k = \lim_{t \to \infty} \frac{1}{t} \ln \parallel \delta_k(t) \parallel < 0 \text{ for all } k$$

[Eq. 7.2]

(iii) Volatility destabilization: There exists a critical volatility threshold σ^{crit} such that for $\sigma_k^{\beta} > \sigma^{\text{crit}}$, the system may become unstable even when the deterministic system is stable.

(iv) Expected convergence rate: Perturbations decay according to:

$$\mathbb{E}[\parallel \delta_k(t) \parallel] \leq \parallel \delta_k(0) \parallel \cdot \exp\left(-\bar{\beta}_k t + \frac{1}{2} \text{Var}(\beta_k) t\right)$$

[Eq. 7.3]

Proof.

Step 1 (Stochastic Lyapunov analysis). Define the Lyapunov function $\mathcal{V}(t) = \| \delta\mathbf{d}(t) \|^2$. Apply Itô's lemma to account for both the drift from $M_B(t)$ and the diffusion from stochastic $\beta_k(t)$.

Step 2 (Expected dynamics). Taking expectations and using the independence of rate fluctuations from perturbation direction yields the expected Lyapunov dynamics involving $\bar{M}_B$.

Step 3 (Oseledets theorem). Almost-sure stability follows from the multiplicative ergodic theorem (Oseledets, 1968), which establishes the existence and properties of Lyapunov exponents for random dynamical systems.

Step 4 (Volatility effect). The Itô correction from $d[\mathbf{B}^{-1/2}]$ introduces a positive-definite term that can overcome negative drift when volatility exceeds the critical threshold. ∎

Corollary 7.3.1 (Domain-Specific Stability). High-$\bar{\beta}$ domains (Cash Flow, Risk Management) are more stable than low-$\bar{\beta}$ domains (Estate, Investment) under rate volatility.

Corollary 7.3.2 (Deterministic Limit). Setting $\sigma_k^\beta = 0$ for all k, the stability condition reduces to $\rho(M(\Lambda)) < 1$, recovering the original Theorem 7.3.

Theorem 7.4 (Stochastic Separability).Domains $\mathcal{D}_i$ and $\mathcal{D}_j$ can be optimized independently without loss of optimality under stochastic discount rates if and only if the following conditions hold:

(i) Reward decomposition: $r(d, a) = r_i(d_i, a_i) + r_j(d_j, a_j) + r_{-ij}(d_{-ij}, a_{-ij})$

(ii) Transition factorization: $P(d' \mid d, a) = P_i(d_i' \mid d_i, a_i) \cdot P_j(d_j' \mid d_j, a_j) \cdot P_{-ij}$

(iii) Structural coupling vanishes: $\mathcal{T}_{ij}(d) = 0$ for all d

(iv) Rate independence: $\mathrm{Cov}(d\beta_i, d\beta_j) = 0$ for all t

(v) Temporal compatibility: $| \bar{\beta}_i - \bar{\beta}_j | < \Delta\beta^{\text{crit}}$

Proof.

Necessity of (i)–(iii). These follow from the original separability theorem; structural coupling through rewards, transitions, or the tensor creates interdependence.

Necessity of (iv). Correlated rate movements create implicit coupling: a shock that increases both $\beta_i$and $\beta_j$simultaneously changes the relative present values of both domains, requiring coordinated response.

Necessity of (v).Large baseline rate differentials create intertemporal arbitrage opportunities. If $\bar{\beta}_i \gg \bar{\beta}_j$, transferring resources from domain i(high urgency, steep discounting) to domain j (low urgency, patient) creates value through temporal mismatch, coupling the domains.

Sufficiency. Under (i)–(v), the value function separates: $V(x, \boldsymbol{\beta}) = V_i(x_i, \beta_i) + V_j(x_j, \beta_j) + V_{-ij}(x_{-ij}, \boldsymbol{\beta}_{-ij})$, and the HJB system decouples. ∎

Corollary 7.4.1 (Reduced Separability). The set of separable domain pairs under stochastic rates is a subset of separable pairs under deterministic rates: $\mathcal{S}^{\text{stoch}} \subseteq \mathcal{S}^{\text{det}}$.

Corollary 7.4.2 (Systemic Events Destroy Separability).If domains i and $j$share a common systemic event e with $\{i, j\} \subseteq \mathcal{D}_e$, then $(i, j) \notin \mathcal{S}^{\text{stoch}}$.

Theorem 7.5 (Priority Sensitivity). The sensitivity of optimal actions to priority changes is:

$$\frac{\partial a^*}{\partial w_k} = -[H]^{-1} \cdot \left(\frac{\partial R_k}{\partial a}\right)$$

[Eq. 7.1]

where $H = \nabla^2_{aa} Q$ is the Hessian of the Q-function.

Proof. Implicit function theorem applied to the first-order conditions $0 = \nabla_a Q(s, a^*; \Lambda)$. Differentiating with respect to w_k:

$$0 = \frac{\partial^2 Q}{\partial a\, \partial w_k} + \left(\frac{\partial^2 Q}{\partial a^2}\right)\left(\frac{\partial a^*}{\partial w_k}\right)$$

Since $\partial Q / \partial w_k = R_k$, I have $\partial^2 Q / \partial a\, \partial w_k = \partial R_k / \partial a$. Solving: $\partial a^* / \partial w_k = -H^{-1} \cdot (\partial R_k / \partial a)$. ∎

Corollary 7.5.1 (Rate Sensitivity). Under stochastic discount rates, the sensitivity of optimal actions to rate changes is:

$$\frac{\partial a^*}{\partial \beta_k} = -[H]^{-1} \cdot \left(\frac{\partial^2 V_k}{\partial a\, \partial \beta_k} - \frac{1}{\beta_k}\frac{\partial R_k}{\partial a}\right)$$

[Eq. 7.4]

The second term captures the direct effect of changed discounting on domain k's contribution; the first term captures indirect effects through the value function.

Proof. Differentiate the first-order condition $0 = \nabla_a Q(s, a^*; \Lambda, \boldsymbol{\beta})$ with respect to β_k. The domain $k$contribution to $Q$is $w_k \int_0^\infty e^{-\beta_k t} R_k\, dt$, whose derivative with respect to β_k yields the $-1/\beta_k$factor. ∎

Remark 7.2 (Comparing Priority and Rate Sensitivity). The sensitivity $\partial a^* / \partial w_k$ (Theorem 7.5) and $\partial a^* / \partial \beta_k$(Corollary 7.5.1) capture different response channels:

- Priority sensitivity reflects response to importance changes: if domain $k$becomes more important, actions shift toward that domain's objectives.
- Rate sensitivity reflects response to urgency changes: if domain $k$becomes more urgent, actions shift toward immediate outcomes in that domain, even if overall importance is unchanged.

Remark 7.1. The robustness of this sensitivity result and its validity across a range of model specifications, rather than depending on precise functional forms, connects to the robust comparative statics literature. Acemoglu and Jensen (2015) develop general methods for establishing qualitative comparative statics in dynamic economies. Theorem 7.5 provides an explicit formula in the financial planning context that inherits their robustness properties when the regularity conditions are satisfied.

Theorem 7.6 (Temporal Coupling). Under time-varying priorities, optimal current actions depend on expected future priority paths:

$$a_t^* = a^*(x_t, \Lambda_t, \mathbb{E}[\Lambda_{t+1}], \mathbb{E}[\Lambda_{t+2}], \dots)$$

[Eq. 7.2]

Proof. The value function under stochastic priorities satisfies:

$$V(x, t; \Lambda_t) = \sup_a \{r(x, a; \Lambda_t)dt + e^{-\beta dt}\mathbb{E}[V(x', t + dt; \Lambda_{t+dt}) \mid \Lambda_t]\}$$

The expectation over future Λ creates dependence on the entire expected priority path. ∎

Corollary 7.6.1 (Joint Temporal Coupling). Under stochastic priorities and stochastic discount rates, optimal current actions depend on expected future paths of both:

$$a_t^* = a^*(x_t, \Lambda_t, \boldsymbol{\beta}(t), \mathbb{E}[\Lambda_{t+1}], \mathbb{E}[\boldsymbol{\beta}(t + 1)], \dots)$$

[Eq. 7.5]

The dependence on expected future rate paths creates **anticipatory rate management**: optimal actions today account for predictable rate changes (e.g., approaching retirement increases β_{Ret}) and probabilistic rate jumps (e.g., health event risk).

Proof. Extend the value function to the augmented state space:

$$V(x, t; \Lambda_t, \boldsymbol{\beta}(t)) = \sup_a \{r(x, a; \Lambda_t, \boldsymbol{\beta}(t))\, dt + \mathbb{E}[V(x', t + dt; \Lambda_{t+dt}, \boldsymbol{\beta}(t + dt)) \mid \Lambda_t, \boldsymbol{\beta}(t)]\}$$

The expectation over both future Λ and future $\boldsymbol{\beta}$ creates dependence on both expected paths. ∎

Theorem 7.7 (Priority Alignment). The welfare loss from using misspecified priorities $\tilde{\Lambda}$ instead of true priorities Λ is:

$$\text{Loss}(\tilde{\Lambda}, \Lambda) = V(x; \Lambda) - \hat{V}(x; \tilde{\Lambda}, \Lambda) \geq 0$$

[Eq. 7.3]

with equality if and only if $\tilde{\Lambda} = \Lambda$, where $\hat{V}(x; \tilde{\Lambda}, \Lambda)$ denotes the value achieved by using policy optimal for $\tilde{\Lambda}$ but evaluated under true preferences Λ.

Proof. By definition, $V(x; \Lambda) = \sup_\pi \mathbb{E}[\int e^{-\beta t} r(\cdot; \Lambda) dt]$. The policy $a^*(\cdot; \tilde{\Lambda})$ achieves some value when evaluated under Λ, but this cannot exceed the supremum. Equality holds only when $a^*(\cdot; \tilde{\Lambda}) = a^*(\cdot; \Lambda)$, which requires $\tilde{\Lambda} = \Lambda$. ∎

Corollary 7.7.1 (Joint Alignment Loss). The welfare loss from using misspecified priorities $\tilde{\Lambda}$ and misspecified rates $\tilde{\boldsymbol{\beta}}$ decomposes as:

$$\text{Loss}(\tilde{\Lambda}, \tilde{\boldsymbol{\beta}}; \Lambda, \boldsymbol{\beta}) \approx \text{Loss}_\Lambda + \text{Loss}_\beta + \text{Loss}_{\text{interaction}}$$

[Eq. 7.6]

where:

(i) $\text{Loss}_\Lambda = \frac{1}{2}(\tilde{w} - w)^\top \mathcal{H}_\Lambda (\tilde{w} - w)$: Loss from priority misalignment

(ii) $\text{Loss}_\beta = \frac{1}{2}(\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta})^\top \mathcal{H}_\beta (\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta})$: Loss from rate misalignment

(iii) $\text{Loss}_{\text{interaction}} = (\tilde{w} - w)^\top \mathcal{H}_{\Lambda\beta} (\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta})$: Loss from joint misalignment

The interaction term can be positive or negative, indicating that errors in priorities and rates may partially offset or compound.

Proof. Second-order Taylor expansion of the loss function around the true parameters $(\Lambda, \boldsymbol{\beta})$. The Hessian matrices $\mathcal{H}_\Lambda$, $\mathcal{H}_\beta$, and cross-Hessian $\mathcal{H}_{\Lambda\beta}$ characterize local curvature of the value function with respect to each parameter type. ∎

Remark 7.3 (Relative Importance of Alignment). Under typical calibrations, $\text{Loss}_\Lambda > \text{Loss}_\beta$: getting priorities wrong is costlier than getting rates wrong. However, Loss_β is non-negligible, particularly for clients in transitional life stages where rates are changing rapidly.

Theorem 7.8 (Value of Integration). *Let $V^*_{\text{int}}(d; \Lambda)$ denote the value from integrated optimization and $\sum_k w_k V^*_{k,\text{sep}}(d_k)$ denote the priority-weighted sum of domain-separated values. Then:*

$$V^*_{\text{int}}(d;\Lambda) \geq \sum_k w_k\, V^*_{k,\text{sep}}(d_k)$$

with equality if and only if the separability conditions of Theorem 7.4 hold. The integration premium satisfies:

$$V^*_{\text{int}} - \sum_k w_k\, V^*_{k,\text{sep}} \geq \frac{1}{2}\Delta\mathbf{d}^\top \mid \Gamma \mid \Delta\mathbf{d}$$

where $\Delta\mathbf{d}$ is the difference between integrated and separated optimal domain states and $\mid \Gamma \mid$ is the element-wise absolute value of the coupling matrix.

Proof.

Step 1 (Inequality). Integrated optimization maximizes over all feasible joint policies:

$$V^*_{\text{int}} = \max_{\pi} \mathbb{E}\left[\int_0^\infty e^{-\beta t} \sum_k w_k\, R_k(X_t, a_t)\, dt\right]$$

Separated optimization maximizes each domain independently:

$$V^*_{k,\text{sep}} = \max_{\pi_k} \mathbb{E}\left[\int_0^\infty e^{-\beta t}\, R_k\big(X_t, a_{k,t}\big)\, dt\right]$$

The integrated optimum can always replicate separated policies (by ignoring cross-domain information), so $V^*_{\text{int}} \geq \sum_k w_k\, V^*_{k,\text{sep}}$.

Step 2 (Equality condition). Equality holds if and only if the separated policies are jointly optimal, which occurs precisely when the separability conditions of Theorem 7.4 hold.

Step 3 (Integration premium bound). Let $\mathbf{d}^*_{\text{int}}$ and $\mathbf{d}^*_{\text{sep}}$ denote the optimal domain states under integrated and separated optimization. Taylor-expand the value function around $\mathbf{d}^*_{\text{sep}}$:

$$V^*_{\text{int}}(\mathbf{d}^*_{\text{int}}) - V^*_{\text{int}}\big(\mathbf{d}^*_{\text{sep}}\big) \approx \frac{1}{2}\Delta\mathbf{d}^\top D^2 V^*_{\text{int}} \Delta\mathbf{d}$$

where $\Delta\mathbf{d} = \mathbf{d}^*_{\text{int}} - \mathbf{d}^*_{\text{sep}}$. The cross-partial terms (off-diagonal elements of Γ) represent synergies captured by integration but missed by separation. The bound follows from $\mid V_{ij} \mid \leq \mid \Gamma_{ij} \mid$ and the quadratic form structure. ∎

Remark 7.4 (Relationship to Theorem 7.13). Theorem 7.8 establishes the integration premium under constant discount rates. Theorem 7.13 extends this to stochastic rates, decomposing the premium into four components:

$$\mathbb{E}[\Pi^{\text{int}}] = \underbrace{\Pi^{\text{det}}}_{\text{Theorem 7.8}} + \underbrace{\Pi^{\sigma} + \Pi^{J} + \Pi^{\Gamma}}_{\text{Stochastic extensions}}$$

The bound in Theorem 7.8 using $|\Gamma|$ generalizes to $|\Gamma_B|$ under stochastic rates, and the expectation accounts for rate variation:

$$V_{\text{int}}^* - \sum_k w_k V_{k,\text{sep}}^* \geq \frac{1}{2}\Delta\mathbf{d}^\top \mathbb{E}[|\Gamma_B|]\Delta\mathbf{d}$$

[Eq. 7.7]

where the expectation is over rate paths.

Theorem 7.9 (Shock Propagation). *A shock $\delta_i(0)$ in domain i propagates to domain j according to:*

$$\delta_j(t) = \sum_{m=1}^{n} [e^{M(\Lambda)t}]_{jm} \cdot (\mathcal{T}_{\text{eff}})_{mi} \cdot \delta_i(0)$$

where $M(\Lambda)$ is the linearized dynamics matrix under priority weights Λ and $\mathcal{T}_{\text{eff}}$ is the effective interdependency tensor.

Proof.

Step 1 (Linearization). Around equilibrium $\mathbf{d}^*$, the dynamics satisfy:

$$\delta\dot{\mathbf{d}}(t) = M(\Lambda)\,\delta\mathbf{d}(t)$$

where $M(\Lambda) = D_{\mathbf{d}}[f + \Gamma_{\text{eff}}(\Lambda)g + \mathcal{H}]\,|_{\mathbf{d}^*}$ incorporates priority-weighted coupling through the effective tensor.

Step 2 (Shock injection). A shock $\delta_i(0)$ in domain i creates initial perturbation:

$$\delta\mathbf{d}(0) = \mathcal{T}_{\text{eff}} \cdot \mathbf{e}_i \cdot \delta_i(0)$$

where $\mathbf{e}_i$ is the i-th standard basis vector. The tensor $\mathcal{T}_{\text{eff}}$ determines how the shock maps to initial perturbations across domains based on the coupling structure.

Step 3 (Propagation). The solution to the linearized system is:

$$\delta\mathbf{d}(t) = e^{M(\Lambda)t}\,\delta\mathbf{d}(0) = e^{M(\Lambda)t}\mathcal{T}_{\text{eff}}\mathbf{e}_i\delta_i(0)$$

The j-th component is:

$$\delta_j(t) = \sum_{m=1}^{n} [e^{M(\Lambda)t}]_{jm}(\mathcal{T}_{\text{eff}})_{mi}\delta_i(0)$$

as claimed. ∎

Corollary 7.9.1 (Rate Shock Propagation). A shock to domain k's discount rate, $\Delta\beta_k(0)$, propagates to domain values according to:

$$\Delta V_j(t) = \int_0^t [\, e^{M_B(s)(t-s)}]_{jk} \cdot \frac{\partial R_j}{\partial \beta_k} \cdot \Delta\beta_k(s)\, ds$$

[Eq. 7.8]

where $M_B(t)$ is the discount-adjusted dynamics matrix and the convolution captures the time-varying propagation under rate mean-reversion.

Proof. A rate shock $\Delta\beta_k(0)$ affects all value functions through the discount-adjusted coupling Γ_B. The propagation follows the linearized dynamics but with time-varying coefficients due to rate mean-reversion: $\Delta\beta_k(t) = e^{-\theta_k t}\Delta\beta_k(0)$. Integrating over the decay path yields the convolution formula. ▫

Corollary 7.9.2 (Systemic Event Propagation). A systemic life event $e$affecting domains $\mathcal{D}_e$ induces correlated rate jumps $\left\{\gamma_k^{(e)}\right\}_{k\in\mathcal{D}_e}$. The total propagation effect is:

$$\Delta V_j^{(e)}(t) = \sum_{k\in\mathcal{D}_e} \Delta\, V_j^{(k)}(t) + \text{Interaction}_j^{(e)}(t)$$

[Eq. 7.9]

where the interaction term captures non-linear effects from simultaneous rate changes across multiple domains.

Figure 8 illustrates how an effect in one domain influences the system across time. The example shows two specific focuses: a tax-focused client and a retirement-focused client. There is an investment shock at $t = 0$, where the shock has an effect for both tax- and retirement focused clients, and illustrates the propagation throughout the system over time. An additional risk management shock occurs at $t = 6$, and the effect is different in both magnitude and how it moves towards equilibrium.

Theorem 7.10 (Rate Volatility Premium). *Define the rate volatility premium as the value difference between stochastic and deterministic rate environments:*

$$\Pi_\sigma(x, \boldsymbol{\beta}) = V(x, \boldsymbol{\beta}; \sigma > 0) - V(x, \bar{\boldsymbol{\beta}}; \sigma = 0)$$

[Eq. 7.10]

Then:

(i) Sign: $\Pi_\sigma < 0$ generically. Rate uncertainty destroys value.

(ii) Monotonicity: $|\ \Pi_\sigma\ |$ is increasing in each σ_k^β.

(iii) Correlation effect: $|\ \Pi_\sigma\ |$ is increasing in cross-domain rate correlations.

(iv) Integration hedge: $\Pi_\sigma^{\text{int}} > \Pi_\sigma^{\text{sep}}$. Integrated planning loses less value from rate uncertainty.

Proof.

Step 1 (Sign). Rate uncertainty introduces variance in discount factors. By Jensen's inequality and risk aversion (concavity of value in wealth), uncertainty about the effective discount reduces expected value.

Step 2 (Monotonicity). Higher σ_k^β increases variance in domain k's discount factor, increasing the Jensen gap.

Step 3 (Correlation). Correlated rate movements prevent diversification across domains; when all rates move together, the portfolio effect is lost.

Step 4 (Integration hedge). Integrated planning can hedge rate uncertainty by shifting resources toward domains whose rates move countercyclically. Siloed planning cannot exploit these hedging opportunities. ∎

Corollary 7.10.1 (Hedge Value). The value of integration includes a hedge component: $\Pi_\sigma^{\text{int}} - \Pi_\sigma^{\text{sep}} > 0$ when cross-domain rate correlations are imperfect.

Theorem 7.11 (Jump Anticipation Value). Decompose rate jumps into predictable and unpredictable components: $\gamma_k(z) = \gamma_k^P(X_t, z) + \gamma_k^U(z)$. Define the jump anticipation value as:

$$\Pi_J = V^{\text{anticipate}}(x, \boldsymbol{\beta}) - V^{\text{myopic}}(x, \boldsymbol{\beta})$$

[Eq. 7.11]

Then:

(i) Sign: $\Pi_J \geq 0$. Anticipation never destroys value.

(ii) Predictability dependence: Π_J is increasing in the predictable fraction $\|\ \gamma^P\ \|/\|\ \gamma\ \|$.

(iii) Action flexibility: Π_J is increasing in the flexibility of the action space $A(x)$.

Proof.

Step 1 (Non-negativity). The anticipating agent has a larger information set than the myopic agent; value cannot decrease with information (Blackwell, 1953).

Step 2 (Predictability). The anticipating agent can only benefit from the predictable component; larger γ^P expands the scope for beneficial preparation.

Step 3 (Flexibility). The anticipating agent benefits from preparation only if preparatory actions are available; larger $A(x)$ expands options. ∎

Theorem 7.12 (Regime-Dependent Optimal Policy). Under the regime-switching formulation, the optimal policy depends on current state x, current regime r, transition probabilities $\{\lambda_{rs}(x)\}_{s\neq r}$, and rate differentials $\left\{\beta_k^{(s)} - \beta_k^{(r)}\right\}_{k,s}$. Explicitly:

$$a^*(x,r) = \arg\max_a \left\{ \sum_k w_k R_k(x,a) + \mathcal{L}_x^a V^{(r)}(x) + \sum_{s\neq r} \lambda_{rs}(x)[V^{(s)}(x) - V^{(r)}(x)] \right\}$$

[Eq. 7.12]

Proof. The optimal policy follows from the first-order conditions of the regime-switching HJB [Eq. 6.13]. The value differential $V^{(s)}(x) - V^{(r)}(x)$ captures the option value of regime transition, weighted by transition intensity $\lambda_{rs}(x)$. ∎

Corollary 7.12.1 (Same State, Different Actions). Generically, $a^*(x,r) \neq a^*(x,s)$for $r \neq s$. The same financial state warrants different actions in different life-stage regimes.

Corollary 7.12.2 (Regime Anticipation). Optimal policy in regime $r$depends on the value differentials to all accessible regimes s, not just the current regime's parameters.

Theorem 7.13 (Stochastic Integration Premium Decomposition). *The expected integration premium under stochastic rates decomposes as:*

$$\mathbb{E}[\Pi^{\text{int}}] = \Pi^{\text{det}} + \Pi^{\sigma} + \Pi^{J} + \Pi^{\Gamma}$$

[Eq. 7.13]

where:

(i) $\Pi^{\text{det}} = \frac{1}{2}\Delta\mathbf{d}^\top \mid \Gamma_{\bar{B}} \mid \Delta\mathbf{d}$: Deterministic base premium

(ii) $\Pi^{\sigma} = (\Pi_\sigma)^{\text{int}} - (\Pi_\sigma)^{\text{sep}} > 0$: Volatility hedge premium

(iii) $\Pi^{J} = (\Pi_J)^{\text{int}} - (\Pi_J)^{\text{sep}} \geq 0$: Jump coordination premium

(iv) $\Pi^{\Gamma} = \mathbb{E}[\Gamma_B] - \Gamma_{\bar{B}} \geq 0$: Coupling amplification premium

Proof.

Step 1 (Deterministic base). Setting all stochastic parameters to zero recovers the deterministic premium from Theorem 7.8, which depends on the domain deviation $\Delta\mathbf{d}$ and the discount-adjusted coupling at baseline rates.

Step 2 (Volatility hedge). By Theorem 7.10, integrated planning hedges rate volatility better than siloed planning; the differential is Π^{σ}.

Step 3 (Jump coordination). By Theorem 7.11, anticipating rate jumps is more valuable under integration; the differential is Π^{J}.

Step 4 (Coupling amplification). Jensen's inequality applied to the convex function $\beta \mapsto 1/\sqrt{\beta}$ implies $\mathbb{E}[\Gamma_B] \geq \Gamma_{\bar{B}}$ component-wise, creating additional coupling under stochasticity. ∎

Corollary 7.13.1 (Component Ordering). Under typical calibrations: $\Pi^{\text{det}} > \Pi^{\Gamma} > \Pi^{\sigma} > \Pi^{J}$. The deterministic base dominates, but stochastic components contribute 5–25% additional premium.

Corollary 7.13.2 (Amplification Factor Range). The total stochastic amplification satisfies $\mathbb{E}[\Pi^{\text{int}}]/\Pi^{\text{det}} = 1 + \phi$ where $\phi \in [0.05, 0.25]$ under calibrated parameters.

These theoretical results characterize the framework's properties abstractly. Section 8 demonstrates their practical implications through detailed applications featuring sample cases.

8 | Applications: The Value of Integration and the Mechanics of Personalization

The mathematical framework developed in Sections 3 through 7 achieves practical significance only when applied to real planning decisions. This section demonstrates four distinct ways the temporal priority-filtered approach reveals insights unavailable through conventional analysis. First, Maria Delgado illustrates how siloed advice systematically undermines a client's highest-weighted priorities and how the integration premium quantifies this failure. Second, the Okonkwo twins show how identical Structural Tensors yield different optimal strategies when filtered through different Priority Matrices. Third, Victoria Chen and Robert Hamill demonstrate the converse: identical priorities produce different premiums when structural complexity varies. Fourth, Marcus Jones reveals how life shocks propagate through the discount rate structure, dramatically amplifying integration value.

Throughout this section, summary results appear inline while complete computational details, such as domain-specific discount rates, stochastic premium decompositions, and sensitivity analyses are documented in Appendix F.

8.1 | The Value of Integration: Maria Delgado's Coordinated Strategy

Maria Delgado, fifty-three, built Delgado Construction Services from a weekend house-flipping operation into a full-service construction company handling plumbing, electrical, HVAC, and

both commercial and residential projects across Colorado's Front Range. Despite eight-figure wealth, she remains anxious about financial security: "I built this business from nothing. I need to know I'll be okay."

Maria is divorced with two adult children. Her daughter Sofia, thirty-one, serves as the company's top salesperson and wants to eventually run the business. Her son Daniel, twenty-eight, is a cardiology fellow with no interest in construction, though Maria wants him to benefit from her wealth.

Financial Position

With net worth of $19 million exceeding the current federal estate tax exemption of $13.61 million, Maria faces potential estate tax liability on approximately $5.4 million, representing roughly $2.2 million in federal estate taxes at the 40% marginal rate.

The Crisis

Maria planned to retire at sixty-two, transitioning to an advisory role while Sofia assumed operational control. Then the housing market contracted. Within eight months, her publicly traded securities declined 30%, dropping from $4 million to $2.8 million. More troubling, Delgado Construction's enterprise value declined from $10 million to an estimated $7.5 million.

Maria now faces sequence-of-returns risk at precisely the moment she intended to begin drawing on her portfolio, with liquid assets contracted by over $1.2 million and business value down 25%.

Maria's Priority Matrix

Maria's priorities reflect a pattern common among first-generation wealth creators who remember scarcity.

Priority Matrix Λ: w_{CF} = 0.35, w_{Ret} = 0.25, w_{RM} = 0.18, w_{Inv} = 0.12, w_{Tax} = 0.06, w_{Est} = 0.04.

Her cash flow weight (w_{CF} = 0.35) dominates because Maria's deepest fear is running out of money. Retirement (w_{Ret} = 0.25) reflects her desire for security after decades of sixteen-hour workdays. Estate planning (w_{Est} = 0.04) ranks lowest as her children's inheritance matters less than her own security.

The Siloed Advice

Maria consulted five specialists, each expert in their domain:

The Estate Planning Attorney recommended transferring Delgado Construction into a Family Limited Partnership (FLP) with 35% valuation discounts, then gifting 80% of partnership interests to Sofia and Daniel through a GRAT structure. Combined with a $3 million ILIT, estimated estate tax savings: $1.8 million.

The Retirement Specialist showed that delaying Social Security from 62 to 70 would increase Maria's annual benefit from $32,400 to $57,000, representing $180,000 in additional lifetime value.

The Investment Advisor recommended selling remaining bond positions to purchase equities at depressed prices, projecting $400,000 in recovered value within three years.

The Insurance Agent proposed a $5 million second-to-die life insurance policy within the ILIT structure. Annual premiums: $62,000.

Each recommendation was technically sound. Total estimated value from siloed advice: approximately $2.6 million.

The Integration Failure

When mapped against Maria's Priority Matrix, these recommendations create systematic mismatch between advice and values. The Structural Tensor S reveals interconnections the specialists missed.

The Cash Flow Catastrophe. The estate attorney's GRAT requires annuity payments that flow back to Maria's taxable estate if she dies during the trust term. The FLP transfers 80% of business ownership, eliminating $640,000 in annual distributions Maria currently receives. The structural coupling $S_{Est,CF}$ is strongly negative here: estate optimization directly undermines cash flow, which compromises retirement security. For a client with w_{CF} = 0.35 and w_{Est} = 0.04, this recommendation optimizes the wrong objective.

The Social Security Paradox. The retirement specialist optimized for Ret alone, but Maria's portfolio decline changes the calculus. Every year she delays claiming requires portfolio liquidation for living expenses at depressed prices. The coupling $S_{Ret,Inv}$ is relevant: retirement optimization through delayed claiming requires investment liquidation that reduces cash flow and impairs portfolio recovery.

The Integrated Solution

An integrated analysis, filtering the Structural Tensor through Maria's Priority Matrix to compute T_{eff}, reveals a coordinated strategy aligned with her actual priorities:

Social Security: Claim at 62. Despite the actuarial penalty, early claiming provides $32,400 annually in inflation-adjusted income without portfolio liquidation. The "cost" is $180,000 in foregone lifetime benefits; the benefit is preserving $200,000+ in portfolio assets during recovery.

Business Succession: Phased Transition. Rather than immediate 80% transfer, Maria retains 60% ownership for five years, receiving $480,000 annually in distributions. This structure captures approximately $900,000 in estate tax benefits while preserving $855,000 in cumulative cash flow during critical transition years.

Insurance: Right-Sized Coverage. A $2 million policy provides adequate liquidity at $24,000 annually, saving $38,000 per year versus the recommended $5 million policy.

Integration Premium

Table 8.1 summarizes Maria's integration premium. Complete discount rate dynamics and stochastic decomposition appear in Appendix F, Tables F.2–F.3.

Table 8.1: Maria Delgado's Integration Premium Summary

Net Worth	Π^{det}	Π^{total}	% of NW
$19,000,000	$2,200,000	$2,513,000	13.2%

The stochastic analysis reveals that Maria's integration premium is amplified by 14.2% under realistic rate dynamics. The jump coordination component is substantial because the market crisis triggered correlated rate jumps in Cash Flow, Risk Management, and Investment domains simultaneously, precisely the scenario where coordinated response creates the most value.

The integration premium of $2.2 million deterministic (11.6% of net worth) and $2.5 million total (13.2%) exceeds the proportional premium observed in simpler cases. This reflects the optionality inherent in Maria's situation: business succession timing, Social Security claiming, and portfolio allocation all interact through T_{eff} in ways that siloed analysis cannot capture.

8.2 | Same Structure, Different Priorities: The Okonkwo Twins

David and Daniel Okonkwo are fifty-two-year-old twin orthopedic surgeons in Chicago with nearly identical finances: $680,000 annual income, $4.2 million in investments, $1.8 million homes with $400,000 mortgages, three children each. Net worth: $6 million each.

Their Structural Tensor S is identical: same tax brackets, same actuarial realities, same market dynamics. Yet their Priority Matrices diverge dramatically due to different life experiences.

David: Security-Focused

The twins' father died of a heart attack at fifty-eight. David internalizes this: "What I think about is whether I'll live long enough to see my daughter graduate."

Priority Matrix Λ: w_{RM} = 0.30, w_{Ret} = 0.25, w_{CF} = 0.20, w_{Est} = 0.12, w_{Inv} = 0.08, w_{Tax} = 0.05.

Daniel: Growth-Focused

Daniel's children are financially independent, his wife earns $280,000, and their father remains healthy at eighty-one. "I've been conservative my whole life. I'm starting to wonder if I've failed to take advantage of opportunities."

Priority Matrix Λ: w_{Inv} = 0.28, w_{Tax} = 0.22, w_{Ret} = 0.20, w_{CF} = 0.15, w_{Est} = 0.10, w_{RM} = 0.05.

Different Effective Tensors, Different Strategies

The Effective Tensors differ dramatically: $T_{eff}^{David} \neq T_{eff}^{Daniel}$ despite identical S.

Consider disability insurance. Both carry policies costing $18,000 annually. The structural coupling $S_{RM,CF}$ is identical and strongly positive for both. But when filtered through their Priority Matrices:

For David (w_{RM} = 0.30), maintaining comprehensive disability coverage is unambiguously correct. The high priority weight amplifies the structural coupling, making the RM-CF coordination essential.

For Daniel (w_{RM} = 0.05), the $18,000 premium redirected to tax-advantaged investments over a decade would generate approximately $250,000 in additional wealth, which is a better use given his priorities and his wife's income as a buffer.

Table 8.2: Okonkwo Twins Integration Premium Comparison

Twin	Π^{det}	% NW	Π^{total}	Stoch Amp
David (Security)	$470,000	7.8%	$526,400	+12.0%
Daniel (Growth)	$580,000	9.7%	$667,000	+15.0%

The twins demonstrate that identical structural positions can yield different stochastic benefits. Daniel's higher stochastic enhancement (15.0% vs 12.0%) reflects his growth-oriented priorities creating greater exposure to Investment-Tax coupling, which exhibits higher rate volatility. Complete rate dynamics appear in Appendix F, Tables F.4–F.5.

8.3 | Different Structure, Same Priorities: Victoria Chen and Robert Hamill

Victoria Chen and Robert Hamill share remarkably similar priority structures despite vastly different wealth levels. Both are sixty-one, both shaped by career paths that instilled similar values around security and growth balance.

Victoria is a CTO earning $1.2 million annually, holding $8 million in company stock (60% of liquid net worth), $4 million in diversified investments, and $1.2 million in retirement accounts. Net worth: approximately $20 million.

Robert is a retired airline pilot with a $96,000 pension, $1.4 million in retirement accounts, and $400,000 in taxable investments. Net worth: approximately $2.3 million.

Both share identical Priority Matrices:

Priority Matrix Λ: w_{CF} = 0.28, w_{Ret} = 0.24, w_{RM} = 0.18, w_{Inv} = 0.14, w_{Tax} = 0.10, w_{Est} = 0.06.

Yet their Structural Tensors differ fundamentally. Victoria's concentrated stock creates structural interdependencies Robert's diversified portfolio lacks. The coupling $S_{Inv,Tax}$ is strongly negative for Victoria (selling triggers substantial capital gains) but near-zero for Robert (diversified holdings allow tax-loss harvesting flexibility).

For Victoria, optimal strategy involves hedging concentration risk through options or variable prepaid forwards while preserving estate efficiency. For Robert, optimal strategy emphasizes Roth conversions and pension optimization which are structural features irrelevant to Victoria.

Table 8.3: Chen vs. Hamill Integration Premium Comparison

Client	Net Worth	Π^{det}	% NW	Stoch Amp
Victoria Chen	$20.0M	$1,180,000	5.9%	+16.0%
Robert Hamill	$2.3M	$161,000	7.0%	+10.0%

Victoria's higher stochastic enhancement (16.0% vs 10.0%) reflects her concentrated stock position creating strong negative coupling that amplifies under volatility. Robert's simpler structure yields a higher percentage premium (7.0% vs 5.9%) despite lower absolute value,

illustrating that integration value does not scale linearly with wealth but with structural complexity and priority mismatch. Complete analysis appears in Appendix F, Tables F.6–F.8.

8.4 | Shock Dynamics: Marcus Jones's Temporal Priority Shifts

Marcus Jones, thirty-four, left emergency medicine for healthcare technology. He earns $185,000 plus equity compensation: 15,000 ISOs (strike $12, FMV $28), 8,000 NSOs, and 5,000 RSUs vesting over four years. Net worth: approximately $800,000.

"I traded guaranteed income for optionality. I need to figure out how to manage that optionality."

Priority Matrix Λ: w_{CF} = 0.30, w_{Inv} = 0.22, w_{Ret} = 0.18, w_{Tax} = 0.15, w_{RM} = 0.10, w_{Est} = 0.05.

Marcus's case demonstrates how life shocks propagate through the Discount Matrix $B(t)$, dramatically transforming the integration premium landscape.

The Health Diagnosis

Marcus receives an unexpected health diagnosis requiring treatment. This triggers correlated rate jumps across multiple domains:

Cash Flow (β_{CF}): 6.0% → 10.5% (+4.5%). Medical expenses and potential income disruption collapse the planning horizon from 11.6 months to 6.6 months half-life.

Estate (β_{Est}): 1.5% → 4.7% (+3.2%). Terminal considerations suddenly become proximate. Half-life drops from 46.2 months to 14.7 months.

Risk Management (β_{RM}): 7.0% → 12.8% (+5.8%). Insurance and protection become immediate concerns. Half-life collapses from 9.9 months to 5.4 months.

Table 8.4: Marcus Jones's Integration Premium with Shock

State	Π^{det}	Π^{total}	Stochastic Amp
Pre-Shock	$60,000	$68,400	+14.0%
Post-Shock	$72,000	$91,800	+27.5%
Change	+20%	+34%	+13.5 pp

The health diagnosis triggered a systemic event affecting CF, Est, and RM domains simultaneously. The integration premium increased by 34% because coordinated response across correlated rate jumps is substantially more valuable than isolated domain responses. The stochastic amplification nearly doubled from 14% to 28%, because the shock topology created precisely the conditions where integration captures maximum value.

This validates Theorem 7.11 (Jump Anticipation Value): when multiple domains experience simultaneous urgency changes, the coordination premium amplifies super-linearly. Complete shock propagation analysis appears in Appendix F, Tables F.9–F.11.

8.5 | Synthesis: What the Personas Reveal

The six personas demonstrate distinct mechanisms through which integration creates value:

Table 8.5: Integration Premium Summary Across All Personas

Persona	Net Worth	Π^{det}	% NW	Π^{total}	% NW
Maria Delgado	$19.0M	$2.20M	11.6%	$2.51M	13.2%
David Okonkwo	$6.0M	$470K	7.8%	$526K	8.8%
Daniel Okonkwo	$6.0M	$580K	9.7%	$667K	11.1%
Victoria Chen	$20.0M	$1.18M	5.9%	$1.37M	6.8%
Robert Hamill	$2.3M	$161K	7.0%	$177K	7.7%
Marcus Jones*	$0.8M	$72K	9.0%	$92K	11.5%

**Post-shock values*

Five key insights emerge:

First, integration creates measurable value. Across all personas, the deterministic premium ranges from 5.9% to 11.6% of net worth, substantially exceeding typical advisory fees and demonstrating that coordination itself generates economic value.

Second, stochastic components amplify premiums by 10–28%. Domain-specific discount rates and correlated shocks create additional coordination value that deterministic analysis misses. The greatest amplification occurs when life events trigger simultaneous rate jumps.

Third, priority heterogeneity drives premium magnitude. Maria's 11.6% premium was highest among the deterministic results, reflecting strong weights on Cash Flow, a domain that siloed advice systematically underweights when optimizing for other objectives.

Fourth, structural complexity creates opportunity. Victoria's concentrated stock generates larger absolute premiums than Robert's diversified portfolio, despite identical priorities. Complex structures create more coupling paths that integration can exploit.

Fifth, shock topology matters. Marcus's correlated rate jump affecting CF, Est, and RM simultaneously increased his premium by 34%. The integration premium responds to the correlation structure of shocks, not just their magnitude.

These applications validate the theoretical framework: the four-tier architecture (S, Λ, $B(t)$, Γ_B) translates abstract mathematics into concrete guidance for financial planners navigating the irreducible complexity of client lives.

9 | Discussion

9.1 | Theoretical Contributions

IIFPT demonstrates four principal theoretical contributions. First, the fiber bundle representation provides rigorous foundations for discussing integration and interdependency. Second, the four-tier decomposition separates structural mechanics from client values and temporal priorities. Third, the stochastic discount rate specification captures how temporal priorities evolve

dynamically, responding to both observable financial state and life events, yielding new phenomena including rate volatility premium, jump anticipation value, and stochastic integration premium amplification of 10-28% after simulation. Fourth, the thirteen theorems provide formal foundations for intuitions practitioners have developed through experience and crucially, provide a roadmap for future research in the field.

9.2 | The Necessity of Human Planners

The implication of IIFPT is that the high dimensionality of integrated financial planning provides rigorous theoretical justification for the financial planning profession itself.

The complete state space includes easily fifteen or more state variables. The curse of dimensionality is not a flaw to be overcome through better algorithms. It is evidence that brute-force optimization cannot solve realistic planning problems in reasonable time.

This is where human financial planners become essential. The planner acts as a heuristic dimensionality reduction operator. Through judgment informed by experience, education, and interaction with a particular client, the planner identifies the active sub-manifold relevant to this client at this time, which are the three or four dimensions that matter given the situation and priorities.

In addition, the relationship between the client and planner (Asebedo, 2019) provides crucial information for computing and adjusting stochastic discount rates. Even if utilizing a regime-switching alternative (Section 6.4), calibration of discount rates within a discrete rate space and standard PDEs necessitates a thorough conversation and understanding of the client, which requires more elicitation.

IIFPT articulates why financial planners provide significant value. This theoretical conclusion aligns with the empirical evidence assembled by Hanna and Lindamood (2010) showing substantial economic benefits from professional planning, and with Warschauer's (2002) documentation of the profession's development as a distinct discipline.

9.3 | Practical Implications

The Structural Tensor S can be pre-computed and updated when regulations change and there are changes structurally to a client's life, while the Priority Matrix Λ must be elicited from clients. This separation dramatically reduces computational burden and allows a financial planner to analyze structures as things change, personalize via Λ repeatedly. The Discount Matrix $\mathbf{B}$ can be computed in a variety of ways, but stochastic, integro-differential are more accurate to real life.

Financial planners can provide value by developing the client relationship to understand potential structural changes, shifting priorities, and urgency.

9.4 | Priority Elicitation in Practice

The Priority Matrix must be inferred from client interactions. Asebedo (2019) posited there are value gains from the interaction between a financial planner and a client. This interaction provides a general recognition that can be applied to eliciting client priorities. There are many methods for deriving client priorities. I briefly discuss three:

Method 1: Direct Ranking. Ask the client to rank the six domains by importance. This yields ordinal but not cardinal information.

Method 2: Pairwise Comparison (AHP). Using the Analytic Hierarchy Process (Saaty, 1987), ask clients to compare domains pairwise on a 1-9 scale. The principal eigenvector of the comparison matrix yields weights.

Method 3: Revealed Preference. Infer priorities from hypothetical scenarios or actual past decisions. "Would you prefer Strategy A (better tax outcome) or Strategy B (better liquidity)?" reveals relative weights.

9.5 | Temporal Structure Monitoring

The stochastic rate framework implies new responsibilities for financial planners: monitoring clients' temporal priority structures and detecting rate changes that warrant strategy revision.

Rate Shift Indicators:

Life Event	Affected Rates	Direction
Liquidity stress	β_{CF}	↑
Health diagnosis	β_{CF}, β_{Est}, β_{RM}	↑, ↑, ↑
Job loss	β_{CF}, β_{Ret}	↑, ↑
Inheritance receipt	β_{CF}, β_{Est}	↓, ↓
Approaching retirement	β_{Ret}, β_{CF}	↑, ↑
Market volatility	β_{Inv}, β_{RM}	↑, ↑

Rate Elicitation Methods: Time-preference scenarios, urgency language analysis, attention allocation tracking, behavioral indicators.

Regime Detection: Recognizing gradual transitions (pre-retirement → early retirement spans 2–5 years), sudden transitions (health crisis is abrupt), reversible transitions (market crisis resolves), and irreversible transitions (spousal death is permanent).

9.6 | Limitations and Research Directions

The IIFPT framework, while providing rigorous theoretical foundations, confronts several practical challenges that define a productive research agenda.

Computational Tractability. Despite the structural-priority separation, solving the priority-weighted HJB equation remains computationally demanding for high-dimensional state spaces. This limitation motivates investigation of machine learning approaches: Irlam (2020) demonstrates that neural network methods can effectively address retirement planning optimization, and extending these techniques to the multi-domain IIFPT setting represents a natural next step. Specifically, deep reinforcement learning architectures that exploit the four-tier decomposition, training separate networks for structural coupling estimation and priority-

weighted synthesis, may achieve tractable approximations while preserving theoretical guarantees.

Priority Elicitation Heterogeneity. Priority elicitation remains nontrivial and may vary across financial planners, introducing measurement error into the Priority Matrix Λ. Validation studies comparing alternative elicitation methods, such as conjoint analysis, direct allocation tasks, and revealed preference from historical planning decisions, would establish which approaches yield priorities most predictive of client satisfaction and plan adherence. Longitudinal studies tracking priority evolution could further inform whether single-point elicitation suffices or whether adaptive updating mechanisms are required.

Rate Calibration Challenges. The stochastic discount rate decomposition $\beta_k(t) = \overline{\beta_k} + f_k(X_t) + \xi_k(t)$ introduces empirical challenges across three dimensions:

Parameter identification: The decomposition has many parameters, creating potential identification problems with standard estimation approaches. Single Case Research (SCR) methods, which compare individual clients against their own optimal benchmarks rather than population averages, offer a promising identification strategy (Kothakota & Wery, 2024). By tracking within-client variation in revealed temporal urgency across planning domains, SCR can isolate endogenous adjustment functions f_k while controlling for unobserved client heterogeneity.

Non-stationarity: Baseline rates $\overline{\beta_k}$ may evolve over multi-decade planning horizons as clients age and circumstances change. Longitudinal panel data following clients through lifecycle transitions would enable estimation of age-dependent baseline functions, transforming this limitation into an empirical question about the functional form of $\overline{\beta_k}$ (age, wealth, health).

Correlation estimation: Systemic event correlations require data on rare life events, including major health diagnoses, market crises, and family transitions, which occur infrequently by definition. Collaborative data-sharing arrangements across planning firms, combined with retrospective analysis of documented planning engagements, could accumulate the event histories necessary for robust correlation estimation while preserving client confidentiality.

Structural Tensor Empirics. The Structural Tensor **S** encodes objective mechanical linkages from tax law, actuarial mathematics, and portfolio theory. While these linkages are in principle computable, empirical estimation of cross-domain coupling strengths from observed planning outcomes would validate theoretical specifications and identify where structural simplifications, such as treating weakly coupled domains as independent, preserve optimality. The development of financial literacy research priorities identified by Bogan et al. (2020) could be advanced by studying how structural interdependency awareness affects planning outcomes.

Software Integration. Translating the four-tier architecture into practitioner software requires balancing computational feasibility with theoretical fidelity. Efficient computational methods, including the regime-switching formulation developed in Section 6.4, offer implementable approximations, while the separability conditions of Theorem 7.4 identify when simpler approaches suffice without sacrificing optimality.

10 | Conclusion

IIFPT separates structural interdependencies from client-specific priorities through a four-tier architecture that distinguishes *what* domains are coupled, *how much* clients care about each

domain, *how urgently* they discount future outcomes in each domain, and *how* these factors combine to generate effective coupling. The framework comprises: Structural Tensor $\mathbf{S}$ (Tier 1) capturing domain interdependencies independent of any client; Priority Matrix Λ (Tier 2a) reflecting client-specific weightings across domains; Discount Matrix $\boldsymbol{B}(t)$ (Tier 2b) encoding domain-specific discount rates β_k that reflect different planning horizons, from Cash Flow's immediate concerns to Estate's terminal orientation; and the discount-adjusted Effective Tensor $\Gamma_B = B^{-1/2}\, T_{eff}\, B^{-1/2}$ that filters structural couplings through both priorities and temporal urgency. This architecture provides foundations for understanding why clients with identical finances may optimally pursue different strategies, and why the same client may optimally pursue different strategies as life circumstances shift their temporal discount rates.

Five key observations result. First, the complexity of financial planning is irreducible since the high dimensionality and non-smooth structure are features of the problem itself. Second, structure, values, and temporal urgency must be separated; the four-tier decomposition provides this separation, recognizing that priorities (what matters) and discount rates (when it matters) operate through distinct channels. Third, human judgment remains necessary. The complexity of a single planning engagement demonstrates that comprehensive optimization is computationally infeasible, while human planners serve as heuristic dimensionality reducers and bulwarks against the uncertainty that stochastic discount rates introduce. Fourth, domain-specific discount rates create differential shock propagation: high-β domains respond quickly while low-β domains exhibit persistent effects, generating coordination value that siloed planning cannot capture. Fifth, the stochastic components Π^{σ}, Π^{J}, and Π^{Γ} amplify integration premiums by 10–28%, demonstrating that realistic rate dynamics substantially increase the value of coordinated planning.

Through the stories of Maria Delgado, the Okonkwo twins, Victoria Chen and Robert Hamill and Marcus Jones, the IIFPT illustrates how abstract mathematics describes concrete guidance. Financial planners translate the structures of the financial life, filtered through client priorities and domain-specific temporal urgency, into decisions about retirement security, family obligation, legacy, and peace of mind.

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Appendices

Appendix A | Extended Proofs

This appendix provides extended derivations for the main theorems, including technical lemmas and verification of regularity conditions.

A.1 | Supporting Lemmas

Lemma A.1 (Convex Hull Preserves Symmetry). If $\{M_n\}$ is a sequence of symmetric matrices and $M = \lim_{n\to\infty} M_n$ exists, then M is symmetric. Moreover, the convex hull of symmetric matrices is symmetric.

Proof. For the limit: $(M)_{ij} = \lim_n (M_n)_{ij} = \lim_n (M_n)_{ji} = (M)_{ji}$.

For the convex hull: Let $M = \sum_k \lambda_k M_k$ where each M_k is symmetric and $\lambda_k \geq 0, \sum_k \lambda_k = 1$. Then $(M)_{ij} = \sum_k \lambda_k (M_k)_{ij} = \sum_k \lambda_k (M_k)_{ji} = (M)_{ji}$. ∎

Lemma A.2 (Convex Hull Preserves Negative Semidefiniteness). The convex hull of negative semidefinite matrices is negative semidefinite.

Proof. Let $M = \sum_k \lambda_k M_k$ where each $M_k \preccurlyeq 0$ and $\lambda_k \geq 0, \sum_k \lambda_k = 1$.

For any vector v:

$$v^\top M v = v^\top \left(\sum_k \lambda_k M_k \right) v = \sum_k \lambda_k (v^\top M_k v) \leq \sum_k \lambda_k \cdot 0 = 0$$

since each $v^\top M_k v \leq 0$ by negative semidefiniteness of M_k. ∎

Lemma A.3 (Clarke Gradient Bound). For locally Lipschitz f with Lipschitz constant L on domain D, every element $\xi \in \partial f(x)$for $x \in D$ satisfies $\| \xi \| \leq L$.

Proof. By definition, $\partial f(x) = conv\{\lim_i \nabla f(x_i)\}$ where $x_i \to x$ and $\nabla f(x_i)$ exists.

At points of differentiability, $\| \nabla f(x_i) \| \leq L$ (since Lipschitz constant bounds gradient norm).

Limits of bounded sequences are bounded: $\| \lim_i \nabla f(x_i) \| \leq L$.

Convex combinations of vectors with norm $\leq L$ have norm $\leq L$:

$$\| \sum_k \lambda_k v_k \| \leq \sum_k \lambda_k \| v_k \| \leq \sum_k \lambda_k L = L$$

Therefore $\| \xi \| \leq L$ for all $\xi \in \partial f(x)$. ∎

A.2 | Extended Proof of Theorem 7.5

I apply the implicit function theorem to the first-order conditions $0 = \nabla_a Q(s, a^*; \Lambda)$. Differentiating with respect to w_k:

$$0 = \frac{\partial^2 Q}{\partial a\, \partial w_k} + \left(\frac{\partial^2 Q}{\partial a^2}\right)\left(\frac{\partial a^*}{\partial w_k}\right)$$

Since $\partial Q / \partial w_k = R_k$, I have $\partial^2 Q / \partial a\, \partial w_k = \partial R_k / \partial a$. Solving:

$$\frac{\partial a^*}{\partial w_k} = -H^{-1} \cdot \left(\frac{\partial R_k}{\partial a}\right)$$

where $H = \nabla^2_{aa} Q$ is the Hessian of the Q-function.

Interpretation: The sensitivity of optimal actions to priority changes is proportional to the gradient of the domain objective (how much the action affects that domain's outcome), scaled by the inverse Hessian (how curved the optimization landscape is).

A.3 | Verification of Regularity Conditions

The main theorems assume regularity conditions on the problem primitives.

Condition (R1): Local Lipschitz Continuity. Each domain objective $R_k(s, a)$ must be locally Lipschitz in (s, a). Tax functions are piecewise linear in income, hence globally Lipschitz on any bounded domain.

Condition (R2): Boundedness. The objectives R_k must be bounded above on any compact state-action region. All objectives are normalized or utility-transformed quantities, bounded by construction.

Condition (R3): Lipschitz Transition Coefficients. The drift $b(s, a)$ and diffusion $\sigma(s, a)$ must be Lipschitz in s. The wealth dynamics have linear drift in W (Lipschitz).

Condition (R4): Compact Action Sets. The feasible action set $A(s)$ must be compact for each s. Action sets in financial planning are typically defined by bounds.

Appendix B | Empirical Calibration Considerations

This appendix discusses methods for empirically estimating the components of the Structural Tensor S.

B.1 | Data Sources

Tax Domain Coupling: IRS Statistics of Income provides aggregate data on how various income sources interact with tax liability. Tax simulation models can directly compute $\partial^2 \tau / \partial y_i\, \partial y_j$.

Retirement Domain Coupling: Social Security Administration data, plan sponsor data, and the Health and Retirement Study (HRS) provide relevant information.

Risk Management Coupling: Insurance claims data and actuarial tables from the Society of Actuaries.

Investment Coupling: Market return data, correlation matrices, and portfolio holdings data.

B.2 | Estimation Methodology

For tax-related coupling, compute derivatives directly from the tax code. For coupling that cannot be computed directly, use regression-based estimation or structural model estimation.

B.3 | Recommended Approach

Compute analytically where possible (tax interactions, actuarial coupling). Estimate empirically for less-structured coupling. Use expert judgment for novel regulatory interactions. Validate by checking whether calibrated S generates realistic effective coupling patterns.

Appendix C | Sub-Domain Element Taxonomy

The following table provides a non-exhaustive list of financial planning sub-domains with related couplings.

Table C.1: Representative Sub-Domain Elements

Fiber Bundle Domain	Sub-Domain Elements	Description & Linkages
Cash Flow (F_{CF})	• Emergency Fund Allocation	Liquid reserves for unexpected expenses; affects insurance deductibles and investment risk capacity [→RM, Inv]
	• Debt Service Management	Principal and interest payments; competes with retirement contributions [→Ret, Inv]
	• Income Timing Decisions	Acceleration or deferral of earned income; affects marginal tax brackets [→Tax]
	• Discretionary Spending Rate	Non-essential consumption proportion; constrains savings [→Ret, Inv, Est]
	• Cash Flow Forecasting	Projection of inflows and outflows; provides constraints for all domains [→All]
Retirement (F_{Ret})	• Social Security Claiming Age	Benefit election between ages 62-70; affects lifetime income and taxation [→Tax, Est, CF]
	• Withdrawal Sequencing	Order of distributions across account types; determines tax liability [→Tax, Inv]
	• Contribution Allocation	Distribution between 401(k), IRA, after-tax; establishes tax diversification [→Tax, Est]
	• Annuitization Decisions	Conversion to guaranteed income; trades longevity protection against legacy [→RM, Est, CF]
	• Retirement Date Selection	Targeted exit age; determines accumulation horizon [→CF, Inv]
Tax (F_{Tax})	• Roth Conversion Strategy	Annual conversion amount; reduces future RMDs [→Ret, Est, CF]
	• Tax-Loss Harvesting	Realization of losses to offset gains [→Inv]
	• Charitable Giving Timing	Strategic bunching including DAF funding [→Est, CF]

Fiber Bundle Domain	Sub-Domain Elements	Description & Linkages
	• Asset Location Optimization	Placement by tax characteristics [→Inv, Ret]
	• Income Bracket Management	AGI control for bracket optimization [→CF, Ret]
Estate (F_{Est})	• Donor-Advised Fund Contributions	Irrevocable charitable gifts [→Tax, CF, Inv]
	• Beneficiary Designation Strategy	Named recipients on accounts [→Ret, Tax, RM]
	• Trust Structure Selection	Choice among trust types [→Tax, RM]
	• Gifting Program Design	Annual and lifetime gift allocations [→Tax, CF]
	• Business Succession Planning	Transfer strategy for business interests [→Tax, Inv, CF]
Risk Management (F_{RM})	• Life Insurance Coverage	Death benefit amount and policy type [→Est, CF, Inv]
	• Disability Income Protection	Replacement ratio and benefit period [→CF, Ret]
	• Long-Term Care Strategy	Insurance, self-funding, or hybrid [→Ret, Est, Tax]
	• Liability Coverage Levels	Umbrella and professional liability [→Inv, Est]
	• Health Insurance Selection	Plan type, deductible, HSA decisions [→CF, Tax, Ret]
Investment (F_{Inv})	• Asset Allocation Policy	Target weights across asset classes [→Ret, RM]
	• Rebalancing Protocol	Return to targets methodology [→Tax, CF]
	• Concentrated Stock Management	Diversification of low-basis shares [→Tax, Est, RM]
	• Factor Exposure Tilts	Size, value, momentum weightings [→Ret]
	• Alternative Investment Allocation	Private equity, real estate, hedge funds [→CF, Tax]

Coupling Intensity Classification:

- **Strong** ($S > 0.5$): Tax ↔Retirement, Tax ↔Investment, Retirement ↔Risk Management
- **Moderate** ($0.2 < S \leq 0.5$): Estate ↔Tax, Cash Flow ↔Retirement
- **Weak** ($S \leq 0.2$): Investment ↔Risk Management, Estate ↔Cash Flow

Appendix D | Computational Implementation

This appendix discusses numerical methods for implementing the priority-filtered planning framework.

D.1 | State Space Discretization

For finite-horizon problems, discretize time $t \in \{0, \Delta t, 2\Delta t, \dots, T\}$, wealth $W \in \{W_1, \dots, W_N\}$ on a grid, and other state variables similarly. Dimensionality limits grid-based methods to $d \leq 4$ state variables.

For wealth, use a log-spaced grid:

$$W_i = W_{\min} \cdot \left(\frac{W_{\max}}{W_{\min}}\right)^{(i-1)/(N-1)}$$

D.2 | Value Function Iteration

For the discretized HJB:

$$V_n(x) = \max_{a \in A(x)} \{r(x, a; \Lambda)\Delta t + e^{-\beta \Delta t}\mathbb{E}[V_{n+1}(X')]\}$$

Iterate backward from terminal condition $V_T(x) = \Phi(x; \Lambda)$.

D.3 | Function Approximation

For high-dimensional problems, approximate $V(x; \Lambda)$ using polynomial basis:

$$\hat{V}(x; \theta) = \sum_{k=1}^{K} \theta_k \, \phi_k(x)$$

where ϕ_k are polynomial basis functions (e.g., Chebyshev polynomials), or neural networks trained by minimizing Bellman residual. Irlam (2020) demonstrates that deep reinforcement learning methods can effectively solve retirement planning problems; extending these techniques to the multi-domain setting represents a promising computational approach.

D.4 | Priority Sensitivity Computation

Given converged $V(\cdot; \Lambda)$ and $a^*(\cdot; \Lambda)$:

1. Perturb $w_k \to w_k + \varepsilon$ (renormalizing other weights)
2. Re-solve for $V(\cdot; \Lambda + \varepsilon e_k)$ and $a^*(\cdot; \Lambda + \varepsilon e_k)$
3. Compute finite difference: $\partial a^* / \partial w_k \approx [a^*(x; \Lambda + \varepsilon e_k) - a^*(x; \Lambda)]/\varepsilon$

For efficiency, use automatic differentiation through the value function iteration.

D.5 | Simplex Projection

When updating priority vectors under the stochastic process of Definition 4.8, the projection Π_{Δ^5} onto the probability simplex must be computed. Duchi, Shalev-Shwartz, Singer, and Chandra (2008) provide an efficient $O(n\log n)$ algorithm for projecting onto the ℓ_1- ball, which can be

adapted for simplex projection. This enables fast updates when simulating priority dynamics or when performing sensitivity analysis across the priority space.

D.6 | Robust Optimization

When priority uncertainty is significant, solve:

$$\max_{a} \min_{\Lambda \in \mathcal{U}} V(x, a; \Lambda)$$

where $\mathcal{U}$ is an uncertainty set around estimated $\hat{\Lambda}$.

For ellipsoidal uncertainty:

$$\mathcal{U} = \{\Lambda : (\Lambda - \hat{\Lambda})^{\top} \Sigma^{-1} (\Lambda - \hat{\Lambda}) \leq \chi^2_{\alpha}\}$$

Appendix E | Notation Glossary

This appendix provides a comprehensive reference for mathematical notation used throughout this paper.

E.1 | Greek Letters and Parameters

Symbol	Definition	First Appearance
Λ	Priority Matrix; diagonal matrix of domain weights representing client preferences	Section 1.2
λ_k	Priority weight for domain k	Section 4.2
β	Discount factor for future utility	Section 6.1
γ	Jump diffusion	Section 2.1
δ	Time step or small perturbation	Section 3.4
ε	Small positive constant	Section 4.3
η	Learning rate or adjustment parameter	Section 4.8
κ	Curvature bound	Section 7.3
μ	Drift coefficient in stochastic dynamics	Section 5.1
ν	Measure or spectral parameter	Section 4.3
π	Policy or strategy	Section 3.4
ρ	Correlation coefficient	Section 5.2
σ	Volatility or standard deviation	Section 5.1
τ	Time to horizon or stopping time	Section 6.2
ω	Shock or perturbation	Section 5.3

E.2 | Tensors and Matrices

Symbol	Definition	First Appearance
S	Structural Tensor; captures objective interdependencies between planning domains	Section 1.3
S_{ijk}	Component of Structural Tensor: sensitivity of domain i objective to domain j decision on domain k objective	Section 4.1
$\mathcal{S}$	Local structure tensor at state s	Section 4.2
$\mathcal{S}_{ijk}(s)$	Component of local structure tensor: element of $\partial^2_{d_i d_j} R_k(s)$	Section 4.2
Λ	Priority Matrix; $\Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_K)$	Section 4.2
T_{eff}	Effective Tensor; $T_{\text{eff}} = S \odot \Lambda$ filters structural interdependencies through priorities	Section 4.4
$\mathcal{T}$	Sensitivity tensor; higher-order derivatives of optimal value with respect to decisions	Section 7.2
$\mathcal{T}_{i_1 \cdots i_\ell}$	Component of sensitivity tensor: $\partial^\ell V^* / \partial d_{i_1} \cdots \partial d_{i_\ell}$	Section 7.2
Γ	Coupling Matrix; $\Gamma_{ij} = \sum_k S_{ijk} \cdot \lambda_k$	Section 4.3
$M(\Lambda)$	Dynamics Matrix; describes evolution of state under priority-weighted system	Section 5.4
$J_n(\lambda)$	Jordan block of size n with eigenvalue λ	Appendix B
H	Hessian matrix of value function	Appendix A
Σ	Covariance matrix	Section 5.2

E.3 | Spaces and Sets

Symbol	Definition	First Appearance
C^k	Space of k-times continuously differentiable functions	Section 4.1
$C^{2,\alpha}$	Hölder space; functions with Hölder continuous second derivatives	Section 3.4
$\mathcal{M}_0$	Primitive manifold; base state space capturing financial states	Section 2.1
Δ^5	Five-dimensional probability simplex for priority weights	Section 4.2
Π	Set of admissible strategies	Section 3.4
$A(x)$	Admissible action set at state x	Section 3.4
$\mathcal{D}$	Domain of value function	Section 6.1
$\mathcal{F}$	Fiber bundle over the primitive manifold	Section 3.2
$\mathcal{U}$	Uncertainty set for robust optimization	Appendix D.6
$\mathcal{Z}$	Jump space; mark space for Poisson random measure	Section 3.5
$\mathbb{R}^n$	n-dimensional Euclidean space	Section 2.1

E.4 | Stochastic Processes and State Variables

Symbol	Definition	First Appearance
W_t	Net worth at time t	Section 3.1
X_t	State process	Section 3.4
X_{t-}	Left-limit of process X at time t (pre-jump value)	Section 3.5
Y_t	Income flow process; also transformed/auxiliary process (context-dependent)	Section 3.1
E_t	Expense flow process; $E_t = E_t^F + E_t^D$ (fixed + discretionary)	Section 3.1
B_t	Standard Brownian motion	Section 3.1
dW_t	Standard Wiener process increment	Section 3.4
$\widetilde{N}(dt, dz)$	Compensated Poisson random measure	Section 3.4
$b(x, a)$	Drift coefficient function	Section 3.4
$\sigma(x, a)$	Diffusion coefficient function	Section 3.4
$\gamma(x, z)$	Jump size function	Section 3.4

E.5 | Functions and Operators

Symbol	Definition	First Appearance
$V(x; \Lambda)$	Value function; optimal expected utility from state x given priorities Λ	Section 6.1
V^*	Optimal value function	Section 6.1
a^*	Optimal action/policy	Section 6.1
$R_k(x, a)$	Objective function for domain k	Section 3.3
$Q(x, a; \Lambda)$	Priority-weighted aggregate objective	Section 6.1
$r(x, a)$	Instantaneous reward function	Section 6.1
$\partial f(x)$	Clarke generalized gradient of f at x	Section 4.5
$\partial^2 f(x)$	Clarke generalized Hessian	Appendix A
$D^- f(x)$	Viscosity subdifferential of f at x	Section 4.5
$D^+ f(x)$	Viscosity superdifferential of f at x	Section 4.5
$\mathcal{L}^a$	Generator of controlled diffusion under action a	Section 6.2
$\mathcal{L}_\sigma$	Diffusion operator; $\frac{1}{2}\mathrm{tr}(\sigma\sigma^\top D^2 u)$	Section 3.5
$\alpha(M)$	Spectral abscissa; $\max\{\Re(\lambda): \lambda \in \mathrm{spec}(M)\}$	Section 7.4
$\mathbb{E}[\cdot]$	Expectation operator	Section 3.5
Π_{Δ^5}	Projection operator onto probability simplex	Appendix D.5
∇	Gradient operator	Section 4.1
Δ	Laplacian operator	Section 6.2

E.6 | Special Operations

Symbol	Definition
$\odot$	Element-wise (Hadamard) product
$\mathrm{conv}\{\cdot\}$	Convex hull of a set
$\mathrm{diag}(\cdot)$	Diagonal matrix formed from vector entries
$\mathrm{spec}(\cdot)$	Spectrum (set of eigenvalues) of a matrix
$\Re(\cdot)$	Real part of a complex number
$\lVert\cdot\rVert$	Euclidean norm (unless otherwise specified)
$\langle\cdot,\cdot\rangle$	Inner product

E.7 | Theorem and Proposition Reference

Theorems

Theorem	Name	Location
Theorem 5.1	Shock Amplification	Section 5.2
Theorem 6.1	Existence and Uniqueness	Section 6.2
Theorem 6.2	Verification	Section 6.3
Theorem 7.1	Interdependency Tensor Structure	Section 7
Theorem 7.2	Priority Filtering	Section 7
Theorem 7.3	Stochastic Local Stability	Section 7
Theorem 7.4	Stochastic Separability	Section 7
Theorem 7.5	Priority Sensitivity	Section 7
Theorem 7.6	Temporal Coupling	Section 7
Theorem 7.7	Priority Alignment	Section 7
Theorem 7.8	Value of Integration	Section 7
Theorem 7.9	Shock Propagation	Section 7
Theorem 7.10	Rate Volatility Premium	Section 7
Theorem 7.11	Jump Anticipation Value	Section 7
Theorem 7.12	Regime-Dependent Optimal Policy	Section 7
Theorem 7.13	Stochastic Integration Premium Decomposition	Section 7

Propositions

Number	Name	Section
Proposition 3.1	Manifold Structure	Section 3.2
Proposition 3.2	Strategy Space Structure	Section 3.4
Proposition 4.1	Coupling Matrix Symmetry	Section 4.2
Proposition 4.2	Properties of Structural Tensor	Section 4.2
Proposition 4.3	Coupling Matrix Eigenstructure	Section 4.3

Number	Name	Section
Proposition 4.4	Properties of Effective Tensor	Section 4.4
Proposition 4.5	Clarke-Viscosity Relationship	Section 4.5
Proposition 5.1	Optimal Response by Uncertainty Type	Section 5.2
Proposition 6.1	HJB Equation	Section 6.2

E.8 | Temporal Priority Structure (Discount Rates)

Symbol	Definition	First Use
$\mathbf{B}(t)$	Discount Matrix; $\mathrm{diag}(\beta_{\mathrm{CF}}(t), \dots, \beta_{\mathrm{Inv}}(t))$	Section 4.3b
$\beta_k(t)$	Domain-specific discount rate for domain $k$at time t	Section 4.3b
$\bar{\beta}_k$	Baseline (long-run mean) discount rate for domain k	Section 4.3b
$f_k(X_t)$	Endogenous rate adjustment function for domain k	Section 4.3b
$\xi_k(t)$	Exogenous shock component for domain k	Section 4.3b
θ_k	Mean-reversion speed for domain $k$rate	Section 4.3b
σ_k^{β}	Diffusion volatility for domain $k$rate	Section 4.3b
$\gamma_k(z)$	Jump size function for domain k	Section 4.3b
ν_k	Lévy measure for domain $k$jumps	Section 4.3b
$N_k(dt, dz)$	Poisson random measure for domain k	Section 4.3b
$\widetilde{N}_k(dt, dz)$	Compensated Poisson random measure	Section 4.3b
$\mathcal{E}$	Set of systemic life events	Section 4.3b
$\mathcal{D}_e$	Domains affected by event e	Section 4.3b
$\lambda^{(e)}$	Intensity of systemic event e	Section 4.3b
$\mathcal{R}$	Set of life-stage regimes	Section 6.4
$\lambda_{rs}(x)$	Transition intensity from regime $r$to s	Section 6.4
$\Gamma_B(t)$	Discount-adjusted coupling matrix	Section 4.4

E.9 | Integration Premiums

Symbol	Definition	First Use
Π^{int}	Integration premium	Section 7
Π^{det}	Deterministic base premium	Theorem 7.13
Π^{σ}	Volatility hedge premium	Theorem 7.13
Π^{J}	Jump coordination premium	Theorem 7.13
Π^{Γ}	Coupling amplification premium	Theorem 7.13
Π_{σ}	Rate volatility premium	Theorem 7.10

Symbol	Definition	First Use
Π_J	Jump anticipation value	Theorem 7.11
ϕ	Stochastic amplification factor	Corollary 7.13.2

Appendix F | Computational Methods and Case Study Calculations

F.1 | Overview

This appendix documents the computational methodology for calculating integration premiums under the stochastic discount rate specification with domain-specific temporal structure. The calculations provide quantitative support for the case studies in Section 8 and validate the theoretical predictions of Theorems 7.8–7.13. The methodology incorporates the four-tier IIFPT architecture: Structural Tensor **S** (Tier 1), Priority Matrix Λ (Tier 2a), Discount Matrix $B(t)$ (Tier 2b), and the discount-adjusted Effective Tensor Γ_B.

F.1.1 | Computational Framework

The integration premium decomposition (Theorem 7.13) incorporates domain-specific discount rates:

$$\Pi^{total} = \Pi^{det} + \Pi^{\sigma} + \Pi^{J} + \Pi^{\Gamma}$$

where:

Π^{det}: Deterministic base premium from coordinated optimization, computed as $V^*_{int} - \Sigma\, V^*_{k,sep}$

Π^{σ}: Volatility hedge premium from rate fluctuation hedging via Itô correction

Π^{J}: Jump coordination premium from systemic event response across correlated β_k jumps

Π^{Γ}: Coupling amplification premium from the discount-adjusted coupling matrix $\Gamma_B = B^{-1/2}$ $T_{eff}\ B^{-1/2}$

F.1.2 | Key Assumptions

Linear-quadratic approximation: Domain objectives are approximated as quadratic functions near optimal states, enabling closed-form solutions for value function gradients.

Domain-specific discount rates: Each domain k operates with discount rate β_k(t) = $\bar{\beta}_k + f_k(X_t)$, where $\bar{\beta}_k$ is the baseline rate and f_k captures state-dependent adjustments.

Stochastic rate dynamics: Rates follow jump-diffusion processes with mean-reversion speeds κ_k, volatilities σ_k, and correlated jump events (calibrated from Tables 4.3, 4.4).

Monte Carlo simulation: 500 rate path simulations per client, 30-year horizon, quarterly time steps.

F.1.3 | Baseline Discount Rate Calibration

The domain-specific discount rates reflect different planning horizons, consistent with Table 8.1:

Table F.1: Domain-Specific Discount Rate Baselines

Domain	β_k	$t_{1/2}$	Interpretation
Cash Flow (CF)	6.0%	11.6 mo	Immediate concerns; high discounting
Risk Mgmt (RM)	7.0%	9.9 mo	Near-term protection focus
Investment (Inv)	5.0%	13.9 mo	Medium horizon; market-linked
Tax	4.0%	17.3 mo	Annual planning cycles
Retirement (Ret)	3.5%	19.8 mo	Long planning horizon
Estate (Est)	1.5%	46.2 mo	Terminal/bequest orientation

Half-life computed as $t_{1/2} = \ln(2)/\beta_k$. Higher-β domains respond quickly to shocks; lower-β domains (Estate) exhibit persistent effects.

F.2 | Deterministic Premium Calculation

F.2.1 |Methodology

The deterministic premium captures the value of coordinating across structurally coupled domains, filtered through client-specific priorities:

$$\Pi^{det} = V^*_{int} - \Sigma_k \, V^*_{k,sep}$$

where V^*_{int} is computed by solving the integrated HJB equation with coupling matrix T_{eff} = S $\odot$ Λ, and each $V^*_{k,sep}$ is computed by solving the single-domain HJB with T_{eff} restricted to domain k.

F.2.2 | Interpretation

The deterministic premium ranges from 5.9% to 11.6% of net worth across our client personas. The premium reflects: (1) structural interdependency strength in S; (2) priority heterogeneity in Λ that creates mismatch between siloed optimization and client values; and (3) complexity of the client's financial situation.

F.3 | Stochastic Premium Components

F.3.1 | Volatility Hedge Premium (Π^σ)

Calculation:

1. Simulate 500 rate paths under jump-diffusion dynamics for each β_k

2. Compute coupling strength $|\Gamma_B|$ at each time step

3. Calculate coefficient of variation: CV = $\sigma(|\Gamma_B|)$ / E[$|\Gamma_B|$]

4. Volatility premium: Π^σ = 0.30 × CV × Π^{det}

Interpretation: Integration captures approximately 30% of the hedging value from rate volatility. The Itô correction term creates optionality that integrated planning can harvest.

F.3.2 | Jump Coordination Premium (Π^J)

Calculation:

For each systemic event e affecting domains D_e:

$$\Pi_e^J = \lambda_e \times |D_e| \times (|D_e| - 1) / 2 \times E[\Delta\beta_k \mid \text{jump}] \times \Pi^{det} / 100$$

Total: $\Pi^J = \Sigma_e \, \Pi_e^J$

Interpretation: Events affecting multiple domains create pairwise coordination opportunities. The premium scales with event intensity λ_e, number of affected domains, and jump magnitudes. This is typically the largest stochastic component.

F.3.3 | Coupling Amplification Premium (Π^Γ)

Calculation:

1. Simulate rate paths and compute $\beta_k^{-1/2}$ for each domain

2. Compare $E[\Gamma_B]$ to $\Gamma_B|\beta=\bar{\beta}$ (baseline)

3. Jensen gap: $E[\beta^{-1/2}] > \bar{\beta}^{-1/2}$ (convexity)

4. Premium: $\Pi^\Gamma = (E[|\Gamma_B|] / |\Gamma_B|\text{baseline} - 1) \times \Pi^{det}$

Interpretation: Jensen's inequality ensures $E[\beta^{-1/2}] \geq \bar{\beta}^{-1/2}$; rate volatility amplifies expected coupling through the discount-adjusted matrix Γ_B. Different β_k cause shock propagation speeds to vary by domain.

F.4 | Case Study Results

F.4.1 | Section 8.1: Maria Delgado

Client Profile:

Age: 53 | Net Worth: $19,000,000 | Situation: Approaching retirement during market crisis

Table F.2: Maria's Domain-Specific Discount Rates

Domain	Baseline	Current	Status	Driver
Cash Flow	6.0%	8.1%	+35%	Market decline elevated concerns
Retirement	3.5%	5.3%	+51%	Approaching retirement age
Risk Mgmt	7.0%	8.5%	+21%	Sequence risk during drawdown
Investment	5.0%	5.8%	+16%	Portfolio decline
Tax	4.0%	4.3%	+8%	Minor adjustment

Estate	1.5%	1.3%	−13%	Deferred during crisis

Table F.3: Maria's Integration Premium Decomposition

Component	Value	% of Total	% of Π^{det}
Deterministic Premium	$2,200,000	87.5%	—
Volatility Hedge (Π^{σ})	$66,000	2.6%	3.0%
Jump Coordination (Π^{J})	$198,000	7.9%	9.0%
Coupling Amplification (Π^{Γ})	$49,000	2.0%	2.2%
Stochastic Total	$313,000	12.5%	14.2%
Total Integration Premium	$2,513,000	100%	—

Key Metrics: Premium as % of wealth: 13.2% | Deterministic %: 11.6% | Stochastic amplification: 14.2% | Elevated domains: 4 (CF, Ret, RM, Inv)

F.4.2 | Section 8.2: The Okonkwo Twins

Comparison of identical twins with *identical* financial positions but different Priority Matrices:

Table F.4: Okonkwo Twins Comparison

Metric	David (Security)	Daniel (Growth)
Net Worth	$6,000,000	$6,000,000
Age	52	52
Primary Focus	RM/Ret/CF	Inv/Tax/Ret
Top Priority Weight	wRM = 0.30	wInv = 0.28
Siloed Value (Σ V*k,sep)	$800,000	$865,000
Integrated Value (V*int)	$1,270,000	$1,445,000
Deterministic Premium	$470,000	$580,000
Premium as % NW	7.8%	9.7%
Stochastic Enhancement	12.0%	15.0%
Total Premium	$526,400	$667,000
Total as % NW	8.8%	11.1%

Insight: Daniel's higher stochastic enhancement (15.0% vs 12.0%) reflects his growth-oriented priorities creating more exposure to Investment-Tax coupling, which exhibits higher rate volatility. Identical Structural Tensor S, filtered through different Priority Matrix Λ, yields different T_{eff} and thus different optimal strategies.

F.4.3 | Section 8.3: Victoria Chen and Robert Hamill

Comparison of different structural situations with *identical* Priority Matrices:

Table F.5: Chen vs. Hamill Comparison

Metric	Victoria Chen	Robert Hamill

Net Worth	$20,000,000	$2,300,000
Age	61	61
Situation	CTO, concentrated stock	Retired pilot, pension
Key Structural Feature	$8M company stock (60%)	$96K pension, diversified
Siloed Value (Σ V*k,sep)	$1,420,000	$142,000
Integrated Value (V*int)	$2,600,000	$303,000
Deterministic Premium	$1,180,000	$161,000
Premium as % NW	5.9%	7.0%
Stochastic Enhancement	16.0%	10.0%
Total Premium	$1,368,800	$177,100
Total as % NW	6.8%	7.7%

Insight: Victoria's higher stochastic enhancement (16.0% vs 10.0%) reflects her concentrated stock position creating strong negative coupling SInv,Tax < 0 (selling triggers tax). Robert's diversified structure has weaker structural couplings. Same Λ but different S yields ~7× the absolute premium for Victoria.

F.4.4 | Section 8.4: Marcus Jones — Shock Impact Analysis

Client Profile: Age: 34 | Net Worth: $800,000 | Situation: Healthcare tech, equity compensation

Table F.6: Marcus's Pre-Shock Discount Rates (Baseline)

Domain	Baseline	Current	Status
Cash Flow	6.0%	6.0%	At baseline
Retirement	3.5%	3.5%	At baseline
Risk Mgmt	7.0%	7.0%	At baseline
Investment	5.0%	5.0%	At baseline
Tax	4.0%	4.0%	At baseline
Estate	1.5%	1.5%	At baseline

Table F.7: Marcus's Post-Shock Discount Rates (Health Diagnosis)

Domain	Pre	Post	Jump	$t_{1/2}$ Change
Cash Flow	6.0%	10.5%	+4.5%	11.6 → 6.6 mo
Estate	1.5%	4.7%	+3.2%	46.2 → 14.7 mo
Risk Mgmt	7.0%	12.8%	+5.8%	9.9 → 5.4 mo
Retirement	3.5%	3.5%	0	No change
Investment	5.0%	5.0%	0	No change
Tax	4.0%	4.0%	0	No change

Table F.8: Marcus's Premium Comparison (Pre vs. Post-Shock)

Component	Pre-Shock	Post-Shock	Change

Deterministic Premium	$60,000	$72,000	+20%
Volatility Hedge (Π^{σ})	$2,400	$4,320	+80%
Jump Coordination (Π^{J})	$4,200	$10,800	+157%
Coupling Amplification (Π^{Γ})	$1,800	$4,680	+160%
Stochastic Total	$8,400	$19,800	+136%
Total Premium	$68,400	$91,800	+34%
% of Net Worth	8.6%	11.5%	+2.9 pp
Stochastic Enhancement	14.0%	27.5%	+13.5 pp

Key Findings: The health diagnosis triggered correlated rate jumps across CF, Est, and RM domains simultaneously. Total premium increased by 34%. Stochastic enhancement nearly doubled from 14% to 28%. The jump coordination component showed the largest relative increase (+157%) because the shock affected 3 domains with strong structural couplings. This validates Theorem 7.11 (Jump Anticipation Value): integration becomes dramatically more valuable when multiple domains experience simultaneous urgency changes.

F.5 |Summary Statistics

F.5.1 | Cross-Client Comparison

Table F.9: Integration Premium Summary Across All Personas

Client	NW	Π^{det}	% NW	Π^{stoch}	Π^{total}	% NW
Maria Delgado	$19.0M	$2.20M	11.6%	$313K	$2.51M	13.2%
David Okonkwo	$6.0M	$470K	7.8%	$56K	$526K	8.8%
Daniel Okonkwo	$6.0M	$580K	9.7%	$87K	$667K	11.1%
Victoria Chen	$20.0M	$1.18M	5.9%	$189K	$1.37M	6.8%
Robert Hamill	$2.3M	$161K	7.0%	$16K	$177K	7.7%
Marcus Jones*	$0.8M	$72K	9.0%	$20K	$92K	11.5%
RANGE	$0.8–20M	$72K–2.2M	5.9–11.6%	$16–313K	$92K–2.5M	6.8–13.2%

**Marcus Jones post-shock values shown*

F.5.2 | Validation Against Theoretical Predictions

Table F.10: Theoretical Validation

Theoretical Prediction	Computed Range	Status
Π^{det} 5–15% of wealth (Theorem 7.8)	5.9%–11.6%	✓ Confirmed
Stochastic enhancement 10–30% (Theorem 7.13)	10.0%–28.0%	✓ Confirmed
Π^J increases with correlated jumps	+157% for Marcus shock	✓ Confirmed
Same S, different Λ → different strategies	Okonkwo twins	✓ Confirmed
Same Λ, different S → different premiums	Chen vs. Hamill	✓ Confirmed
Higher β_k → faster shock response	$t_{1/2}$ = ln(2)/β_k verified	✓ Confirmed

F.6 | Computational Parameters

F.6.1 | Stochastic Rate Parameters (from Tables 4.3, 4.4)

Mean-Reversion Speeds (κ_k): CF: 2.0 yr^{-1}, RM: 2.5 yr^{-1}, Inv: 0.8 yr^{-1}, Tax: 1.0 yr^{-1}, Ret: 0.5 yr^{-1}, Est: 0.3 yr^{-1}

Rate Volatilities (σ_k): CF: 1.5%, RM: 2.0%, Inv: 1.2%, Tax: 1.0%, Ret: 0.8%, Est: 0.5%

Systemic Events:

Health diagnosis: λ = 0.05 yr^{-1}, affects CF/Est/RM

Job loss: λ = 0.03 yr^{-1}, affects CF/Ret

Market crisis: λ = 0.08 yr^{-1}, affects RM/Inv/CF

F.6.2 | Simulation Parameters

Number of simulations: 500

Time horizon: 30 years

Time step: 0.25 years (quarterly)

Random seed: Varied for Monte Carlo estimation

F.7 Code Availability

The computational framework is implemented in Python and available in the supplementary materials:

iifpt_core.py: Main IIFPT classes including Structural Tensor, Priority Matrix, Discount Matrix, and premium calculation

iifpt_stochastic.py: Stochastic rate dynamics, jump-diffusion simulation, and Γ_B computation

case_studies.py: Section 8 persona analysis with full premium decomposition

Dependencies: NumPy, SciPy